WRRA Neuropsychology

Exploring WRRA's Potential for Integration Across Disciplines

Wonsik Choi · Core-linked bilingual edition v0.9 · 22 October 2026

This book examines the explanatory and computational contributions WRRA may make to connecting different disciplines. Psychology and neuroscience provide the first case: accounts of experience and behavior, the nervous system and the body are connected through a common structure.

Starting from the evidence in each field, readers can identify the connections actually made within WRRA, their outputs, and the questions still requiring investigation. The central question is what these connections add to understanding and computing phenomena.

Preface

This book begins with a question. Can WRRA connect the ways different disciplines explain the same phenomenon, and generate meaningful explanations and computations through those connections? We chose psychology and neuroscience as the first case for exploring this possibility.

Here, integration across disciplines means connecting explanations from fields with different objects of study and units of measurement. We preserve the respective meanings of experience reports, behavior, neural data, and bodily data while specifying their relationships through states, transformations, and observation rules. We distinguish descriptions rewritten in a common vocabulary from connections that actually enter an explanation or computation.

The book is for readers interested in mind and behavior and researchers exploring connections between disciplines. It starts with familiar phenomena—memory and learning, emotion and relationships—examines what each field has established, and asks what connecting them through WRRA can additionally explain. The familiar topical organization of general psychology provides a guide to this inquiry.

WRRA's purpose and starting point

WRRA stands for Wonsik Reality-Renderer Architecture. It is a framework for explaining a single, nonprivileged universe through the common method of minimal computation (termed Minimum Computation in Core 1.0), common carriers, and phenotypes. This book examines whether that method is also useful for connecting phenomena studied by different disciplines. The explanation below draws on Wonsik Choi's WRRA Core 1.0 and Wonsik Choi and Jeongin Choi's *How Life Runs*. Appendix A presents the standard definitions. (Choi, 2026 W1; Choi and Choi, 2026 W2)

WRRA asks not only what result occurred but also what it received as input and through which relationships and rules it was executed. Distinguishing input, present state, the influence of previous experience, transmission and updating, rendering and observation allows us to specify where disciplinary explanations connect and where links are missing. Applications in particular fields make this execution structure concrete for the object being studied.

Minimal computation is a criterion for preserving required function

Minimal computation first specifies the functions to preserve and the errors that can be tolerated, then uses the states and processing needed to meet those criteria. A reduction in time that sacrifices accuracy or safety does not count as performing the same function at lower cost. The

margin and redundancy needed to preserve function also enter the cost accounting. What was reduced and what was preserved must both be recorded.

A common carrier asks what is shared

A common carrier is a transmission or updating structure reused for distinguishable information or operations. Even where the underlying infrastructure is shared, information content, ownership, transmission paths, and purposes remain distinct. Networks reused by several functions are possible candidates in neuropsychology. What is actually shared requires investigation; the term common carrier alone does not identify a particular brain region or neurotransmitter.

Distinguishing phenotype from observable

A phenotype is an implemented result produced from an internal state through execution and rendering rules under boundary conditions. An observable is a value read from that result using an instrument and a procedure. This distinction separates behavior generated by a model from behavior actually recorded, and a person's reported experience from a signal measured by a researcher. Rules must connect what was observed to the internal state inferred from it.

Connecting present state and residue through execution structure

Present state comprises the conditions involved in execution now. Residue is the part of a previous event's influence that remains active in the present; it can include learned skills, familiarity, and trust. The past enters execution through present states, residues, or accessible records. A model that reads records must specify their present location and access paths. Residue may be part of present state, so the definitions must prevent counting the same information twice.

A concrete application specifies inputs and their sources, then states how relationships and rules change the state. Boundaries define the system's scope and permissible flows. Updating rules determine the next state; rendering and observation rules determine what can be read from it. The ledger records ownership, units, times, sources, and transformations. Actual structures may contain feedback and parallel processing: phenomena are not assumed to pass just once through a single linear sequence.

Reading one hypothetical scene

Consider a hypothetical study of a person who stops speaking just before a presentation. The current question and the audience's reactions are candidate inputs. Fatigue and task

understanding are candidate present conditions. The influence remaining from previous presentations is a candidate residue; speaking time and authority are boundaries of the task and relationships. This is an illustration of constructing research variables, not a finding that establishes the cause of silence.

A WRRA application specifies how to measure these candidates and the rules connecting them, then explains or computes how changes in conditions affect the timing or choice of speaking. Connecting psychological reports and behavioral data to bodily and neural data also requires connecting measurement times and errors. If a neural pathway corresponding to a common carrier has not been measured, that pathway remains a question for subsequent research. Reproduction of a hypothetical computation and fit to actual people receive separate verdicts.

In this book, neuroscience and brain science refer to overlapping fields. The research focus is on connections between psychology and neuroscience that actually contribute to explanation and computation. Readers can apply these questions to the chapters on memory and learning, emotion and relationships.

Criteria for assessing integration's contribution

Claims are assessed in the order validation inputs → WRRA-specific transformation → output → falsification condition. We identify the evidence from each field, the structures and rules by which WRRA connects it, what the connection explains or computes, and the results that would require revising it. An external theory is recorded as integrated only to the extent that it is actually connected to and executed in computations within WRRA.

Calibration using verified observations and reproduction of known values through the framework's own rules count as legitimate construction, reality consistency, and explanatory performance. Originality is assessed in the structure of connections, transformations, and explanations. An unmeasured quantity generated by a fixed model is recorded as that model's prediction. Independent prediction is not a mandatory condition of integration's contribution or originality.

What this book provides

Chapters 1–39 cover existing research and its proposed WRRA correspondences. Chapter 40 onward examines model construction, computation, and the conditions for integration. The hypothetical-input computations in Chapters 41 and 42 can be reproduced using the code in Appendix D. The conceptual connections and hypothetical computations presented here establish the basis for identifying subsequent tasks involving actual behavioral and neural data.

Each chapter follows eight stages: scene and question; starting points in established research; validation inputs; WRRA-specific transformation; outputs and execution record; falsification conditions and interpretive limits; current verdict; and next questions and evidence. Author–year citations point to existing research; WRRA interpretations and computational records identify the author's work. Everyday scenes are hypothetical illustrations unless presented as experimental examples with sources. References are in Appendix E and the common writing contract in Appendix F.

Readers who want to follow a connection from beginning to end may first read the speaking example in Section 4.2 of Chapter 45. It shows how the scene in Chapter 1 connects to the hypothetical computation in Chapter 42 and where links remain missing.

Common rules for reading chapter specifications

Author–year citations refer to the existing research in the immediately preceding sentence or paragraph. Subsequent WRRA transformations, hypothetical scenes, and proposed studies are the author's work. Support for existing research does not substitute for empirical validation of the WRRA application. This rule applies throughout the book.

The planned inputs in Section 3 and planned outputs in Section 5 are specifications for future investigation. Without a separate execution record, they should not be read as results collected, analyzed, or executed for this manuscript. The terminology correspondences in Section 4 are a starting point for constructing execution rules; they do not by themselves establish completed computation or integration of neural mechanisms.

Empirical tests in Section 6 whose tolerances have not been specified are OPEN. When such tests are undertaken, comparison criteria must be fixed before examining results, in light of measurement error and the research purpose. INFERRED in Section 7 marks a proposed explanatory structure; OPEN marks unresolved empirical fit. Appendix F explains the verdicts and writing principles in detail.

Reference identifiers in Section 8 lead to the bibliographic details, DOIs, and URLs in Appendix E. Core definitions are in Appendix A; numerical results and reproduction code are in Appendix D. Shared explanations are consolidated here, leaving each chapter's particular inputs, transformations, outputs, and falsification conditions in place.

This book is a Domain Profile applying the frozen Core 1.0. Appendix A11 connects the paper to this application. Core 1.0 DOI · Official full text.

Contents

Part 1 Foundations of Psychology and the WRRA Research Perspective

Chapter 1 The Subject Matter and Main Perspectives of Psychology

Chapter 2 The Basic Structure of WRRA Neuropsychology

Chapter 3 Research Methods and Causal Inference

Chapter 4 Measurement, Statistics, Reproducibility, and Research Ethics

Part 2 Biological Bases of Mind and Behavior

Chapter 5 Neurons, Synapses, and Neural Signal Transmission

Chapter 6 Brain Structure and Functional Networks

Chapter 7 The Body, Endocrine System, and Interoception

Chapter 8 Heredity, Development, Experience, and Neuroplasticity

Part 3 Sensation Perception Attention and Consciousness

Chapter 9 Sensation

Chapter 10 Perception

Chapter 11 Attention

Chapter 12 Consciousness, Nonconscious Processing, Sleep, and Dreams

Part 4 Learning and Memory

Chapter 13 Conditioning and Reinforcement

Chapter 14 Observational Learning, Skill Acquisition, and Habits

Chapter 15 Memory Encoding, Retention, and Retrieval

Chapter 16 Forgetting, Memory Reconstruction, and Updating

Part 5 Thinking Language and Intelligence

Chapter 17 Concepts, Reasoning, Problem Solving, and Creativity

Chapter 18 Judgment, Decision Making, Uncertainty, and Cognitive Bias

Chapter 19 Language Comprehension, Production, and Acquisition

Chapter 20 Intelligence, Cognitive Abilities, Individual Differences, and Measurement

Part 6 Motivation Emotion and Self Regulation

Chapter 21 Needs, Rewards, Goals, and the Direction of Action

Chapter 22 Emotional Experience, Bodily Responses, and Expression

Chapter 23 Emotion Regulation, Self-Control, and Action Selection

Chapter 24 Stress, Fatigue, Recovery, and Available Resources

Part 7 Development Personality and Self

Chapter 25 Infancy and Childhood

Chapter 26 Adolescence and Adulthood

Chapter 27 Later Life

Chapter 28 Temperament, Personality, and Persistent Individual Differences

Chapter 29 Self-Concept, Identity, and Sense of Agency

Part 8 Society Relationships and Culture

Chapter 30 Social Perception, Attribution, Attitudes, and Persuasion

Chapter 31 Attachment, Intimacy, Family, and Interpersonal Relationships

Chapter 32 Conformity, Obedience, Groups, Cooperation, and Conflict

Chapter 33 Culture, Norms, Social Environments, and Mind

Part 9 Mental Health Treatment and Practical Applications

Chapter 34 Assessing Mental Health and Psychological Difficulties

Chapter 35 Anxiety-, Obsession-, and Trauma-Related Phenomena

Chapter 36 Depression, Mood Changes, and Low Energy and Motivation

Chapter 37 Addiction-, Psychosis-, Neurodevelopmental-, and Neurocognitive-Related Phenomena

Chapter 38 Psychotherapy, Biological Treatment, and Environmental Intervention

Chapter 39 Applications in Education, Work, Health, and Technology Environments

Part 10 WRRA Computational Models and Integrative Research

Chapter 40 Constructing State Variables and Transformation Rules

Chapter 41 Computational Example One: Maintaining a Memory Trace

Chapter 42 Computational Example Two: Failure Experience and the Next Choice

Chapter 43 Connecting Neural Activity, Bodily Responses, Experience Reports, and Behavior

Chapter 44 Model Calibration, Reproduction, Prediction, and Falsification

Chapter 45 Achievements and Open Research Questions

Appendix A Standard Explanation of WRRA 1.0

Appendix B Terminology Correspondences

Appendix C Measurement and Variable Specifications

Appendix D Numerical Results and Reproduction Code

Appendix E References and Thematic Reading

Appendix F Common Writing Contract for the WRRA Academic Series

Part 1 Foundations of Psychology and the WRRA Research Perspective

Chapter 1 The Subject Matter and Main Perspectives of Psychology

1 Scene and question

Someone opens the door and a conversation stops. One person feels pleased, another becomes tense, and someone else notices no change. Psychology studies these differences through reports of experience and data on behavior, the body, and the environment. Its starting point is to specify both what needs explaining and how to measure it. (Spielman et al., 2020, §1.1 R1)

1.1 Working backward from a presentation situation

Suppose a hypothetical researcher wants to understand a participant's speaking in a meeting. The researcher first records the number of opportunities, the choice to speak, the time until speaking begins, and earlier feedback. Task understanding, current fatigue, and the relationship with a supervisor are then examined. This sequence does not prejudge silence as a personality trait.

An explanation involving earlier criticism can be investigated by asking whether differences remain when present conditions are matched. If guaranteeing a turn to speak reduces the difference, an environmental boundary may matter. Both may operate together. The researcher revises the structure in response to observations while recording which evidence prompted each change.

Chapter question: Why does the same event lead to different experiences and behavior in different people?

2 Starting points in established research

Behaviorism emphasizes observable learning relationships; cognitive approaches emphasize the selection and processing of information; biological approaches examine the nervous system and the body. Psychodynamic approaches have focused on unconscious motives and conflicts, and humanistic approaches on meaning and growth. A perspective's historical importance must be distinguished from empirical support for particular claims. Membership in a school does not establish every claim it makes. (Spielman et al., 2020, §1.2 R1)

2.1 What psychology seeks to explain

This book distinguishes four research tasks: describing what happened, explaining the conditions to which it is related, deriving outcomes under specified conditions, and investigating how outcomes change when conditions change. These tasks support one another but are different

achievements. Some studies first need accurate observations; others connect established facts within a common structure.

For someone who stops speaking before a presentation, the frequency and duration of silence can be recorded. To investigate why, information about task difficulty, audience reactions, speaking opportunities, the person's thoughts, and bodily state must be added. Renaming silence as anxiety does not complete an explanation. An examinable connection is needed between observed behavior and the process proposed to produce it.

2.2 Why experience and behavior both matter

A person's statement that they felt uncomfortable is experience-report data. A researcher takes that report seriously while documenting what was asked, when, and how. Behavioral records are another kind of data. Someone may feel confident but answer inaccurately, or feel uncomfortable while performing accurately. Neither kind of evidence invariably substitutes for the other. (Spielman et al., 2020, §2.2 R1)

This book connects mind and behavior while preserving their distinct indicators. Studying experience requires specifying reporting conditions; studying behavior requires specifying possible choices and task rules. If neural activity is measured, a separate account is needed of the processes the indicator reflects. Disagreement among the three kinds of data is itself a research finding to investigate.

2.3 Questions connecting the main perspectives

Behavioral approaches ask which stimuli and consequences change responses. Cognitive approaches examine attention and memory, expectation and judgment. Biological approaches ask how these processes are implemented in the nervous system and body. Sociocultural approaches examine how relationships and institutions change inputs and choices. Each provides important distinctions at a different level. (Spielman et al., 2020, §1.3 R1)

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: current stimuli, experience reports, and behavioral records. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: distinguish present state from the residue of past influences.

WRRA neuropsychology addresses these questions through connections among present state, experiential residue, relationships and boundaries, resources, updating, and phenotype. Restating them in a shared grammar is the starting point; the ultimate aim is to identify the connections that contribute to explanation and computation. Existing findings serve as inputs and comparison material, while the transformations actually performed in WRRA are recorded separately.

4.1 Keeping explanatory levels distinct

Cellular changes, circuit activity, personal judgments, and group norms have different observational scales. A variable describing a group rule cannot simply be identified with a substance in a neuron. Conversely, the absence of neural measurements does not make a relational explanation meaningless. What was explained and which levels were connected must both be stated.

If participation increases after speaking rights are expanded, the explanatory performance of a social-boundary change can be evaluated. The brain circuits mediating that change remain a further question. This book records integration only for connections that actually enter computation. Citing a neuroscience paper does not establish completed circuit implementation.

4.2 The research position of WRRA neuropsychology

Our subject is the process in which experience, body, and environment jointly constitute present responses and their consequences remain in the next present. WRRA's distinctiveness lies in connecting fields through a common transformation grammar and making that structure concrete as executable rules.

Reproducing a known phenomenon through its own rules counts as consistency and explanatory performance. Research does not have to begin with a new number. If unmeasured conditions can subsequently be computed, predictive performance can be added. Existing theories and WRRA are evaluated using the same data and criteria.

4.3 From existing research to a WRRA application

WRRA connects these questions through a common recording format. What residue remains from reinforcement history? In which present state is expectation measured? Which path did a neural signal follow? How did an institution define the boundary of possible action? These questions preserve differences among schools while showing which part of the same phenomenon each claim explains.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned output: a structure explaining differences in responses across situations.

6 Falsification conditions and interpretive limits

Falsification condition: revise the explanation if the specified variables and rules fail to reproduce the observed differences.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Explaining differences in responses requires recording experience, behavior, and their conditions together.

8 Next questions and evidence

Next research question: Do response differences persist in repeated observations that separate presentational inputs from experiential history?

8.1 Questions to check after reading

When observing behavior, can you distinguish direct observations from inferred causes? If experience reports differ from bodily responses, what else should be measured? When a familiar psychological explanation is restated in terms of present input, residue, relationships, boundaries, and rendering, what connection has actually been added? Answering these questions prepares the reader for the next chapter.

Chapter references: R1, R2, R7, R21.

Chapter 2 The Basic Structure of WRRA Neuropsychology

1 Scene and question

Chapter question: How can a psychological phenomenon be specified through WRRA's execution structure?

2 Starting points in established research

Established neuropsychology studies relationships among brain function, behavior, and cognition. The distinction below between present state and residue is defined for this book's application of Core 1.0; it is not presented as standard terminology throughout established neuropsychology. (APA, n.d. R2)

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: the observational target, measurement scope, and execution conditions. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: map Core elements to variables and observation paths in the field.

WRRA uses the name Wonsik Reality-Renderer Architecture. This book follows Core 1.0, frozen on 7 September 2026. Minimal computation reduces unnecessary states and processing while preserving required function and stability. A common carrier allows distinguishable information to share reusable transmission and updating structures. A phenotype is a result actually realized from an internal state under particular conditions.

4.1 Reading a scene as an execution structure

Consider a hypothetical learner viewing a problem and selecting an answer. The displayed problem is input. Currently maintained information and bodily conditions constitute present state. Learning from earlier problems and the influence of failures can be treated as residue active now. The time limit and available answers are boundary conditions. The answer actually selected and the time taken become observed outcomes under those conditions.

Memory is recorded separately from present state to distinguish rapidly changing components from those carrying prior experience, not because they necessarily occupy physically separate locations. If residue is already included in the complete present state, the same information is not added again. Before adding variables, we must explain what new distinction they introduce.

4.2 Defining minimal computation alongside function

Suppose one model stores every earlier trial, while another retains only a summary needed for the next choice. If both meet specified accuracy and stability criteria, memory, processing time, and the number of required relationships can be compared. Where no model minimizes every cost, several qualifying trade-offs may remain.

Minimal computation is distinct from the claim that the actual brain finds a global optimum at every moment. The design principle here is to seek a minimal implementation satisfying a particular task's requirements. A different task may require different memory. What is minimal must therefore be interpreted alongside the declared task and system boundary.

4.3 What reusing a common carrier means

The same transmission network may process different information without erasing its identity. A proposed common carrier in this book must specify both the shared structure and the rules distinguishing channels. Observing a region involved in several functions does not establish the entire carrier structure.

One approach is to investigate interference between two tasks requiring shared resources. However, interference may also arise from instruction comprehension or response conflict. Experiments must distinguish its location. Common carrier is a principle for specifying sharing and differentiation together, rather than a label explaining all differences through one cause.

4.4 Separating state updating from rendering

What has been learned and what can be demonstrated now may differ. Separate rules for the state update following learning and for reading current state as behavior can express this difference. Memory may persist while response time is insufficient; changed behavior does not by itself establish lasting learning.

An observation path also separates neural state from an experience report. Selecting a neural indicator as a candidate residue does not make that indicator the memory content itself. Connections among the indicator, learning conditions, and actual behavior require examination. Missing rules between explanatory levels remain research questions.

4.5 Specifying resource dimensions and boundaries

Time, sleep, metabolic energy, attentional task capacity, and social support have different units. A study selects relevant dimensions and measures them directly or specifies valid proxies. Defining resource insufficiency solely from poor performance creates circularity by incorporating the outcome into the definition of its cause.

Boundaries also require differentiation. Physical boundaries created by bodily membranes or sensory organs differ in implementation from task response restrictions or workplace authority. If changing authority improves outcomes, the effect of a social boundary can be evaluated. It is not automatically recorded as an increase in an individual's internal resources.

4.6 A ledger of records and ownership

Specifying who owns an experience, who owns an observation, and which model performed a computation makes the source of each achievement visible. A person's memory differs from a researcher's log, and an on-screen hint differs from internally maintained information. Use of external records requires including their access costs and availability among present conditions.

An execution ledger records input sources, rule versions, how coefficients were determined, outputs, and errors. If rules were adopted from an existing theory, this is stated along with the parts actually executed through WRRA's state and observation connections. Shared knowledge and computations performed within the framework can both be made transparent.

4.7 From existing research to a WRRA application

In this application, present state means neural, bodily, and cognitive conditions involved in the current response. Residue is the influence of earlier events that remains active in the present. They are not physically isolated boxes. When the complete present state is recorded, residue may be a component, so definitions must prevent double counting. Records and their access paths must also exist in the present if they are to be read.

4.8 Specifying an application against Core 1.0

The reference is WRRA Core 1.0, frozen on 7 September 2026. Its three invariants—minimum computation, the common carrier, and phenotype—and eleven required execution elements are retained. Minimum computation is Pareto-minimal cost subject to a declared function and error tolerance, rather than unconditional reduction of processing. [W1, §§0–3]

This revision specifies a neuropsychology Domain Profile without changing the Core. It prioritizes state-dependent interpretation, the perception–action loop, and the distinction between state estimation and control. History enters through presently instantiated state, residue, or accessible records, with ownership and duplication specified. A common carrier requires an explicit account of which channels share what; a proposed neural implementation requires separate testing. Appendix C2 provides the mapping, and C4 records the origins of suggestions.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned output: a variable-definition document for neuropsychological applications.

6 Falsification conditions and interpretive limits

Falsification condition: rewrite variable definitions if correspondences are ambiguous or duplicated.

Relationships describe connections of influence; boundaries specify the inside and outside of a system and conditions of passage. Resources are made concrete as time, energy, available devices, support, and similar dimensions. One resource score does not replace them all. Changing a coefficient or measurement method in neuropsychology does not automatically change Core 1.0 definitions. Appendix A provides the detailed standard explanation.

6.1 Conditions for completing an explanatory contract

Interpretation, computation, prediction, and control are distinct tasks in this book. A model explaining current data is assessed by which data it reproduces. Actual outputs derived from inputs and rules establish computational performance. Outputs for unmeasured conditions from a fixed model count as predictions; effects of interventions on state are validated separately.

A missing connection is left open. This separates what has been established from what needs further research rather than abandoning the whole explanation. Changes to field-specific coefficients or hypotheses leave minimal computation, common carriers, phenotypes, and Core execution principles intact unless separately revised.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: A WRRA application makes the common Core concrete through task-specific states, updating rules, and observation rules.

8 Next questions and evidence

Next research question: Can the units and observation paths of state, residue, and resources be fixed for one task?

8.1 Practicing a variable specification

Choose an action frequently repeated during a day and separately list input, present conditions, traces of experience, possible actions, and observed results. Mark which were measured directly and which were inferred. Finally, specify which output would change if a particular input changed. Explaining or computing that connection produces a concrete application.

Chapter references: R2, R7.

Chapter 3 Research Methods and Causal Inference

1 Scene and question

Suppose people who speak less in meetings also have higher anxiety scores. Anxiety may have caused silence, but an organization withholding speaking opportunities might have produced both. Covariation and the claim that changing one condition changes another require different evidence.

Chapter question: How can the influence of past experience be distinguished from present conditions?

2 Starting points in established research

Experiments manipulate and compare conditions; observations and surveys examine everyday variation that is difficult to manipulate. Longitudinal studies follow the same targets over time. Random assignment reduces systematic differences between comparison conditions but does not automatically eliminate attrition or measurement error. Sample scope and research setting must also enter interpretation. (Spielman et al., 2020, §2.2 R1)

2.1 From a question to a research design

A research question must become a measurable contrast. Asking whether failure reduces challenge-taking requires defining failure, the choice constituting a challenge, and the comparison time. Actual failure differs from believing one has failed; encountering a difficult task can also be distinguished from receiving negative feedback.

First specify the outcome and competing explanations. Earlier failure residue might influence it, or participants might rationally assess the next task's success probability as lower. Separating current difficulty from feedback information helps distinguish these paths. A useful research question shows which comparison separates explanations.

2.2 Observation, interviews, surveys, and case studies

Everyday observation reveals contexts difficult to capture in a laboratory. Interviews explore participants' meanings; surveys examine response distributions to common questions. Case studies document rare or complex pathways in detail. Each method accesses some evidence and may miss other evidence. (Spielman et al., 2020, §2.2 R1)

No method guarantees causality by its name. Observational studies may strengthen causal inference using temporal ordering and natural changes in conditions, while experiments may become difficult to interpret if manipulation differs from participants' actual experience. (Spielman et al., 2020, §2.3 R1)

2.3 What random assignment does

Random assignment sets participants' probabilities of entering comparison conditions according to the design. It helps reduce confounding in sufficiently large samples, although chance imbalances can remain in small samples. Recruitment and assignment are separate: good assignment does not remove generalization limits arising from recruitment from one particular population. (Spielman et al., 2020, §2.2 R1)

Task instructions, experimenter behavior, equipment, and measurement times should be matched, with assessor blinding considered where possible. Checks are also needed on how participants understood the manipulation. A manipulation check may itself influence subsequent behavior, so its timing belongs in the design. (Spielman et al., 2020, §2.2 R1)

2.4 Between-person and within-person comparisons

Between-group designs assign different people to different conditions. Within-person designs expose the same person to several conditions. The latter reduce individual differences but may mix practice, fatigue, and carryover into later conditions. This motivates varying order and specifying intervals between trials. (Spielman et al., 2020, §2.2 R1)

2.5 Longitudinal data and mediation paths

Measuring a candidate cause before an outcome is important but insufficient. A common cause may affect both processes. Repeated measurements also require intervals appropriate to the processes' speeds: daily records cannot directly establish the direction of second-scale connections. (Spielman et al., 2020, §2.3 R1)

Suppose an intervention changes bodily responses and behavior. Explaining the bodily response as a mediator requires additional assumptions.

3 Validation inputs

Literature-supported design principles are distinguished from planned inputs. Planned inputs: experiential histories, manipulated conditions, and comparison-group data. The score example uses hypothetical values—Group A, 70→75; Group B, 50→60—not actual survey data.

4 WRRA-specific transformation

Transformation specification: estimate residue's contribution while controlling present conditions.

A WRRA claim about residue requires identifying past influence distinct from other present conditions. Comparing people with and without earlier failure does not identify its residue if current task difficulty and success probability also differ. Before experimentation, specify which term each manipulation changes and how other changes will be checked.

4.1 Observation, interviews, surveys, and case studies — WRRA interpretation

The WRRA ledger records how data were collected and which conclusions that method permits.

In WRRA, order effects may be both noise to control and the residue under investigation. Decide beforehand which influence to control and which to measure. If an earlier condition continues to

affect performance after the task changes, it can enter the model as part of present state. Residue to be removed and residue to be investigated are not treated as identical.

4.2 Longitudinal data and mediation paths — WRRA interpretation

WRRA specifies intervention locations, intermediate states, output times, and observation paths, and compares alternatives. Drawing a path and identifying causality are separate achievements.

5 Outputs and execution record

Execution record: the hypothetical final-level difference is 75−60=15 points. Changes are 5 points for A and 10 for B. This arithmetic example shows how the question depends on the comparison quantity. Planned output: estimated condition differences and causal effects. No actual causal effect was estimated.

5.1 A hypothetical example of score differences and changes

Suppose A's mean changes from 70 to 75 and B's from 50 to 60. A finishes 15 points higher, but its change is 5 points versus B's 10, so the difference in change points in the opposite direction. The calculations are simple; the comparison questions differ.

This does not prove that B's intervention caused the difference. Initial levels, measurement error, regression to the mean, and other experiences must be examined. Final level, improvement, and intervention effect call for different analyses. The comparison quantity appropriate to the purpose should be chosen before examining data.

6 Falsification conditions and interpretive limits

Falsification condition: narrow the causal claim if confounding factors or reverse-direction explanations account for the result.

6.1 Distinguishing calibration, exploration, and confirmation

Revising a model while examining data is important exploration. Coefficients and structure may be calibrated against verified findings. Results used for calibration must, however, be distinguished from those evaluated with subsequently fixed rules. If the same data serve both roles, that scope is stated.

Preregistration records when questions and analysis plans were determined. Revisions and further exploration remain possible if reasons are documented. (Center for Open Science, n.d. R22) Registration itself guarantees neither a theory's truth nor an experiment's adequacy. WRRA research can evaluate explanatory consistency, implementation consistency, and failure conditions independently of whether it produces new predictions.

7 Current verdict

The arithmetic of the hypothetical values is COMPUTATIONAL PASS. It reproduces statistics and comparison quantities, not empirical validation of a WRRA-specific psychological mechanism. Correspondences in the research specification are INFERRED; evaluation using actual data is OPEN.

Chapter in one sentence: Covariation and outcomes caused by changing a condition require different causal evidence.

8 Next questions and evidence

Next research question: With present difficulty controlled, does changing only the order of earlier experience change the result?

8.1 Specifying a first behavioral experiment

A possible design independently manipulates earlier feedback and current difficulty in a neutral, nonclinical task. Outcomes are challenge choice and reaction time; learning history and instruction comprehension are recorded. Sample size, tolerances, and exclusions are determined by purpose and expected variation. This book does not report executing that design.

The study could examine whether earlier feedback makes a difference when present information is matched, and how well a residue rule reproduces it. If present difficulty is sufficient, the residue claim is reduced. The point is to specify which result changes which connection, rather than constructing an explanation compatible with every possible outcome.

Chapter references: R1, R3, R22.

Chapter 4 Measurement, Statistics, Reproducibility, and Research Ethics

1 Scene and question

Assigning anxiety a number does not measure the whole mind. Questionnaire responses are important data but depend on item meanings, timing, language, and context. Reliability concerns consistency; validity asks whether evidence supports the intended interpretation of a measurement. (Spielman et al., 2020, §7.5 R1)

Chapter question: Which observables measure psychological concepts, and how should reproducibility be assessed?

2 Starting points in established research

Mean differences should be reported alongside individual differences, effect sizes, and uncertainty. Treating repeated measurements from one person as independent people can inflate

evidence. Data used for fitting and evaluation should be separated by appropriate units, such as person or time, to prevent leakage. Replication research examines the conditions under which psychological findings hold. (Lazic, 2010 R49; Vabalas et al., 2019 R50; Open Science Collaboration, 2015 R3)

2.1 Turning psychological concepts into observables

An operational definition specifies how a concept is treated in a particular study. An anxiety score results from particular questions and scoring rules; reaction time is measured between defined starting and ending events. Clear definitions allow other researchers to use the same procedure. No single operational definition exhausts a concept. (Spielman et al., 2020, §2.2 R1)

The measurement record states ownership, time, unit, instrument, error, and missingness. A shorter reaction time may require additional tests to distinguish changes in reading speed, selection time, and hand movement.

2.2 Reliability and validity

Reliability concerns score consistency under specified conditions. Validity concerns support for interpretations and uses of scores. Repeated identical values may fail to measure the intended target. Conversely, when the target changes rapidly, not every change should be interpreted as error. (Spielman et al., 2020, §7.5 R1)

A stable-personality test and a same-day fatigue measure have different expectations over time. Item similarity, relationships to other measures, and connections to actual functioning must be assessed for the intended purpose. Applications across languages or populations also require examining whether the same score has the same meaning. (Putnick and Bornstein, 2016 R44)

2.3 Why means and variation are both needed

If five hypothetical participants change by 4, 2, 6, 2, and 6 points, mean change is 4. Reporting only the mean hides individual variation. The sample standard deviation is 2 points; assuming the five changes are independent, the standard error of the mean is approximately 0.894427 points. These values were directly computed as a learning exercise.

Standard deviation describes dispersion among observations; standard error concerns uncertainty in a specified estimator. A small standard error does not mean everyone responds similarly. Standardized effect sizes can compare outcomes in different units, but the reference variance must be identified. (Altman and Bland, 2005 R55)

2.4 Repeated measurements and information leakage

Five people completing twenty trials each produce one hundred trials but still only five people. Within-person records may share personal characteristics and influences from preceding trials.

Analyzing them as one hundred people can underestimate uncertainty in person-level conclusions. Research and analysis units must be specified. (Lazic, 2010 R49)

Training and evaluation splits must avoid placing nearly identical records from the same person on both sides when inappropriate to the purpose. Generalizing to new people and predicting the same person's future require different splitting units. Preprocessing and variable selection must also be determined without consulting evaluation data for fair performance assessment. (Vabalas et al., 2019 R50)

2.5 What a replication project investigated

The Open Science Collaboration attempted to replicate 100 studies from three psychology journals. It compared several indicators, including effect size and statistical significance, without using one success rate to judge all psychology as true or false. Its contribution was a procedure for measuring published findings again under the same research questions. (Open Science Collaboration, 2015 R3)

Scope of evidence: data, procedures, and comparison indicators should be recorded together. Reproducing a WRRA computation and replicating findings in people are separate tests.

3 Validation inputs

Research-specification inputs: measurements, sampling, and preprocessing records. The statistical exercise uses the hypothetical independent observations 4, 2, 6, 2, and 6. They are not interpreted as estimates of an effect in actual people.

4 WRRA-specific transformation

Transformation specification: estimate variables while including errors and the repeated-measurement structure.

The WRRA measurement ledger specifies variable ownership, units, times, instruments, missing-data treatment, and exclusion criteria. Research collecting memories and emotions includes explanation, consent, withdrawal, and confidentiality as part of its procedures. The hypothetical computations here involve no actual participants. Future human research requires separate review and participant consent.

4.1 Turning psychological concepts into observables — WRRA interpretation

WRRA states and residues follow the same principles. Naming them while postponing measurement makes connection to actual computation difficult.

5 Outputs and execution record

Execution record: the hypothetical changes yield a mean of 4 points, sample standard deviation of 2, and standard error of 0.894427. The standard error assumes independent observations. Planned outputs: effect sizes, uncertainty, and replication results. These plans are distinct from the arithmetic exercise.

5.1 Statistical judgments and substantive meaning

A p-value concerns data incompatibility under a specified statistical model; it does not directly state the probability that a hypothesis is true or the effect's size. The American Statistical Association statement emphasizes these distinctions. (Wasserstein and Lazar, 2016 R23) Interpretation requires considering estimated differences, uncertainty, measurement precision, and practical importance together.

A nonsignificant result alone does not establish equality between conditions. A claim of negligible difference requires a purpose-appropriate equivalence range or sufficient precision. Conversely, large samples can detect very small differences precisely. Their functional or explanatory importance remains a separate judgment.

6 Falsification conditions and interpretive limits

Falsification condition: limit the scope if conclusions reverse when measurement method or sample changes.

6.1 A ledger for reproduction and robustness

Reproduction code checks whether the same inputs and settings produce the same results. Replication in different samples and conditions asks a separate question. Robustness checks examine how conclusions change under modest preprocessing changes. These results are not combined into one undifferentiated pass.

The WRRA ledger records data identifiers, coefficient sources, random procedures, program versions, exclusions, and failures. Conditions of breakdown are recorded alongside successful cases. Comparisons with external theories match data, costs, tolerances, and interpretive criteria.

6.2 Designing research that respects participants

People must receive explanations allowing them to understand and choose participation. Collection purposes, procedures, expected burdens, withdrawal, and confidentiality should be clear. The Belmont Report identifies respect, beneficence, and justice as basic principles of human research. (National Commission, 1979 R24) These principles do not replace a particular country's legal procedures.

The APA ethics code also addresses consent, deception, debriefing, and related procedures. (APA, n.d. R25) Sensitive experiences should not be requested unnecessarily, and data access and retention must be defined. Using past failure or suffering requires examining scientific necessity and burden. Actual research review and consent are separate from executing hypothetical calculations.

6.3 Theoretical strengths and participant rights

WRRA's ownership principle helps document whose experience supplied data as well as who performed calculations. However, the model's technical definition of ownership differs from legal privacy rights. Access to data does not imply permission to use them for every purpose.

Readers should now be able to consider measurement meaning and error, the limits of statistical judgments, and participants' rights together. More precise explanation and protection of human dignity can both be addressed within research design.

7 Current verdict

The arithmetic of the hypothetical values is COMPUTATIONAL PASS. It reproduces statistics and comparison quantities, not empirical validation of a WRRA-specific psychological mechanism. Correspondences in the research specification are INFERRED; evaluation using actual data is OPEN.

Chapter in one sentence: Measurement interpretation, statistical uncertainty, and participants' rights belong within the same research procedure.

8 Next questions and evidence

Next research question: Do measurement interpretations and effect uncertainty hold when data are split by person?

Chapter references: R1, R3, R22, R23, R24, R25, R44, R49, R50, R55.

Part 2 Biological Bases of Mind and Behavior

Chapter 5 Neurons, Synapses, and Neural Signal Transmission

1 Scene and question

Many transmission processes occur between hearing speech and moving a hand. Electrical activity and chemical and electrical interactions between cells provide their physical basis. Cell types, connection direction, and temporal activity patterns must be examined together. Glia and the surrounding environment also belong within the research scope. (Spielman et al., 2020, §3.2 R1)

Chapter question: How do momentary neural signals differ from lasting changes in connections?

2 Starting points in established research

2.1 Electrical state changes and action potentials

Ion distributions differ across neuronal membranes; channels and transport processes influence membrane potential. Action potentials are rapid changes involving voltage-dependent channels. Stronger stimuli do not make individual action potentials increase in proportion to the intensity of a mental experience. Timing, frequency, and which cells are active together must also be considered. (Spielman et al., 2020, §3.2 R1)

2.2 What changes at a synapse

At chemical synapses, released neurotransmitters affect the next cell through receptors and subsequent processes. Effects depend on receptors, cell state, and connection location as well as the transmitter's name. Excitation and inhibition operate together in circuits; they are not simply good and bad signals. (Spielman et al., 2020, §3.2 R1)

2.3 Fast signals and lasting changes

An episode of activity and subsequent changes in connection properties occur on different timescales. Molecular and synaptic memory research provides concrete candidates for changes persisting after learning. (Kandel, 2001 R26)

2.4 Animal experiments on connection changes

Bliss and Lømo stimulated the perforant path in anesthetized rabbits and recorded synaptic responses in the dentate gyrus. They observed a lasting increase in transmission responses after strong stimulation. What was directly measured was transmission in a particular pathway, not recall of memory content or human experience reports. (Bliss and Lømo, 1973 R4)

Scope of evidence: lasting neural changes can be proposed as candidates for implementing residue. Whether a candidate explains the behavioral effects of particular learning requires separate input and observation rules.

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: cell activity, stimulation, and connection properties. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: distinguish shared transmission paths from channel-specific effects.

A WRRA common carrier is not defined by naming a single substance. A transmission network or updating structure reused by several functions can be a concrete candidate. Sharing a structure does not mean every signal has the same meaning. What is transmitted, through which path, and which state it affects must be differentiated.

4.1 Electrical state changes and action potentials — WRRA interpretation

WRRA defines states while distinguishing the magnitude of a single value from an activity pattern.

4.2 What changes at a synapse — WRRA interpretation

WRRA transmission rules specify the source and destination cells and the relevant state dependence.

4.3 Fast signals and lasting changes — WRRA interpretation

Fast activity can be treated as part of present state and maintained properties affecting later trials as candidate residues. Their actual contribution to behavior must be connected; persistence alone does not identify them as the owner of a particular memory.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned output: computation of signal transmission and response changes.

6 Falsification conditions and interpretive limits

Falsification condition: investigate another implementation if the effect persists when the proposed pathway is controlled.

Classic demonstrations of persistent changes in synaptic transmission support changes in connection properties after experience. (Bliss and Lømo, 1973 R4) Identifying a particular transmission change as the content of a particular memory requires further evidence. WRRA residue provides a question for locating that connection, not permission to call all neural changes memory.

6.1 A research example and its specification

Suppose neural responses and behavior differ before and after repeated stimulation. Conditions must distinguish adaptation, changes in sensory reception, motor fatigue, and learning. Candidate transmission rules can be tested by changing stimulation location or intervals. This chapter does not execute a new neuron model; it provides design criteria for a concrete model connecting cells and behavior.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Distinguishing momentary transmission from persistent change is necessary to connect neural signals with the influence of experience.

8 Next questions and evidence

Next research question: Can immediate responses and delayed connection changes be distinguished through separate observations?

Chapter references: R1, R4, R5, R26.

Chapter 6 Brain Structure and Functional Networks

1 Scene and question

Naming a brain region differs from explaining a psychological function. Increased activity in a region does not establish that a person necessarily experienced one particular emotion, because the region may participate in several tasks. This problem of reverse inference is important in interpreting neuroimaging. (Poldrack, 2006 R6)

Chapter question: What level of function can be inferred from brain locations and networks?

2 Starting points in established research

2.1 Why learn structure?

The cerebral cortex, subcortical structures, cerebellum, and brainstem are interconnected parts of the nervous system. Lobes help locate structures but do not divide complex mental functions into an equivalent number of boxes. Several connections may participate in a task. Readers should use a brain map while also tracing inputs, outputs, and interacting pathways. (Spielman et al., 2020, §3.4 R1)

2.2 Which instrument reads what?

EEG records electrical signals at the scalp. The BOLD signal in functional MRI is an indirect indicator related to changes in blood oxygenation. Temporal and spatial resolution, noise, and preprocessing differ. (Logothetis et al., 2001 R35)

2.3 Learning from lesions and interventions

Changes in performance after brain damage provide important evidence about functional roles. However, lesion extent, changes in other connections, and recovery may be mixed together. Preserved performance in one task alongside difficulty in another motivates examining their separation, while alternatives such as differences in task difficulty must also be checked. (Scoville and Milner, 1957 R10)

2.4 Original research comparing BOLD with neural activity

Logothetis and colleagues compared BOLD responses, local field potentials, and single- and multi-unit firing in monkey visual cortex. The comparison shows that BOLD reads neural activity through a hemodynamic pathway and is not identical to firing counts. Interpreting an indicator requires specifying the task and measurement path. (Logothetis et al., 2001 R35)

Scope of evidence: a function connecting WRRA states to imaging observables specifies measurement paths and delays. Findings in monkey visual cortex do not by themselves complete connections to every human emotion task.

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: neuroimaging, task performance, and experience reports. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: distinguish and connect observation paths at each measurement level.

WRRA records relationships alongside locations. It asks which inputs arrived, which connections transmitted change, and what conditions existed in other circuits and the body. Statistical relationships obtained through functional connectivity are not automatically restated as direct anatomical connections or causal influences.

4.1 Criteria for connecting circuits and psychology

Accurately classifying a subsequent choice from circuit activity does not establish that the circuit caused the choice. Classification on external evaluation data, actual differences between options, and causal interventions are reported separately. Output differences between a model with a circuit connection and one without it show dependence within that model. The same dependence in a natural circuit requires separate validation.

4.2 Which instrument reads what? — WRRA interpretation

The two measurements do not read the same state in the same way. WRRA specifies measurement procedures and temporal alignment, and does not equate indicator changes with neural firing itself.

4.3 Learning from lesions and interventions — WRRA interpretation

WRRA treats lesion location, state transition, and observation as separate terms.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned output: correspondences between circuit and behavioral levels.

6 Falsification conditions and interpretive limits

Falsification condition: restrict a single-function interpretation if the same imaging pattern occurs in other tasks.

Connecting brain science and psychology from this perspective requires observation paths across levels. Imaging signals, task performance, and personal reports are connected through their respective measurement models. Even when all three covary, the conditions under which agreement holds must be examined.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Function is assessed through connections, tasks, and observation paths rather than established from a region's name alone.

8 Next questions and evidence

Next research question: Does adding network data to regional activity expand explanatory scope?

Chapter references: R1, R5, R6, R7, R10, R35.

Chapter 7 The Body, Endocrine System, and Interoception

1 Scene and question

Hunger or a racing heart can change how the same situation is experienced. The body is more than a background to psychological research. Interoception involves detecting and interpreting signals inside the body; autonomic and endocrine changes provide research questions linking mind and body. (Garfinkel et al., 2015 R36; Spielman et al., 2020, §3.5 R1)

Chapter question: How do bodily changes and interpretations of them jointly influence experience?

2 Starting points in established research

2.1 Timescales in the nervous and endocrine systems

Neural transmission and hormonal regulation influence bodily states across several timescales. Autonomic activity may differ by situation and organ. Without knowing which indicators respond quickly or slowly, measurement order can be mistaken for causal order. (Spielman et al., 2020, §3.5 R1)

2.2 Interoception and interpretation

Interoceptive performance accuracy, the subjective tendency to consider oneself sensitive to bodily signals, and metacognitive awareness of agreement between performance and confidence can be measured separately. Garfinkel and colleagues investigated relationships among these dimensions using studies including heartbeat tasks. The proposal to measure detection and threat interpretation separately is this book's research design. (Garfinkel et al., 2015 R36)

2.3 Approaching the experience of pain

Pain experience relates to signals concerning damage, but observed damage alone cannot replace all reported suffering. Research considers bodily state, attention, expectations, and context together. Self-report is central evidence about a person's experience and is not invalidated merely by differing from a physiological indicator. (Raja et al., 2020 R37)

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: physiological data such as heart rate, subjective experience, and context. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

3.1 Separating measurement conditions

Similar heart rates immediately after exercise and before an evaluation may occur in different contexts and behaviors. Alternative conditions should be compared rather than inferring an emotional cause from a bodily indicator alone. Repeated measurements tracing which reports and behaviors follow bodily changes can investigate interactions between current inputs and residue. This chapter provides no new diagnostic accuracy estimates for clinical biomarkers.

4 WRRA-specific transformation

Transformation specification: separate bodily inputs, interpretation, and behavioral paths.

WRRA distinguishes input from the body from its interpretation. Heart-rate changes and anxiety reports are different variables even when they co-occur. Exercise, pain, medication, and situational interpretations must be examined. Shortcuts such as judging personality or intention from one hormone level are avoided.

4.1 Timescales in the nervous and endocrine systems — WRRA interpretation

WRRA separately records input time, bodily response time, and report time.

4.2 Approaching the experience of pain — WRRA interpretation

WRRA's phenotype connection respects experience reports while specifying their measurement paths.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned output: correspondences between bodily responses and experience across conditions.

6 Falsification conditions and interpretive limits

Falsification condition: narrow the proposed mediation path if physiological changes do not explain the psychological outcome.

Including bodily state makes questions about individual responses more concrete. Where do current fatigue and earlier experience each act? Does changing bodily conditions change responses? If not, other residue or relational conditions can be investigated. This chapter proposes possible connections without establishing any biomarker as a dedicated WRRA indicator.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Tracing experiential differences requires distinguishing bodily state from the conditions under which it is interpreted.

8 Next questions and evidence

Next research question: With the same bodily indicator, do experience reports differ according to situational interpretation?

Chapter references: R1, R7, R12, R36, R37.

Chapter 8 Heredity, Development, Experience, and Neuroplasticity

1 Scene and question

People begin with biological conditions preceding experience, but these do not uniquely specify every later response. Genetic differences, developmental environments, learning, and current situations must be considered together. Population-level relationships must not be converted directly into an individual's destiny. (Spielman et al., 2020, §3.1 R1)

Chapter question: How can genetic and experiential influences be connected along developmental time?

2 Starting points in established research

2.1 Genes and expression

Genetic information manifests in function through expression, proteins, cells, and connections. In the nervous system, when and where a gene acts matters. (NINDS, n.d. R27)

2.2 Not translating heritability into an individual percentage

Heritability concerns statistical decomposition of variation in a particular population and environment. It does not mean that a stated percentage of an individual's characteristic was genetically produced. Estimates may change with the distribution of environments. (Spielman et al., 2020, §3.1 R1)

2.3 Experience timing and conditions for change

Similar experiences can have different outcomes with different timing, repetition, rest, and context. A sensitive period is a research concept concerning conditions of heightened influence from particular inputs; it does not establish complete impossibility of change at other times. (Spielman et al., 2020, §9.3 R1)

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: initial characteristics, developmental timing, and experiential data. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: distinguish initial conditions from experience-dependent updating.

WRRA separates starting conditions and subsequent updates. Initial connections and bodily conditions constrain possible changes; actual experience changes which paths develop within those possibilities. Neuroplasticity is important here, but not every change is beneficial or equally possible at every time.

4.1 Explaining plasticity and stability together

If every experience greatly changed every connection, learned functions would be hard to maintain. With no change, adaptation to new environments would be difficult. WRRA addresses this by separately specifying maintenance and updating conditions. Comparing persistent traces with rapidly changing responses helps determine the number of required states and their timescales.

4.2 Genes and expression — WRRA interpretation

WRRA separates inherited information from current execution conditions. Discovering a genetic association and computing a path to behavior are different achievements. Measurement and modeling must fill the connection between them.

4.3 Not translating heritability into an individual percentage — WRRA interpretation

Distinguishing initial conditions from subsequent updating also requires keeping population-level variance separate from an individual's actual history.

4.4 Experience timing and conditions for change — WRRA interpretation

Updating coefficients may be specified as development-dependent, but values not measured are marked as hypothetical settings.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned output: a model of developmental paths and possibilities for change.

6 Falsification conditions and interpretive limits

Falsification condition: add condition dependence if the same coefficients do not hold in different environments.

Research separately records genetic information, developmental timing, experience type and intensity, and observation intervals. Persistence of residue differs from difficulty changing it. Examining whether an old response changes in a new environment can allow maintenance and updating rules to be estimated separately.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Heredity and experience connect through concrete implementations and conditions for change along developmental time.

8 Next questions and evidence

Next research question: Can lasting changes be identified when developmental timing and exposure history are recorded?

Chapter references: R1, R4, R26, R27.

Part 3 Sensation Perception Attention and Consciousness

Chapter 9 Sensation

1 Scene and question

Sensation begins the processing of physical events such as light and sound by the nervous system. A stimulus's presence must be distinguished from a report of detecting it. Reporting a faint sound may involve confidence and response criteria as well as sensory sensitivity. (Spielman et al., 2020, §5.1 R1)

Chapter question: How can sensory detection be separated from differences in response criteria?

2 Starting points in established research

2.1 Sensory transduction and modalities

Sensory reception connects a stimulus's physical properties to nervous-system activity. Light, sound, and mechanical pressure involve different input types and receptors, so even the phrase stimulus intensity requires modality-specific units. (Spielman et al., 2020, §5.1 R1)

2.2 Thresholds and adaptation

Detection thresholds are estimated under specified response criteria and procedures. Responses near a threshold can vary across trials, so one correct response does not establish a threshold. Adaptation under sustained stimulation is investigated separately from changes in input. Equating reported desensitization with receptor changes requires further evidence about transmission. (Spielman et al., 2020, §5.1 R1)

2.3 Four response outcomes

Reporting a present signal is a hit; reporting its absence is a miss. Reporting a signal when none is present is a false alarm; reporting absence correctly is a correct rejection. An increase in hit rate alone may reflect a strategy of always reporting presence. All four outcomes help distinguish sensitivity from criterion. (Spielman et al., 2020, §5.1 R1)

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: stimulus intensity, signal presence or absence, and detection responses. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: separate sensory sensitivity, response criteria, and effects of previous trials.

WRRA distinguishes external stimuli, transmission, present state, and response rules. If detection rates change for the same stimulus, sensory transmission or the decision criterion for avoiding misses may have changed. Without separating them, input changes and interpretive changes can be confused.

4.1 Sensory transduction and modalities — WRRA interpretation

SOURCE specifies the values actually presented, distinct from internal states after reception and transmission. Equal physical stimuli do not automatically imply equal sensations.

4.2 Four response outcomes — WRRA interpretation

WRRA specifies where the influence of the preceding trial enters the response rule.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned outputs: hit and false-alarm rates.

6 Falsification conditions and interpretive limits

Falsification condition: withdraw a sensory-change claim if a changed response criterion sufficiently explains the result.

A starting task intermixes signal-present and signal-absent trials and records hits and false alarms together. Adding residue requires distinguishing the preceding outcome's influence on the next response from current stimulus intensity. Separating sensory limits from decision criteria is central here.

6.1 A hypothetical detection task

Suppose participants detecting faint sounds are told that missing one has a high cost. More detection reports do not establish improved auditory reception. False alarms, confidence, and reaction time are compared together. This illustrates investigating how residue and boundary rules may change observations; performance is judged once the corresponding data and computations exist.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Detection outcomes reflect both sensory discrimination and response criteria.

8 Next questions and evidence

Next research question: How do detection outcomes change when stimulus intensity and response criterion are manipulated separately?

Chapter references: R1.

Chapter 10 Perception

1 Scene and question

We may mistake a distant stranger for someone familiar and correct the mistake on approaching. Perception can be studied as a process connecting input features, context, and previous experience. Illusions are both perceptual errors and tools for investigating the cues contributing to interpretation. (Spielman et al., 2020, §5.1 R1)

Chapter question: Why does the same sensory input yield different perceptual judgments?

2 Starting points in established research

2.1 Organizing parts into wholes

Perception research examines figure–ground separation and grouping by proximity or continuous contours. Such organization determines what is treated as a single object. (Spielman et al., 2020, §5.6 R1)

2.2 Cues for depth and motion

Several cues are available for judging distance and motion. Restricting inputs helps identify what can be judged from the remaining cues. Choices when cues conflict help investigate combination rules. (Spielman et al., 2020, §5.3 R1)

2.3 Research on expectation and perception and comparison theories

Kok, Jehee, and de Lange (2012) reported expectation-related differences in primary visual cortex representations using fMRI in a human visual task. This task-specific result motivates considering expectation; it does not establish identical effects for all memory, attention, or bodily states. [R65]

Rao and Ballard's (1999) predictive-coding model addresses higher-level predictions and ascending errors. Friston et al. (2017) connect perception and action through generative models and active inference. Comparison with WRRA requires explicit alignment of variables, timescales, objectives, and observation rules. Prediction error is not automatically WRRA's historical RESIDUE; variational

free energy is not automatically physical energy or the Core's minimum cost. These are comparison theories, not a completed computational integration. [R66, R67; suggestion provenance I1, I2]

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: identical or controlled stimuli, contexts, and judgments. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: distinguish input features from the weighting of experiential residue.

WRRA separates information already in a stimulus from residue the observer currently carries. Explaining rapid detection of an expected object in a familiar place requires manipulating visual features and expectations separately. Naming expectation does not specify an actual transformation.

4.1 Combining expectations and current input

Reading an ambiguous letter differently across contexts illustrates investigating experience. Actual stimulus differences, contextual information, and response demands must be distinguished. Context can improve accuracy or reinforce mistaken expectations. Minimal computation evaluates rapid interpretation from limited input alongside misidentification costs under the same functional criteria.

4.2 Organizing parts into wholes — WRRA interpretation

WRRA specifies input arrangements and relationships. Collecting features does not itself establish object meaning; the connection from grouping to actual judgment must be shown.

4.3 Cues for depth and motion — WRRA interpretation

Transformations distinguish cue availability from weighting and record how weights are determined so they cannot be freely changed to match each result.

4.4 State-dependent expression and the action loop

Testing holds the physical stimulus constant while varying conditions such as expectation or attention. Instructions and cues are additional SOURCE inputs: an identical sensory stimulus does not imply identical total information. The specification defines present state and residue, fixes the RENDERER and boundary, and produces condition-specific distributions of judgment, choice, and

response time. A changed-renderer hypothesis is a separate comparison model. This transformation specification is the author's WRRA application design. [W1, §1.3, §2.1; R65]

Actions change the conditions of subsequent input. UPDATE maps current SOURCE and state to the next state; the RENDERER generates an action phenotype; environmental and sensory pathways return the next SOURCE. OBSERVABLE consists of measured stimuli, actions, and neural signals; RECORD preserves their timing and provenance. The shared transmission/update scaffold and channel ownership require explicit specification. This is a task-level application of Core grammar, not a claim to have implemented active inference's mathematics. [W1, §§0–3; R67; I1, I2]

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned outputs: perceptual judgments, accuracy, and confidence.

6 Falsification conditions and interpretive limits

Falsification condition: limit a claim of changed sensory experience if changes in verbal responses sufficiently explain the outcome.

An experiment might provide different prior information for the same ambiguous stimulus. Interpretation, accuracy, and confidence can be measured while the stimulus is held constant. Claiming changed sensory experience from changed interpretation requires additional supporting evidence.

6.1 Learning rules from illusions

Rather than dismissing illusions as exceptions unrelated to normal perception, they can test which relationships and cues change outcomes. Size or contrast arrangements are varied systematically and judgments measured. WRRA evaluates whether one rule computes several arrangements. If every arrangement needs a separate rule, the scope of a common explanation narrows.

6.2 Prespecified failure conditions

A state-dependence test first specifies the conditions, effect sizes to explain, and tolerances. Absence of an effect cannot refute a rule that permits no effect; distinguishable predictions must therefore be specified first. Loop tests assess delays and errors between input timing, updating, action, and subsequent input. Failure to meet the declared reconstruction thresholds calls for revision of the relevant application rule. These additions are INFERRED designs; empirical adequacy remains OPEN.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Perceptual differences are investigated through combinations of present input, context, and experience while distinguishing report from experience.

8 Next questions and evidence

Next research question: With identical stimuli, do prior-information effects remain separately in accuracy and confidence?

Chapter references: R1.

Chapter 11 Attention

1 Scene and question

A notification may interrupt reading even while one's eyes remain on the same line. Attention research asks which information is selected and maintained and how other information interferes. Resource is a useful term here but does not immediately identify a single measurable substance. (Pashler, 1994 R38)

Chapter question: How do selection and interference conditions change accuracy and reaction time?

2 Starting points in established research

2.1 Selection, maintenance, and switching

Dual-task research studies processing interference when two tasks are performed together. Pashler's review compares explanations at input, central-processing, and response stages. (Pashler, 1994 R38)

2.2 Bottlenecks and interference

Dual-task decrements may arise at several points in input, judgment, or response. Varying intervals between tasks or response devices can distinguish candidate bottlenecks. (Pashler, 1994 R38)

2.3 Trading speed against errors

Emphasizing speed may reduce response times while increasing errors. Comparing mean times alone could then suggest improved function incorrectly. Accuracy and reaction-time distributions should be examined together under fixed task criteria. (Heitz, 2014 R39)

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: task demands, time limits, and performance records. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: construct information-selection and maintenance rules under task-specific constraints.

WRRA first specifies required distinctions and available conditions. Available time, simultaneous items, and stimulus competition make resources concrete. Merely invoking insufficient resources after observing poor performance is circular. Resource indicators must be defined independently of outcomes.

4.1 Selection, maintenance, and switching — WRRA interpretation

Selection, sustained attention, and switching are specified as different tasks; their distinction itself is not evidence for one resource substance.

4.2 Bottlenecks and interference — WRRA interpretation

Explaining reuse of a common carrier requires testing which paths are actually shared. Observed interference motivates investigating that connection; it does not establish a particular shared circuit.

4.3 Trading speed against errors — WRRA interpretation

Minimal computation reduces costs while preserving the criteria. It differs from lowering the criteria by changing instructions.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned output: joint changes in accuracy and reaction time.

6 Falsification conditions and interpretive limits

Falsification condition: improved speed outside the required accuracy criterion does not count as successful minimal computation.

Accuracy and reaction time can be compared for single-task and dual-task performance. A structure reducing processing or time while maintaining the same success rate can count as minimal-computation performance. Increased speed with more errors is assessed against the predefined functional criteria.

6.1 Notifications and work

In a notification-heavy environment, researchers examine both interruption frequency and information needed to resume work. Recording the preceding goal may support a design hypothesis that resumption becomes easier. The external record's contribution is not counted as increased individual memory capacity. Required state and cost are compared within the combined person–device boundary.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Attentional costs are evaluated through speed, accuracy, and conditions for returning to a task together.

8 Next questions and evidence

Next research question: Are speed and accuracy criteria satisfied when notification timing and resumption costs change?

Chapter references: R1, R38, R39.

Chapter 12 Consciousness, Nonconscious Processing, Sleep, and Dreams

1 Scene and question

We cannot directly report every processing operation. Evidence that information influenced behavior and a claim that it was consciously experienced require different tests. Distinguishing consciousness, attention, and reporting ability is a starting point. (Spielman et al., 2020, §4.1 R1)

Chapter question: How can processing, conscious experience, and later reporting be distinguished?

2 Starting points in established research

Sleep involves several physiological states rather than simply suspending wakefulness. Relationships among sleep, memory, and everyday functioning are important neuroscience topics. (NINDS, n.d. R8)

2.1 Consciousness and reporting ability

A stimulus influencing behavior does not establish awareness without separate measurement. Conversely, failure to report does not easily establish absence of experience.

2.2 Observations distinguishing sleep states

Sleep research distinguishes states through indicators such as EEG, eye movements, and muscle activity. Biological rhythms and time spent awake are investigated alongside them. (Spielman et al., 2020, §4.3 R1)

2.3 Dreams and the timing of recall

Questions about dreams read post-awakening reports rather than directly accessing the experience. When someone was awakened, the delay before questioning, and question wording all enter the observation. Numbers or meanings of unremembered dreams are not supplied without evidence.

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: arousal conditions, sleep records, reports, and behavior. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: separate state changes, memory accessibility, and reporting paths.

4.1 From existing research to a WRRA application

WRRA can compare present-state and residue changes before and after sleep. This book does not assume a single rule according to which sleep deletes all unnecessary memories.

4.2 Consciousness and reporting ability — WRRA interpretation

WRRA distinguishes stimulus processing, accessibility, and response rules. Preserving the distinction between philosophical questions about consciousness and task-measured access clarifies the scope of findings.

4.3 Observations distinguishing sleep states — WRRA interpretation

Before–after sleep changes do not alone establish causes; elapsed time, new experiences, and testing times are compared. In data using real time units, model update intervals are distinguished from arbitrary iteration counts.

4.4 Dreams and the timing of recall — WRRA interpretation

WRRA treats candidate dream experience, currently accessible traces, and rendered reports as separate layers.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned output: differences in performance and experience reports across states.

6 Falsification conditions and interpretive limits

Falsification condition: restrict an interpretation as a difference in consciousness itself if reporting ability explains the results.

Dream reports also have observation paths. The time of experience is distinguished from recall after awakening, and unreported is not automatically judged unexperienced. This book does not claim to solve the nature of consciousness. It begins with what can be computed about information leading to reports and behavior under specified states.

6.1 Researchable questions and remaining problems

Even without resolving consciousness's nature, we can study which information is used in reports and behavior under different conditions. The same tasks can be presented across states, accuracy and confidence compared, and possible neural indicators connected. These are partial achievements in consciousness research. The extent actually computed in an explanation connecting neural signals and subjective experience is stated separately.

7 Current verdict

The WRRA correspondences distinguishing consciousness, reporting, and processing are INFERRED. Empirical validation is OPEN.

Chapter in one sentence: Conscious experience, information processing, and later reports have separate observation contracts.

8 Next questions and evidence

Next research question: Do processing differences between states remain after controlling changes in reporting ability?

Chapter references: R1, R8.

Part 4 Learning and Memory

Chapter 13 Conditioning and Reinforcement

1 Scene and question

An animal responding to a sound previously followed by food differs from behavior changing after an action is rewarded. The first centers on relationships between stimuli; the second on relationships between action and consequence. Reinforcement and punishment must be defined by actual behavioral change. (Spielman et al., 2020, §6.2 R1; §6.3 R1)

Chapter question: How do stimuli, actions, and consequences on earlier trials change the next response?

2 Starting points in established research

2.1 Learning relationships between stimuli

Classical conditioning concerns not merely co-occurrence but the information one stimulus provides about another event. Timing, event probabilities, and prior learning can be distinguished. (Spielman et al., 2020, §6.2 R1)

2.2 Four kinds of behavioral consequence

Reinforcement is a relationship in which subsequent consequences increase behavior; punishment decreases it. Positive and negative mean adding and removing a stimulus. Negative reinforcement is therefore not another name for bad punishment. If an action stopping an unpleasant sound increases, it can be analyzed as reinforcement by removal. (Spielman et al., 2020, §6.3 R1)

2.3 Extinction and context

Researchers examine decreasing responses when learned consequences are no longer provided. Whether the decrease means complete deletion of the earlier relationship requires tests after delays or in other contexts. (Spielman et al., 2020, §6.2 R1)

2.4 Evidence distinguishing extinction from complete deletion

Bouton's review compares the return of extinguished responses after changes in time or context in Pavlovian extinction research. Such findings support context-dependent performance after extinction and challenge a simple reversal-of-learning account. This is a review of behavioral research, not a direct measurement of persisting memory molecules. (Bouton, 2004 R56)

Scope of evidence: this book examines explanations separating maintained states and current expression. A claim that residue actually remains requires comparing behavioral evidence under return conditions with candidate maintenance rules.

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: trial-level records of stimuli, actions, and consequences. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: update residue from earlier relationships according to outcomes.

Changes left by earlier trials act in the next trial's present. Updating based on the difference between expected and actual outcomes is one possible implementation. Its rule must be specified and computed within WRRA; citing an established learning model does not complete implementation.

4.1 Existing learning equations and WRRA

An existing expectation–outcome difference rule can be implemented within an input–residue–output structure. Its original contribution is credited while the boundaries and observations connected through WRRA are recorded. Actual execution within WRRA counts as computational performance even if the same numbers result. Merely citing an equation does not establish integration.

4.2 Learning relationships between stimuli — WRRA interpretation

Relationships are specified as states remaining active now. Naming sound and food is insufficient; the update changing the next response must be determined.

4.3 Four kinds of behavioral consequence — WRRA interpretation

The WRRA ledger records how behavior actually changed.

4.4 Extinction and context — WRRA interpretation

WRRA distinguishes maintained residue from current expression when designing tests. Conditions are specified to discriminate maintenance hypotheses rather than explaining every possible outcome through hidden memory.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned outputs: choice and response on the next trial.

6 Falsification conditions and interpretive limits

Falsification condition: narrow a claim of different learning amounts if current performance conditions explain the difference.

Faster responses in a more-rewarded group do not alone establish more learning. Satiety, motor capacity, and task comprehension may differ. Separating learning residue from current expression conditions distinguishes what was learned from what is displayed now.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Separating learning changes from current performance conditions allows assessment of what trial outcomes leave behind.

8 Next questions and evidence

Next research question: With performance conditions controlled, do earlier outcomes affect the next trial?

Chapter references: R1, R56.

Chapter 14 Observational Learning, Skill Acquisition, and Habits

1 Scene and question

People learn by observing others as well as through their own trial and error. Bandura and colleagues' classic experiment investigated relationships between a model's observed behavior and children's subsequent behavior. (Bandura et al., 1961 R9) Findings from individual studies should not be generalized to imitation or personality change in every situation.

Chapter question: How can what was observed, what was learned, and what was actually executed be distinguished?

2 Starting points in established research

2.1 Observation and actual performance

Someone may carefully observe and remember behavior but lack the capacity or opportunity to perform it now. Separating acquisition from expression allows investigation of this difference. Understanding instructions, executing movements, and displaying learning under changed rewards are separate questions. (Spielman et al., 2020, §6.4 R1)

2.2 Practice arrangement and feedback

Performance during practice differs from delayed retention. Roediger and Karpicke compared delayed memory after rereading material and after taking recall tests. That paper does not also validate every claim about practice spacing and task diversity. (Roediger and Karpicke, 2006 R32)

2.3 Questions for identifying habits

Repeated behavior in a familiar situation may reflect reevaluation of a goal or easy elicitation by a cue. Changing outcome value and action–outcome relationships offers one possible test. One test does not classify all everyday habits. (Wood and Rünger, 2016 R46)

2.4 What the classic observational-learning experiment showed

Bandura and colleagues assigned 72 young children to an aggressive model, a nonaggressive model, or no model. Behavior was then recorded in another play situation without the model; more model-like behavior appeared after observation of the aggressive model. The finding concerns imitation under those specified conditions. (Bandura et al., 1961 R9)

Scope of evidence: this supports studying observational input separately from later expression. Imitation in one experiment does not directly establish lifelong personality or all everyday aggression.

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: observational experiences, practice, cues, and outcome values. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: separate acquired residue from current execution conditions.

Observed behavior is input; retained representations or action tendencies are residue; actual performance is phenotype passing through current capacities and permissible conditions.

Observation, inferred acquisition, and observed execution remain distinct. Skill acquisition also requires specifying practice arrangement and feedback.

4.1 Benefits and costs of automatization

Skills requiring less attention may leave capacity for other tasks. However, continued familiar behavior after rules change may increase error costs. Minimal computation considers both rapid familiar execution and revision under new conditions. Checks and reserve resources needed for stability and flexibility also count as costs.

4.2 Observation and actual performance — WRRA interpretation

Unobserved behavior does not directly establish absent learning.

4.3 Practice arrangement and feedback — WRRA interpretation

WRRA distinguishes successful performance with present assistance from structures retained for independent use. Removing assistance and conducting delayed tests can investigate the difference.

4.4 Questions for identifying habits — WRRA interpretation

WRRA separates cues, goals, and consequences to specify the required update.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned outputs: imitation, skilled performance, and habitual choice.

6 Falsification conditions and interpretive limits

Falsification condition: reconsider a strong habit interpretation if behavior responds flexibly to changed outcome values.

Repetition alone does not sufficiently define habit. Researchers can examine maintenance under changes in contextual cues and outcome value. Low-attention execution may connect to minimal computation, but automatic does not by itself mean optimal or adaptive.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Observation, acquisition, and execution are different stages, and automatization's benefits are evaluated within task criteria.

8 Next questions and evidence

Next research question: Can acquisition and performance be distinguished by separating model observation, practice quantity, and present execution conditions?

Chapter references: R1, R9, R32, R46.

Chapter 15 Memory Encoding, Retention, and Retrieval

1 Scene and question

Quickly forgetting a newly heard name may differ from knowing a name but being unable to retrieve it. Memory research distinguishes encoding during experience, retention over time, and retrieval when needed. Working and long-term memory also differ in roles and measurement tasks. (Baddeley, 2003 R40; Spielman et al., 2020, §8.1 R1)

Chapter question: Does poor memory performance originate in encoding, retention, or retrieval?

2 Starting points in established research

2.1 Working and long-term memory

Working memory concerns maintaining and using information needed for a current task. Long-term-memory research examines knowledge, events, skills, and other contents. Distinctions in a model are not equated with physically isolated brain boxes. (Baddeley, 2003 R40; Spielman et al., 2020, §8.1 R1)

2.2 Recall, recognition, and relearning

Uncued recall, recognition of presented items, and changes during relearning are different tests. Failure on one does not establish that every trace has disappeared. Conversely, correct recognition does not establish recall of all content and context. (Spielman et al., 2020, §8.1 R1)

2.3 Conditions of encoding and retrieval

Performance can depend on attention during learning and cues during retrieval. Explaining these processes requires separately manipulating learning-time and test-time conditions. (Spielman et al., 2020, §8.1 R1)

2.4 The scope of functional inference from lesions

Scoville and Milner reported cases of severe impairment in memory for recent experiences after bilateral medial temporal surgery. Clinical observations connecting surgical extent and memory performance provided evidence for investigating these structures' importance in memory

formation. This differs from concluding that all memory is stored in one place. (Scoville and Milner, 1957 R10)

Scope of evidence: the WRRA application separates post-learning state changes from test outcomes. The retention rate 0.85 in Chapter 41 is a hypothetical setting, not a neural constant estimated from these cases.

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: learning items, delays, and retrieval cues. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: distinguish encoding, retention, and access paths.

WRRA treats memory as structures remaining in the present and contributing to subsequent processing. This does not combine every memory mechanism into one. Classic difficulties in recent-experience memory after medial temporal damage, including the hippocampus, are important evidence. (Scoville and Milner, 1957 R10) They do not establish storage of all memory at one location.

4.1 Physical implementation and levels of memory

Circuit and molecular research provides concrete bases for investigating retention mechanisms. (Kandel, 2001 R26) Correspondences between particular contents and circuit changes must be validated for the task and measurement. WRRA specifies trace ownership and timescale. Chapter 41's single decay value illustrates a minimal maintenance rule rather than a constant representing this complex implementation.

4.2 Working and long-term memory — WRRA interpretation

WRRA preserves relevant research distinctions while specifying which information is accessible to present processing.

4.3 Recall, recognition, and relearning — WRRA interpretation

An observation function must specify separately which state each test reads.

4.4 Conditions of encoding and retrieval — WRRA interpretation

WRRA distinguishes differences in generated traces from differences in access paths. Equal recall rates do not establish identical underlying paths.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned outputs: recall, recognition, and reaction time.

6 Falsification conditions and interpretive limits

Falsification condition: revise complete-loss interpretations if changed cues restore performance.

Currently inaccessible content may become retrievable with different cues. Low recall must therefore be distinguished as possible residue loss or an access problem. Chapter 41 computes only one aspect of retention; it does not reproduce all actual memory or hippocampal function.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Explaining memory performance requires distinguishing trace retention from paths providing access to the trace.

8 Next questions and evidence

Next research question: How much does low recall recover when retrieval cues change?

Chapter references: R1, R10, R26, R40.

Chapter 16 Forgetting, Memory Reconstruction, and Updating

1 Scene and question

Retelling a memory may fill gaps or change emphasis. Later information, current questions, and memory for the original context may influence reports. Confidence and factual accuracy require separate examination. (Johnson et al., 1993 R41)

Chapter question: How can trace changes be distinguished from present reconstruction?

2 Starting points in established research

2.1 Separating time and interference

Other experiences occur as time passes. Poorer delayed memory may mix decay, interference, and changed retrieval cues. Controlling intervening activity or comparing interference conditions narrows candidate causes. (Spielman et al., 2020, §8.3 R1)

2.2 Remembering information sources

Familiarity with content differs from knowing where it was encountered. Misattributing a source can produce confident beliefs inconsistent with actual events. Content accuracy, source accuracy, and confidence are measured separately. (Johnson et al., 1993 R41)

2.3 Testing memory updating

If later reports change after memory is reused, researchers must compare added information and altered access paths. Reactivation and a changed trace are not treated as identical. (Johnson et al., 1993 R41)

2.4 Scope of a review on source judgments

Johnson and colleagues reviewed the source-monitoring framework and related experiments about judging where experiences originated. They distinguished content familiarity, source judgments, and the experiential properties and criteria used in those judgments. The review compared phenomena such as eyewitness reports and misattributed familiarity within one framework. (Johnson et al., 1993 R41)

Scope of evidence: ownership distinctions in WRRA's ledger are model specifications preventing source confusion, not themselves the neural mechanisms of human source memory.

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: original events, subsequent information, and recall reports. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: connect retained traces with present reconstruction rules.

Residue is not a perfect recording. Reports arise through currently usable traces and reconstruction rules. Memory reconstruction does not make every report equally uncertain, however. Independent records and repeated tests should identify what can be checked.

4.1 Separating time and interference — WRRA interpretation

WRRA records actual intervening inputs rather than explaining all forgetting solely by update counts.

4.2 Remembering information sources — WRRA interpretation

Ledger ownership supports these distinctions as a research principle; it is not assumed to perform the same function as psychological source memory.

4.3 Testing memory updating — WRRA interpretation

WRRA specifies existing traces, new inputs, and maintenance and output rules. Claims about a particular updating mechanism require experiments distinguishing change timing and conditions.

4.4 Separating content provenance and change history

Drawing on source-monitoring research, distinguish the original experience, later information, recalled content, and the source attributed by the person recalling it. Score content errors and source confusions separately; preserve change history in a presently accessible RECORD maintained by the researcher. Event identifiers, versions, modification times, and ownership are computational design choices, not claims that the brain stores those literal IDs or ledgers. A source-preserving model must still be revised if it cannot reconstruct condition-specific confusion distributions within tolerance. [R41; W1, §§2.1–2.2]

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned outputs: distributions of accurate recall, errors, and confidence.

6 Falsification conditions and interpretive limits

Falsification condition: leave a particular memory-mechanism verdict open if no discriminating test is available.

Computing forgetting requires distinguishing elapsed time, interference from other learning, and changed cues. Several causes can generate the same recall curve, so one curve does not identify a mechanism. WRRA connects residue maintenance and expression without continually adding unobserved traces to explain every failure.

6.1 Everyday memory and independent records

Calendars and message records can supplement memory as external resources. Where they conflict with recall, the times and contexts each reflects should be examined. External records may also be incomplete or edited and are not automatically absolute standards. Independent evidence can narrow factual possibilities, but confidence does not fill evidential gaps.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Recall changes are interpreted by comparing trace change, present reconstruction, and independent records together.

8 Next questions and evidence

Next research question: Compared with independent records, how much does subsequent information change errors and confidence separately?

Chapter references: R1, R41.

Part 5 Thinking Language and Intelligence

Chapter 17 Concepts, Reasoning, Problem Solving, and Creativity

1 Scene and question

Recognizing an unfamiliar object as a chair requires grouping shared features. Problem solving requires defining goals, available information, and permissible actions. Familiar solutions may help or obstruct new ones. (Spielman et al., 2020, §7.3 R1)

Chapter question: How does reusing experience contribute to problem solving and new outputs?

2 Starting points in established research

2.1 Forming categories and concepts

Concepts allow objects to be treated through common properties or relationships. Similarity to representative examples, remembered individual instances, and explicit rules may all participate. Research compares classification and judgment times for unfamiliar objects. (Spielman et al., 2020, §7.1 R1)

2.2 Problem representation changes solutions

How goals and permissible actions are represented changes the search space. Drawing relationships or externally recording intermediate results can change the conditions for handling a problem. (Spielman et al., 2020, §7.3 R1)

2.3 Analogy and reuse of experience

Problem-solving research distinguishes circumstances in which prior knowledge and familiar solutions help or hinder solving a new problem. (Spielman et al., 2020, §7.3 R1)

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: problem conditions, knowledge, and search records. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: search and combine available relationships under constraints.

Concepts and knowledge are currently usable relational structures; problem solving connects them with present input under constraints. Minimal computation does not simply mean stopping thought quickly. It evaluates reduced unnecessary search while preserving the distinctions the task requires.

4.1 Forming categories and concepts — WRRA interpretation

WRRA describes the retained relationships allowing a new input to enter a category. Identical classifications do not establish identical internal representations.

4.2 Problem representation changes solutions — WRRA interpretation

WRRA distinguishes this change from increased internal individual ability. A combined person–recording-tool boundary allows evaluation of where information was maintained and which costs fell.

4.3 Analogy and reuse of experience — WRRA interpretation

Relationships from earlier problems can be specified as presently usable residue. Tracing the connections actually contributing to solutions, while including superficially similar problems requiring different solutions, is this book's proposed study design.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned outputs: solution rates, search costs, and evaluations of products.

6 Falsification conditions and interpretive limits

Falsification condition: limit general-creativity claims if success holds only by changing evaluation criteria.

Creative products require criteria for contextual appropriateness as well as novelty. Many answers differ from good answers. A WRRA model records memory, relationships, and search costs; outputs are assessed with separate criteria. Whether reuse of familiar connections and formation of new ones can actually be distinguished must also be checked.

6.1 Evaluating creative outputs

Number, diversity, novelty, and appropriateness are different indicators. Evaluators and criteria should be disclosed and agreement examined. Even if WRRA structures generate new combinations, their value is judged by the relevant field's standards. Connecting computation and evaluation is needed to investigate when low cost and high creativity coincide.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Making problem representation and experiential reuse explicit allows search costs and output quality to be compared together.

8 Next questions and evidence

Next research question: Does changing problem representation improve both search cost and solution quality?

Chapter references: R1.

Chapter 18 Judgment, Decision Making, Uncertainty, and Cognitive Bias

1 Scene and question

People must choose even without sufficient evidence. Readily recalled examples or familiar forms may serve as judgment cues. Tversky and Kahneman systematically studied heuristics and biases in judgment. (Tversky and Kahneman, 1974 R11) Bias observed in one task should not be expanded into irrationality in every situation.

Chapter question: Does the same information lead to different choices under different reference points and constraints?

2 Starting points in established research

2.1 Probability, value, and choice

In constructing this book's choice models, outcome probability and magnitude, waiting time, and current resources are separate candidate variables. This is the author's design for specifying comparisons, not the conclusion of one experiment. Expected mean outcomes, tolerable risk, and personal goals are distinguished; choice-value definitions and rules transforming them into behavior are recorded separately.

2.2 Reference points in judgment

The book examines hypotheses in which previous expectations or present circumstances serve as comparison standards. Reference points must be determined independently of outcomes, or competing reference-point models compared. Changing them afterward to fit every choice makes explanatory performance hard to assess. Treating WRRA residue as a candidate reference point is the author's proposal, not an established identity between the concepts.

2.3 Bias and environmental fit

Judging from readily recalled examples may be fast and useful in one environment and produce errors in another. Evaluation requires the actual information distribution and the goal demanded. (Tversky and Kahneman, 1974 R11)

2.4 Evidence connecting choice observations to theory

Kahneman and Tversky examined preference patterns under risk that departed from expected utility theory's descriptive account and proposed prospect theory. Its structure distinguishes a value function evaluating changes relative to a reference point from probability-related decision weights. This constitutes theory construction for explaining human choice. (Kahneman and Tversky, 1979 R57)

Scope of evidence: WRRA residue's involvement in reference-point formation is a separate hypothesis. Connections between the accounts are assessed only to the extent that reference points and weights were set beforehand and used to compute and compare choice data.

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: options, probabilities, outcome histories, and time limits. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: fix rules through which current information and residue influence choice value.

WRRA records information actually accessible at choice time, residue, and time and action boundaries. Computation is reconstructed under those conditions rather than condemning choices with hindsight. This does not assume every choice was optimal: goals and costs are specified beforehand and used for assessment.

4.1 Bias and environmental fit — WRRA interpretation

WRRA compares computation saved and errors produced within one contract. Researchers must not assume participants knew everything the researchers know.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned outputs: choice probabilities, losses, and reaction times.

6 Falsification conditions and interpretive limits

Falsification condition: reject a rule if observations repeatedly run in the opposite direction after conditions are fixed.

Explaining previous losses requires separating current wealth, risk exposure, and information changes. The same loss history may lead one person to avoidance and another to further challenges. Residue's sign and magnitude are research questions; opposite outcomes are not both declared successful afterward.

6.1 Testing changed conditions

The same person can receive different information displays or choice deadlines. Measuring probability comprehension, actual choices, and confidence together can distinguish representation from preference. Rules fitted in one condition are applied to another display format. If they fail, the altered information transformation is revised and its scope stated.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Judgment is evaluated under conditions specifying input information, reference points, and constraints on choice.

8 Next questions and evidence

Next research question: How does the choice distribution change when reference points and constraints are specified?

Chapter references: R11, R57.

Chapter 19 Language Comprehension, Production, and Acquisition

1 Scene and question

The same sentence may be understood differently depending on context. Language research distinguishes sounds and writing, word meaning, sentence structure, and conversational context. Understanding language and expressing one's meaning are not fully revealed by the same test. (Spielman et al., 2020, §7.2 R1)

Chapter question: Which processes are shared or different in comprehension and production?

2 Starting points in established research

2.1 Levels of language

Speech-sound discrimination, word forms and meanings, sentence structure, and utterance intentions can be distinguished. The same words may convey different intentions depending on the interlocutor and preceding context. (Spielman et al., 2020, §7.2 R1)

2.2 Observing comprehension and production

Reading time, comprehension accuracy, speech-onset time, and speech errors are different indicators. Slow word retrieval must be distinguished from failure to understand a sentence.

2.3 Development and multilingual environments

Introductory accounts of language units and development treat acquisition as a topic in cognitive development. (Spielman et al., 2020, §7.2 R1)

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: language stimuli, exposure histories, and contexts. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: connect current language knowledge and conversational residue to interpretation rules.

Auditory or visual language signals are inputs. Currently available vocabulary, grammatical knowledge, and conversational traces provide interpretive conditions. Outcomes are observed as comprehension judgments or speech. Response paths must be separated so that not speaking is not equated with not understanding.

4.1 Computing conversational residue

A possible task tests how a referent in the preceding sentence influences a following pronoun's interpretation. Candidate referents, relationships, and context-retention rates could define a small model. No numerical execution was performed here, so this remains a design. Later inputs of actual interpretations and times across contexts could assess the required maintenance rule.

4.2 Levels of language — WRRA interpretation

WRRA distinguishes inputs and relationships at each level and specifies transition rules between them. Language processing is not reduced to one word lookup.

4.3 Observing comprehension and production — WRRA interpretation

Task-specific output functions allow comparison of how the same experiential residue appears in different tests, making access and expression paths concrete.

4.4 Development and multilingual environments — WRRA interpretation

Proposed studies record input quantity and quality, conversational opportunities, and contexts of use alongside age. Comparisons involving bilingual speakers specify which language was used in which situations. The book's interpretive principle is that one test under different language histories should not be translated into differences in general intelligence or personal worth.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned outputs: comprehension accuracy, speech, and reaction time.

6 Falsification conditions and interpretive limits

Falsification condition: outcomes explained by response-mode differences do not establish comprehension-ability differences.

Explaining acquisition requires specifying developmental timing, languages encountered, and interaction types and quantities. This book does not derive all language principles from one residue equation. It proposes first computing preceding-context effects in short conversations and testing changed interpretations when context changes.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Testing comprehension–production connections requires distinguishing shared language processing from separate output paths.

8 Next questions and evidence

Next research question: Can shared states and separate outputs be distinguished in tasks understanding and speaking the same content?

Chapter references: R1.

Chapter 20 Intelligence, Cognitive Abilities, Individual Differences, and Measurement

1 Scene and question

Cognitive tests estimate aspects of ability through tasks performed under particular conditions. Language comprehension, reasoning, memory, and processing speed can be distinguished. Score meanings depend on standardization populations and procedures, measurement error, and intended use. (Spielman et al., 2020, §7.5 R1)

Chapter question: Under what conditions can individual cognitive differences be compared fairly?

2 Starting points in established research

2.1 Common and specific abilities

Correlations among cognitive tests can be analyzed for common factors and domain-specific characteristics. A statistically extracted factor summarizes observed relationships; it does not itself discover a neural organ or substance. (Spielman et al., 2020, §7.4 R1)

2.2 Standardization and norms

Score interpretation requires knowing the population and procedures used to construct comparison standards. Raw scores, normed scores, and measurement-error ranges are considered together. Repeated testing can introduce task familiarity, so timing and previous test experience are recorded. Changed scores are not immediately read as increases in one ability. (Spielman et al., 2020, §7.5 R1)

2.3 Fair comparisons

Giving the same test to people differing in language, sensory conditions, or educational opportunity does not always provide equivalent measurement. Items may carry different meanings or burdens across groups. Validation and accessibility checks appropriate to the comparison are needed. (Putnick and Bornstein, 2016 R44)

2.4 Measurement conditions for comparing equal scores

Putnick and Bornstein review procedures and reporting practices for measurement invariance across groups and times. Presenting identical items and establishing score comparability are separate matters. Permissible comparisons depend on the level of invariance examined. (Putnick and Bornstein, 2016 R44)

Scope of evidence: differences in measurement paths are examined alongside group means rather than translating mean differences directly into internal-state differences. Invariance does not mean every individual must respond identically.

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: standardized tasks, repeated scores, and testing conditions. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: separate persistent characteristics, present state, and task experience.

WRRA seeks to distinguish relatively persistent individual differences, test-day states, and task familiarity. Lower scores on a fatigued day do not establish a change in overall ability. Conversely, present state is not used to dismiss every persistent difference. Repeated measurements and relationships between tasks are needed.

4.1 Common and specific abilities — WRRA interpretation

WRRA distinguishes test scores, latent characteristics, and neural implementation. A relationship between a common factor and common carrier is a separate connecting hypothesis.

4.2 Fair comparisons — WRRA interpretation

Personal states, external support, and task boundaries specify the conditions whose outcomes were compared.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned outputs: task performance and latent-characteristic estimates.

6 Falsification conditions and interpretive limits

Falsification condition: restrict direct comparison if measurement meanings differ across languages or populations.

Culture, language, and educational opportunity enter interpretation. Representing all a person's diverse capacities with one score requires correspondingly broad evidence. Nor does completing a test quickly establish high intelligence merely because it resembles minimal computation.

6.1 Individual explanations and limits of use

Using tests for selection or educational support requires distinguishing research-level correlations from individual-level error. Predictive ability does not mean accurate classification of every person. WRRA models also assess purposes and misclassification costs separately. Explanatory simplicity or computational speed alone cannot justify consequential judgments about individuals.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Cognitive comparisons require norms, measurement invariance, and intended purposes together.

8 Next questions and evidence

Next research question: Do interpretations of individual differences hold after matching norms and language conditions?

Chapter references: R1, R44.

Part 6 Motivation Emotion and Self Regulation

Chapter 21 Needs, Rewards, Goals, and the Direction of Action

1 Scene and question

Understanding action requires examining sought and avoided outcomes and personally important goals. The same activity may involve different experiences and choices when undertaken from interest or for external evaluation. Motivation is not fully measured by the amount of behavior. (Spielman et al., 2020, §10.1 R1)

Chapter question: How do goals, expectations, and costs change action initiation and persistence?

2 Starting points in established research

2.1 Distinguishing needs, goals, and action

Hunger, a goal of passing an examination, and actually starting to study are related but distinct. Goals may conflict, or costs may prevent action despite a goal. (Spielman et al., 2020, §10.1 R1)

2.2 Expectations and costs

Ajzen's theory of planned behavior distinguishes attitudes, subjective norms, perceived behavioral control, intentions, and behavior. It is not read here as one theory of every effort cost. (Ajzen, 1991 R4790020-T))

2.3 External rewards and intrinsic motivation

Deci and colleagues' meta-analysis examined relationships with intrinsic motivation by reward type and delivery conditions. Its findings cannot be reduced to all external rewards always increasing or decreasing motivation. (Deci et al., 1999 R54)

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: goal reports, reward conditions, and choice costs. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: connect goals, residue, and costs to action priorities.

Goals enter WRRA as conditions setting action priorities. Goal-directed expectations, residues of earlier outcomes, current costs, and available actions are distinguished. Reward responses, actual enjoyment, and willingness to persist are not combined into one variable.

4.1 Intrinsic reasons and external rewards

Activities sustained by interest and meaning may differ experientially from those seeking external evaluation, but these motives are not always exclusive. Reward form, delivery, and choice must be investigated together. WRRA examines personal goals rather than computing all motivation from external reward magnitude alone.

4.2 Distinguishing needs, goals, and action — WRRA interpretation

WRRA separates goal presence from execution conditions, recording what was sought and which inputs and resources were actually used.

4.3 Expectations and costs — WRRA interpretation

Expectations, effort, time, and failure burdens are candidate variables with defined units and measures. Connections from experiential residue to expectation and from current conditions to execution are hypotheses for this book to test.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned outputs: changes in initiation, persistence, and choice.

6 Falsification conditions and interpretive limits

Falsification condition: redefine the objective term if assumed goals do not explain the person's choices.

Researchers must establish participants' goals. Maximizing the researcher's score is not necessarily their sole purpose. Minimal computation is assessed under explicit goals and maintenance conditions; optimization ignoring other goals is limited accordingly.

6.1 Interpreting persistence and stopping

Long persistence does not establish good goal adaptation. Stopping an activity whose value has fallen or cost risen may also be rational. Duration, outcomes, and alternative values are considered together. Minimal computation evaluates reduced unnecessary expenditure under goal-achievement and maintenance criteria.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Persistence is examined through expectations, costs, and permissible conditions alongside goals.

8 Next questions and evidence

Next research question: How does persistence change when expected rewards and effort costs vary separately?

Chapter references: R1, R47, R54.

Chapter 22 Emotional Experience, Bodily Responses, and Expression

1 Scene and question

Emotion includes reported experience, bodily responses, facial expressions, and behavior. These need not change in identical directions or magnitudes. People may suppress anger's expression or perform required actions while tense. (Gross, 1998 R42; Barrett et al., 2019 R53)

Chapter question: When do experience reports, bodily responses, and emotional expression change together?

2 Starting points in established research

2.1 Emotion categories and dimensions

Emotions can be classified into categories such as anger and fear or described along dimensions such as pleasantness and arousal. Each offers different distinctions. Their suitability for the research question and the meaning of reporting language must be checked. (Spielman et al., 2020, §10.4 R1)

2.2 Appraising a situation's meaning

The same loss may be experienced differently depending on whether it was expected, recoverable, or attributed to someone. Appraisals are candidate transformations to study, not explanations freely appended afterward to fit outcomes. (Spielman et al., 2020, §10.4 R1)

2.3 Expression and relationships

Facial, vocal, and postural meanings require context and relationships. Expressions can be regulated, and similar expressions can arise from different experiences. Faces alone do not establish every internal emotion without the person's report. (Barrett et al., 2019 R53)

2.4 Evidence for inferring emotion from facial movements

Barrett and colleagues review consistency, distinctiveness, and generalization in emotion inference from facial movement. Correspondences between observed movement and emotion categories require attention to context and measurement. A particular facial pattern cannot simply be assumed to mean one emotion in every person and situation. (Barrett et al., 2019 R53)

Scope of evidence: the WRRA application separates experience reports, expression choices, and observation. A facial classifier's output is recorded as a candidate observation to compare with personal experience.

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: emotion reports, physiology, expression, and context. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

3.1 Temporal alignment in emotion research

An immediate bodily response and an emotion described much later are not simultaneous records. Time windows must specify connections between fast responses and later interpretation. Distinguishing onset, maintenance, and regulation helps investigate which indicators relate to which stages. One average should not erase all temporal structure.

4 WRRA-specific transformation

Transformation specification: separate observation paths for experience, the body, and behavior.

WRRA treats emotion as processes connected to situational meaning and readiness for action. Present input, experiential residue, bodily state, and relational conditions are made concrete. Anger or sadness is not predetermined to correspond one-to-one with one circuit or substance.

4.1 Emotion categories and dimensions — WRRA interpretation

Emotion names, physiological values, and action directions remain distinct observations.

4.2 Appraising a situation's meaning — WRRA interpretation

Appraisal terms are specified beforehand, then investigated for how inputs, residue, and present goals enter them.

4.3 Expression and relationships — WRRA interpretation

WRRA can separately study paths forming experience and paths selecting expression in social situations.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned outputs: responses along each path, agreements, and disagreements.

6 Falsification conditions and interpretive limits

Falsification condition: restrict single-indicator interpretations if one measurement path cannot reliably infer another.

Research aligns emotion-report timing, physiological windows, and behavioral contexts. Increased heart rate does not identify a particular emotion, nor does expression establish internal experience. Disagreements motivate investigation of differing paths rather than dismissal as simple error.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Emotional experience, bodily response, and expression are not assumed to occur at the same time in the same way.

8 Next questions and evidence

Next research question: Are common changes observed after aligning reports, physiology, and expression in time?

Chapter references: R1, R7, R42, R53.

Chapter 23 Emotion Regulation, Self-Control, and Action Selection

1 Scene and question

Feeling an unpleasant emotion differs from how it is handled. Reinterpretation, shifting attention, and modifying expression may act at different stages. Their suitability depends on goals and situations. (Gross, 1998 R42)

Chapter question: Do regulation effects arise from personal rules or environmental changes?

2 Starting points in established research

2.1 Changing situations and changing interpretations

Avoiding or changing an unpleasant situation acts at a different point from understanding it differently. Attention shifts and expression regulation can also be distinguished. (Gross, 1998 R42)

2.2 Flexible regulation

Reducing discomfort is not every situation's goal. Important action may require tolerating discomfort or expressing a legitimate concern. Research considers personal goals and actual function together.

2.3 Assumptions about self-control

Reduced performance after one task does not establish depletion of a single store of willpower. Hagger and colleagues' multilaboratory preregistered study found no strong replication evidence for ego depletion under its procedure. This does not establish absence of every self-regulatory phenomenon. (Hagger et al., 2016 R43)

2.4 Replication results for a resource model

Hagger and colleagues' preregistered replication involved 23 laboratories and 2,141 participants. The combined effect under the standardized sequential-task procedure was d=0.04, with a 95% confidence interval of −0.07 to 0.15, including zero. This procedure did not provide clear evidence for the decrement expected by a self-control resource-depletion model. (Hagger et al., 2016 R43)

Scope of evidence: this single estimate does not determine results for all other self-regulation tasks. WRRA resource variables must likewise be tested through independent indicators and manipulations distinguishing alternatives.

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: regulation strategies, task

conditions, and multiple outcomes. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: designate beforehand where the intervention acts.

WRRA specifies whether intervention changes input, relationships, residue activation, current resources, or expression paths. Lower emotion reports may coexist with unchanged physiology or function, requiring multiple outcomes. Changed necessary actions despite persisting emotion are also a distinct achievement.

4.1 The role of environmental design

Reducing distractions or reordering choices may change the same person's actions. This differs from improved internal capacity. WRRA specifies person–environment boundaries and records functions performed by external devices. Testing persistence after assistance is removed distinguishes lasting learning from immediate support.

4.2 Changing situations and changing interpretations — WRRA interpretation

Specifying intervention location expresses how the same outcome can arise through different paths. Which method is preferable requires validation against goals and conditions.

4.3 Flexible regulation — WRRA interpretation

Maintenance criteria may include task functions and relational conditions alongside emotion scores.

4.4 Assumptions about self-control — WRRA interpretation

Resource variables require independent indicators and effects distinguishing alternative explanations.

4.5 Comparing accuracy and costs within the same task

Shenhav et al.’s (2013) expected-value-of-control theory provides a comparison framework involving payoff, control investment, and cognitive-effort cost. This does not make effort and physical energy the same quantity. WRRA fixes task function and tolerances while recording time, operation count, and memory separately. Physical energy requires its own measurement and units; subjective fatigue is a reported scale. Combining different units requires explicit normalization and weights. A cheaper model that loses required function does not satisfy Core minimum computation. [R69; W1, §1.1]

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned outputs: experiential, behavioral, and functional changes.

6 Falsification conditions and interpretive limits

Falsification condition: revise the mechanism claim if the designated path is unchanged while only other conditions change.

Explaining self-control as one quantity of willpower makes measurement ambiguous. Rewards, distractions, available choice time, and alternatives should first be examined. This book's resource concept does not automatically adopt a particular willpower-depletion theory.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Regulation is assessed by which paths personal rules and situational changes alter.

8 Next questions and evidence

Next research question: Can situational-change and reappraisal effects be distinguished using the same evaluation criteria?

Chapter references: R1, R42, R43.

Chapter 24 Stress, Fatigue, Recovery, and Available Resources

1 Scene and question

Bodily responses may help meet emergencies, while prolonged demands introduce other burdens. McEwen discussed both protective and damaging effects of stress mediators. (McEwen, 1998 R12) Stress is therefore more than a quantity of bad stimulation.

Chapter question: How can fatigue and recovery be specified without one resource score?

2 Starting points in established research

2.1 Demands and controllability

Having much to do differs from appraising those tasks as difficult to manage. Control, predictability, and support may change the meaning of identical demands. (Spielman et al., 2020, §14.1 R1)

2.2 Dimensions of fatigue

Sleepiness, mental fatigue, and muscle fatigue are different indicators. Questionnaires, tasks, and physiological measurements may read different things. Sleep or activity records can better differentiate current conditions. (NINDS, n.d. R8)

2.3 Recovery paths

Improvement after rest may combine elapsed time, reduced new input, physical recovery, and changed expectations. Comparison with other activities over the same period helps narrow pathways. Recovery rates may differ across people and functions; not all residues receive one decay rate. Chapter 41's single ratio is a small comparison example.

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: demands, sleep, fatigue, and support conditions. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: separate load, residue, and recovery conditions.

WRRA separates demand magnitude, interpretation, coping means, remaining influence, and recovery conditions. Sleep duration, subjective fatigue, and physiological data measure different things. Combining them into one resource value requires explaining weights and units.

4.1 Conditions reducing load

Reducing environmental demands or increasing authority and support differs from regulating individual responses, though both can be combined. The ledger distinguishes externally reduced costs from internally changed states. Improved outcomes count within their scope; further measurement assesses which connection was established.

4.2 Demands and controllability — WRRA interpretation

External demands, personal interpretation, and available responses are recorded separately. Discomfort alone is not a reason to assume the environment is safer than it actually is.

4.3 Dimensions of fatigue — WRRA interpretation

Resource indicators are selected for the purpose; combining indicators requires grounds for their units and weights.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned outputs: functional changes and recovery paths.

6 Falsification conditions and interpretive limits

Falsification condition: deny explanatory credit if resource indicators are not defined independently of outcomes.

Available resources are not defined retrospectively from behavior. Independently measured conditions must explain subsequent performance. Chapter 42's dimensionless resource values are hypothetical inputs illustrating a principle, not measurements of actual fatigue. Recovery research also separates mere elapsed time from actual environmental change.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Explaining recovery requires specifying resources as concrete dimensions such as sleep, time, and support.

8 Next questions and evidence

Next research question: Can recovery paths be identified by separating sleep, time pressure, and support?

Chapter references: R1, R8, R12.

Part 7 Development Personality and Self

Chapter 25 Infancy and Childhood

1 Scene and question

Early development combines bodily and neural change, interactions with others, language, and play. Similar ages need not mean identical rates or environments. Developmental stages organize observations rather than placing every individual's changes on one timetable. (Spielman et al., 2020, §9.1 R1)

Chapter question: How can developmental changes be separated from task understanding and measurement differences?

2 Starting points in established research

2.1 Developmental domains and interactions

Physical and motor development, perception and language, emotion and social functioning are related but require different observations. The rate in one domain does not determine a child's overall ability. (Spielman et al., 2020, §9.3 R1)

2.2 Observing attachment and exploration

Caregiver research can jointly record proximity seeking, exploration, separation, and reunion. Responses in one scene should not be read as lifelong relationship destiny. (Spielman et al., 2020, §9.3 R1)

2.3 Developmentally appropriate measurement

Before sufficient verbal responding is possible, gaze, behavior, and play records may be used. Indicator meanings require task-specific validation. Looking longer does not directly establish complete understanding or preference. (Spielman et al., 2020, §9.3 R1)

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: developmental timing, interactions, and behavioral observations. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: connect initial conditions, repeated experience, and reciprocal updating.

WRRA asks which residues and relationships repeated interactions form and how they change subsequent exploration and action. Caregiver characteristics and children's responses may influence each other; neither is automatically the sole cause. Bodily growth and task understanding also enter measurement.

4.1 Developmental domains and interactions — WRRA interpretation

WRRA decomposes the structures and resources on which new behavior depends. Increased movement changing exploration and thereby learning is also a researchable connection.

4.2 Observing attachment and exploration — WRRA interpretation

Relational conditions and the child's state are specified, then tendencies persisting across environments are examined.

4.3 Developmentally appropriate measurement — WRRA interpretation

Clear observation–state links allow competing explanations for the same behavior to be compared.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned outputs: changes in exploration, learning, and relational responses.

6 Falsification conditions and interpretive limits

Falsification condition: avoid direct comparison on one scale if measurement meanings differ across ages.

Adult questions and assumed adult meanings may mismeasure children. Developmentally appropriate indicators include play, gaze, and behavioral observation. This account does not claim that early experience determines lifelong outcomes. Updating under new conditions receives equal research attention.

6.1 Change after early experience

Recognizing early experience's importance also requires studying later environments and support. Persistent difficulties and recovery under new conditions are tracked together. WRRA separates maintenance rates for lasting residue from updating through new experience, connecting repetition and context instead of explaining every later outcome through one event.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Interpreting developmental differences first requires task understanding and age-appropriate observations.

8 Next questions and evidence

Next research question: Do developmental differences remain after establishing task understanding and age-appropriate measurement?

Chapter references: R1.

Chapter 26 Adolescence and Adulthood

1 Scene and question

Adolescence changes roles in school, peer groups, and family, alongside available choices. Adulthood combines work, caregiving, intimacy, and social responsibility in different ways. Biological changes and changing social opportunities must be considered together. (Steinberg, 2008 R52; Spielman et al., 2020, §9.3 R1)

Chapter question: How do previous experience and new choices combine in life transitions?

2 Starting points in established research

2.1 Development and social transitions

Adolescent and adult change involves school, work, family, and peer expectations alongside biology. People of the same age can have different possible actions and responsibilities. (Spielman et al., 2020, §9.1 R1)

2.2 Risk and available choices

Research on novel choices must examine actual alternatives and failure costs. Giving adults and adolescents the same explanation does not equalize information understanding or social pressure. (Steinberg, 2008 R52)

2.3 Life events and individual differences

Employment changes, parenting, caregiving, and relationship beginnings or endings may not produce changes in the same direction. Timing, expectations, financial support, and personal goals are recorded together. (Spielman et al., 2020, §9.3 R1)

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: age, life events, relationships, and resources. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: separately update biological conditions and changes in social boundaries.

WRRA also views development as changing boundaries of possible action and relationships. It tracks how existing residue operates as new experiences expand and how changed choice and support alter responses. Brain immaturity alone does not explain every behavior at a given age.

4.1 Development and social transitions — WRRA interpretation

Age is not used merely as one causal variable: changed structures and boundaries are specified. More concrete explanatory units produce clearer testable questions.

4.2 Risk and available choices — WRRA interpretation

Choice computations distinguish present information, experiential residue, and relational expectations. Group means do not automatically classify an individual's judgment.

4.3 Life events and individual differences — WRRA interpretation

WRRA focuses on the concrete inputs and boundaries changing state rather than event names, investigating divergent paths after nominally identical events.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned outputs: changes in roles, goals, and choices.

6 Falsification conditions and interpretive limits

Falsification condition: limit developmental interpretations if cohort differences explain apparent age effects.

Age-group comparisons can mix cohort and educational differences. Within-person follow-up and group comparisons should be interpreted together. Unemployment or parenting is not treated as an identical stimulus for everyone; income, support, personal meaning, and choice are recorded.

6.1 Stability and persistence of change

Immediate responses to a new role differ from later adaptation. Repeated observations help separate rapid state change from lasting habits. Comparisons with previous generations may mix educational and social differences. WRRA treats within-person change and between-group differences as separate research contracts.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Life-transition changes are investigated as combinations of past experience, new environments, and available choices.

8 Next questions and evidence

Next research question: How do within-person and environmental changes co-occur before and after a life transition?

Chapter references: R1, R52.

Chapter 27 Later Life

1 Scene and question

Not all abilities change in the same direction or at the same rate in later life. Processing new information and using long experience may show different patterns. Health, sensory function, living conditions, and social support can also influence performance. (Spielman et al., 2020, §9.3 R1)

Chapter question: How can performance changes in later life be distinguished from effects of supportive environments?

2 Starting points in established research

2.1 Typical change and functional difficulties

Distinguishing average aging-related change from conditions affecting everyday functioning requires multiple sources. Trajectories and changes in daily life matter alongside a single score. (NIA, n.d. R28)

2.2 Speed, knowledge, and strategies

Rapid new-information processing and use of accumulated knowledge impose different demands. Slow responses need not be wrong judgments; they may reflect longer examination preserving accuracy. (Heitz, 2014 R39; Spielman et al., 2020, §9.3 R1)

2.3 Assistive tools and boundaries of everyday life

Notes, reminders, and accessible environments can support daily functioning. Conditions of availability and learning costs are examined together. (NIA, n.d. R28)

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: repeated cognitive tests, sensory function, health, and everyday functioning. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: separate retained knowledge, access, and present bodily conditions.

WRRA distinguishes accumulated experiential residue, current access paths, and bodily conditions. Slow name retrieval and impaired everyday judgment are not merged into one phenomenon. Lower scores under insufficient sensory input are not immediately interpreted as changed cognitive ability.

4.1 Typical change and functional difficulties — WRRA interpretation

Function is explained through bodily state, sensory and access paths, retained knowledge, and environmental support. Age itself is not one deficiency indicator.

4.2 Speed, knowledge, and strategies — WRRA interpretation

Time, error, and actual goals enter evaluation together. Strategy differences are distinguished from impairment of overall ability.

4.3 Assistive tools and boundaries of everyday life — WRRA interpretation

Maintaining function with support differs from unaided performance. Independence is defined for the research purpose rather than exclusively as doing everything alone.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned output: correspondences between task changes and everyday function.

6 Falsification conditions and interpretive limits

Falsification condition: reconsider cognitive-decline interpretations if sensory or testing conditions explain the difference.

Distinguishing typical individual variation from disease-related changes requires professional evaluation. This book supplies no criterion diagnosing causes from one memory test; it proposes a research structure combining repeated measurement, functional change, and health.

6.1 Cohorts and repeated tests

Age-group differences may mix education, health, and technological experience. Repeated tests also introduce practice and attrition bias. Different designs must inform interpretation together. The WRRA ledger records changes alongside recruitment and retention conditions to assess explanatory scope.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Later-life function is evaluated through personal performance, assistive tools, and living environments together.

8 Next questions and evidence

Next research question: Do test scores and everyday functioning change together when assistive tools are provided?

Chapter references: R1, R28, R39.

Chapter 28 Temperament, Personality, and Persistent Individual Differences

1 Scene and question

Some people approach unfamiliar situations immediately; others observe first. Understanding these tendencies requires repetition across situations and times. One response may reflect that day's state or a particular relationship. (Spielman et al., 2020, §11.7 R1)

Chapter question: How do persistent differences coexist with situation-dependent responses?

2 Starting points in established research

2.1 Theories of personality

Trait approaches summarize persistent differences; social-cognitive approaches examine interpretation and learning. Psychodynamic and humanistic perspectives have posed other questions about conflict and self-experience. Historical influence differs from support for individual propositions. (Spielman et al., 2020, §1.2 R1)

2.2 Temperament and personality

Early response tendencies can be studied as temperament without excluding subsequent experience and development. Personality is not assumed to become fixed and manifest identically in every situation. (Spielman et al., 2020, §11.7 R1)

2.3 Consistency across situations

Someone may respond differently among friends and under evaluation while retaining repeatable situation-specific tendencies. Repeated observations across situations distinguish individual differences from situational effects. (Spielman et al., 2020, §11.4 R1)

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: behavior across situations and repeated personality measures. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: separate persistent traits, states, and environmental selection.

WRRA treats personality as relatively persistent interpretive and response tendencies. It differentiates contributions from initial conditions, accumulated residue, and frequently encountered environments. Existing trait scales provide comparison and measurement material; their names are not automatically translated into neural mechanisms.

4.1 Theories of personality — WRRA interpretation

WRRA specifies which aspects of present state, residue, relationships, and goals a perspective explains.

4.2 Temperament and personality — WRRA interpretation

Separating starting conditions, repeated experiences, and frequently chosen environments supports models permitting both maintenance and change.

4.3 Consistency across situations — WRRA interpretation

Rather than stopping at one coefficient per person, WRRA investigates which conditions activate which rules.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned output: response tendencies persisting across situations.

6 Falsification conditions and interpretive limits

Falsification condition: revise a single-coefficient interpretation if the same trait has different meanings across situations.

Traits and states are not mutually exclusive. Stable differences allow situational change. Person-specific coefficients require sufficient repeated data; one questionnaire cannot establish every updating rule. Personality types should not become labels restricting human possibilities.

6.1 Uses of personality measurement

Self-reports, others' ratings, and behavior estimate personality through different paths. Disagreement prompts examination of observed situations and periods rather than automatically declaring one wrong. Convenient classifications may inadequately represent complexity. Simplicity in WRRA is judged as minimality preserving actual distinctions.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Repeated observations can distinguish persistent differences and situational changes within one model.

8 Next questions and evidence

Next research question: Can stable coefficients and situational changes be distinguished in repeated person-level observations?

Chapter references: R1.

Chapter 29 Self-Concept, Identity, and Sense of Agency

1 Scene and question

We understand ourselves as particular people and connect action outcomes with our choices. Self-concept includes self-knowledge and evaluation; identity may involve relationships, social roles, and life narratives. Feeling that one initiated an action is a separate research topic. (Haggard, 2017 R45; Spielman et al., 2020, §9.2 R1)

Chapter question: How do self-description and sense of agency relate to actual controllability?

2 Starting points in established research

2.1 Information used to describe the self

Judgments of ability and worth, bodily feelings, social roles, and personal stories contribute to self-understanding. Some information persists while evaluations may change rapidly with context. (Spielman et al., 2020, §9.2 R1)

2.2 Others and self-interpretation

Others' reactions can enter self-evaluation, but their meaning depends on relationships and experience. Receiving criticism differs from interpreting it as a judgment of one's entire worth.

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: self-reports, action outcomes, and control conditions. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: connect self-related residue, bodily inputs, and relational feedback.

WRRA treats self-related information as currently usable structures. It asks how memory continuity, bodily sensation, and others' responses enter self-interpretation. The whole self is not defined as one storage location or residue value.

4.1 Agency and actual control

Experiencing a choice as one's own and actually being able to change an outcome can be examined separately using altered delays or feedback. WRRA distinguishes possible-action boundaries from reports. Increased perceived control does not alone establish increased external control.

4.2 Information used to describe the self — WRRA interpretation

WRRA separates present states and residues relevant to the question rather than reducing the whole self to one number.

4.3 Others and self-interpretation — WRRA interpretation

The other's behavior, inferred intention, and self-related judgment are distinct terms. Locating change makes intervention hypotheses more concrete.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned outputs: changes in self-evaluation, agency, and behavior.

6 Falsification conditions and interpretive limits

Falsification condition: changed reports under unchanged control conditions do not establish increased objective control.

Self-reports, behavior, and others' evaluations are observed separately. Subjective agency and objective controllability can also be measured separately. This structure does not claim to resolve philosophical free will; investigation begins with manipulable control conditions and reports.

6.1 Continuity and change

Stories linking past and present selves involve memory and present interpretation. A changed narrative and changed everyday life are checked separately. WRRA treats self-narratives as presently accessible records and tracks their revised interpretations in subsequent action. It does not claim to resolve every philosophical question about the self.

7 Current verdict

Correspondences between self-related residue and control conditions are INFERRED. Validation of actual observation paths is OPEN.

Chapter in one sentence: Self-description, agency, and objective control are distinct but connectable research targets.

8 Next questions and evidence

Next research question: How do agency reports and behavior change when objective control conditions change?

Chapter references: R1, R45.

Part 8 Society Relationships and Culture

Chapter 30 Social Perception, Attribution, Attitudes, and Persuasion

1 Scene and question

A delayed reply may suggest indifference, yet the other person may be busy or disconnected. Social perception interprets people and situations from limited information; attribution asks where behavior's causes are located. (Spielman et al., 2020, §12.1 R1)

Chapter question: How can the other's actual state differ from the observer's interpretation?

2 Starting points in established research

2.1 Locating causes

Explaining behavior through personal characteristics or circumstances depends on information and observational perspective. Studies separately present actor and situational information and examine changed judgments. (Spielman et al., 2020, §12.1 R1)

2.2 Connecting attitudes and behavior

Reported agreement and actual action may pass through different conditions, including opportunity, cost, social pressure, and habit. (Ajzen, 1991 R4790020-T))

2.3 Information and relationships in persuasion

Message evidence, source credibility, attention, understanding, and personal goals can be investigated together. A famous speaker does not make a message accurate. (Spielman et al., 2020, §12.3 R1)

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: messages, relational histories, judgments, and actions. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: distinguish the other's actual conditions from the observer's interpretive model.

WRRA separates the other's actual state from the observer's model of it. Strong observer residue may give ambiguous behavior a particular meaning. Independent data must assess accuracy. The record should not imply direct access to another person's mind.

4.1 Locating causes — WRRA interpretation

Observer residue, present inputs, and causal models are separate. Inferred intentions are not recorded as the other's actual state.

4.2 Connecting attitudes and behavior — WRRA interpretation

Attitude reports, intentions, possible actions, and actual choices remain distinct. Changed reports do not establish all subsequent behavior has changed; persistence and context are checked separately.

4.3 Information and relationships in persuasion — WRRA interpretation

WRRA distinguishes which judgments involve information content and relational residue. Persistence with another speaker or after a delay helps assess explanatory scope.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned outputs: attributions, attitude reports, and choices.

6 Falsification conditions and interpretive limits

Falsification condition: changed responses alone do not establish changed behavior or lasting beliefs.

Persuasion research distinguishes messages, speakers, prior knowledge, and social conditions. Agreement and later choices may differ. WRRA's contribution is tracking which combinations of input and residue alter which expression paths.

6.1 Research on prejudice and behavior

Generalized beliefs, evaluations, and actual discriminatory actions are related but not identical indicators. Differences between explicit responses and automatic tasks also require interpreting the tasks. WRRA separates entry points of group cues, personal experience, and paths by which rules and authority change outcomes. Morally judging a person from one test exceeds its measurement scope.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Checking attribution and persuasion accuracy requires separating the observer's interpretation from the other's actual state.

8 Next questions and evidence

Next research question: Does independent information about the other's state revise attribution and choice?

Chapter references: R1, R47.

Chapter 31 Attachment, Intimacy, Family, and Interpersonal Relationships

1 Scene and question

A close person's words may matter more than a stranger's because shared experience enters current expectations. Relationship research does not explain trust, intimacy, conflict, and support through one person's personality alone. (Cohen and Wills, 1985 R51)

Chapter question: How can relationships in which each person's action becomes the other's next input be recorded?

2 Starting points in established research

2.1 A relationship is a two-person process

One person's report can assess relationship quality but is not the complete interaction record. Recording both people's expectations, interpretations, and actions can reveal different experiences.

Cohen and Wills distinguished direct support effects from stress-buffering hypotheses. Their review does not imply that every close relationship has the same effect. (Cohen and Wills, 1985 R51)

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: both parties' reports and interaction records. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: combine person-specific residues with inputs given to each other.

WRRA jointly records each person's present state, residue, and interaction. One person's action becomes the other's input, whose response feeds back as input. The same event need not leave identical residue in both.

4.1 Paths of repeated conflict

One hypothesis is that silence is interpreted as rejection, provoking pressure that increases silence. Testing this requires action timing and each person's interpretation. WRRA combines both states and updates, comparing one-way influences and interaction rather than prejudging one person's personality as the cause.

4.2 Support and autonomy

Assistance may have different outcomes depending on content, received meaning, and choice. Practical burden reduction and emotional support are not one scale. WRRA specifies the provided resources and changed boundaries, distinguishing receiving support from restored function when evaluating effects.

4.3 A relationship is a two-person process — WRRA interpretation

My model of the other and the other's report have different owners. Disagreement is investigated by event and context rather than erased through averaging.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned outputs: recurring patterns of expectations and actions within relationships.

6 Falsification conditions and interpretive limits

Falsification condition: add a joint interaction model if one party's data are insufficient.

Attachment or relationship types may organize tendencies without fixing every relationship into a type. Changes of partner or situation reveal what persists and what changes. Research distinguishes internal individual changes from changed interaction conditions.

6.1 Measuring relationship change

Conflict frequency, recovery time, mutual understanding, and everyday function can be investigated, with target outcomes fixed beforehand. Where safety or autonomy is compromised, conversational skills alone do not explain every problem. WRRA considers actual authority, risks, and external support alongside internal residue.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Relationships require jointly recording two people's states, residues, and feedback.

8 Next questions and evidence

Next research question: Which connection in repeated conflict can be changed when both people's time series are examined together?

Chapter references: R1, R51.

Chapter 32 Conformity, Obedience, Groups, Cooperation, and Conflict

1 Scene and question

Choices may differ between judging alone and speaking before a group. Majority information, belonging motives, authority, and sanctions may each matter. This is why conformity and obedience need not share one cause. (Spielman et al., 2020, §12.4 R1)

Chapter question: How do group rules and information flows change individual choices and collective outcomes?

2 Starting points in established research

2.1 Following information or norms?

Changing judgment because others seem better informed differs from changing a public response to preserve belonging. Separating private and public judgments and manipulating information reliability helps investigate these paths. (Spielman et al., 2020, §12.4 R1)

2.2 Authority and responsibility

Explaining obedience requires examining authority's basis, refusal costs, and responsibility allocation. Institutional boundaries may intervene between moral judgment and actual action. (Spielman et al., 2020, §12.4 R1)

2.3 Cooperation and shared resources

Individual gains may conflict with collective maintenance when resources are shared. Information disclosure, repeated meetings, sanctions, and trust can be varied to study cooperation. (Ostrom, 2009 R48)

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: group rules, information, and individual choices. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: update authority, connection structure, and individual states together.

WRRA specifies group rules and information flows, personal goals and resources, and boundaries of refusal or exit. Cooperation prompts investigation of repeated interaction, responsibility, and disclosure as well as personality. Persistent conflict also requires recording expectations about one another's responses.

4.1 Explaining collective outcomes

Groups of equally capable people may differ because of connections, speaking order, and information sharing. WRRA compares sums of individual states with interaction effects. Adding a network does not itself explain a group phenomenon: the model must compute or the study must measure how outcomes change when particular connections change.

4.2 Following information or norms? — WRRA interpretation

WRRA specifies which outputs involve shared inputs, social boundaries, and individual goals.

4.3 Authority and responsibility — WRRA interpretation

Authority and responsibility owners are recorded. Strong structural pressure differs from a conclusion that every individual action was inevitable.

4.4 Cooperation and shared resources — WRRA interpretation

Individual and common resource ledgers are separated. Transfers of one party's benefit into another's cost must not produce incorrect totals.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned outputs: cooperation rates, group choices, and conflict paths.

6 Falsification conditions and interpretive limits

Falsification condition: reconsider the proposed connection mechanism if structural changes have no effect.

Whether individual-level means fully explain collective outcomes must be checked. A few network positions or decision rules may substantially alter results. Without neural measurements, such cases explain social state transitions, not computed neural mechanisms.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Collective outcomes are investigated through information flow, responsibility, and exit conditions alongside individual averages.

8 Next questions and evidence

Next research question: How do collective outcomes change with public information, responsibility, and exit possibilities?

Chapter references: R1, R48.

Chapter 33 Culture, Norms, Social Environments, and Mind

1 Scene and question

Appropriate emotional expression and responsible choice relate to social context. Culture does not mean that everyone in one country has the same mind. Generational, group, and individual differences exist within societies. (Spielman et al., 2020, §11.8 R1)

Chapter question: How do records retained in cultures and institutions connect to individual experience?

2 Starting points in established research

2.1 Diversity within groups

Groups defined by nationality or language still differ by generation, class, region, and experience. Average differences do not become characteristics of every individual. (Spielman et al., 2020, §11.8 R1)

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: languages, norms, institutions, and group-specific measurements. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: distinguish individual residue, external records, and environmental boundaries.

Norms, institutions, language, and living conditions shape inputs, boundaries, and repeated experience. Personal residue and social records have different owners. A law or custom's existence, a person's knowledge of it, and compliance require separate checks.

4.1 Culture as an environment of inputs and rules

Customs, language, education, and institutions change repeated experiences and possible actions. Cultural explanation does not imply uniform group members. WRRA specifies which norms are known to whom and actually applied. Personal interpretation, public records, and institutional constraints are different variables.

4.2 Cycles between individuals and institutions

Institutions influence people, while actions can change rules and records across different timescales. WRRA distinguishes internal residue from social records and connects transmission paths. Stating which scales have supporting data reduces overgeneralization.

4.3 Diversity within groups — WRRA interpretation

Analysis distinguishes individual and group levels. Distributions may overlap substantially; the conditions increasing differences should be shown alongside that overlap.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned outputs: contextual differences in expression and choice.

6 Falsification conditions and interpretive limits

Falsification condition: score differences do not establish psychological-trait differences if measurement meanings differ.

Cross-group questionnaires first require checking whether identical items mean the same thing. Cultural differences do not establish innate differences. Samples, contexts, and generalization limits are documented.

6.1 Translation and measurement invariance

Using the same translated word does not establish equivalent question meaning. Item interpretation, response modes, and socially permissible expression require examination. Different group-specific observation functions make direct score comparisons difficult. Unestablished invariance becomes an interpretive boundary.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Cultural and institutional records have owners and pathways different from individual experience.

8 Next questions and evidence

Next research question: Do contextual differences remain after establishing measurement invariance?

Chapter references: R1.

Part 9 Mental Health Treatment and Practical Applications

Chapter 34 Assessing Mental Health and Psychological Difficulties

1 Scene and question

Understanding mental health requires considering symptom types and duration, functional change, and personal suffering together. Diagnostic classifications provide communication and research criteria, but identical diagnoses do not establish identical causes. Symptoms can also be studied dimensionally. (NIMH, n.d. R7; NIMH, n.d. R13)

Chapter question: Through which observation paths can symptoms, function, and living conditions be assessed together?

2 Starting points in established research

2.1 Symptoms and function together

Assessment considers what someone experienced, for how long, how life changed, and what help they want. Equal symptom scores may require different assessments under different environments, risks, or bodily conditions. (Spielman et al., 2020, §15.2 R1)

2.2 Classification and dimensions

Diagnostic categories provide common criteria and communication; dimensional measures express differences in degree. They may serve different questions. (NIMH, n.d. R7)

2.3 Assessment paths and alternatives

Fatigue and concentration difficulties may involve psychological experiences, sleep, physical illness, medication, and other pathways. (NIMH, n.d. R17; NINDS, n.d. R8)

2.4 Original research comparing scale scores and clinical interviews

Kroenke and colleagues administered the PHQ-9 to 6,000 primary-care and obstetrics–gynecology patients. Criterion validity was examined against independent mental-health-professional interviews in 580 patients, alongside functioning and symptom-related difficulties. The study validated score meanings by connecting them to other observations. (Kroenke et al., 2001 R58)

Scope of evidence: validation scope must be checked when populations, languages, or uses change. Symptom scores are observables, not automatically individual causes or WRRA latent states.

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: symptoms and course, function, bodily conditions, and living conditions. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: distinguish observed difficulties, candidate causes, and maintenance conditions.

WRRA goes beyond renaming a diagnosis as a new variable. It separates current symptoms, earlier influences, living conditions, physical illness, medication, and support. Calling an unknown cause a residue problem does not advance research without a concrete path to examine.

The clinical chapters are a conceptual manuscript connecting existing research. They introduce neither a dedicated WRRA diagnostic method nor a validated treatment. Actual assessment connects professional interviews, appropriate tests, and medical evaluation where needed. Individual accuracy and misclassification consequences must be checked before applying research findings to individuals.

4.1 Symptoms and function together — WRRA interpretation

Separating outcome labels from maintenance paths allows differences within one diagnosis to be studied.

4.2 Classification and dimensions — WRRA interpretation

WRRA records the classification used and dimensions actually measured. It examines whether categorical boundaries serve the purpose without declaring clinical validity merely because a new system is proposed.

4.3 Assessment paths and alternatives — WRRA interpretation

Phenomenon-specific inputs, states, and external conditions are separated. One neural indicator does not establish a whole person's state; combining evidence still requires individual-level accuracy and error assessment.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned output: research models of individual pathways.

6 Falsification conditions and interpretive limits

Falsification condition: leave causal verdicts open if diagnoses alone do not identify pathways.

6.1 Preserving comprehensive scope

Eating-disorder and personality-disorder phenomena also belong in general psychology. Eating disorders are not reduced to food preferences or willpower, and clinical personality-disorder assessment differs from everyday personality typing. (NIMH, n.d. R33; NIMH, n.d. R34) This manuscript does not compute all these clinical domains; it identifies where later phenomenon-specific pathways and evidence can be expanded.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Mental-health assessment considers symptoms, function, and course together while preserving alternative causal explanations.

8 Next questions and evidence

Next research question: Does adding function and course to symptom scores change assessment's explanatory scope?

Chapter references: R1, R7, R8, R13, R17, R33, R34, R58.

Chapter 35 Anxiety-, Obsession-, and Trauma-Related Phenomena

1 Scene and question

Temporary worry differs from persistent anxiety disrupting life. Obsessive-compulsive disorder may involve recurrent unwanted thoughts or excessive repetitive behavior. Post-trauma responses do not take the same form in everyone exposed. (NIMH, n.d. R14; NIMH, n.d. R15; NIMH, n.d. R16)

Chapter question: How can current danger, experiential residue, and repetitive behavior be distinguished?

2 Starting points in established research

2.1 Threat and current danger

Threat anticipation must be examined in relation to present circumstances. Actual high danger differs from persistent reactions under low danger. (NIMH, n.d. R14)

2.2 Obsessive thoughts and compulsive behavior

Content, discomfort, time spent, and functional effects are recorded separately. Distinguishing repetitive behavior's short-term consequences from later course allows maintenance loops to be investigated. (NIMH, n.d. R15)

2.3 Paths after trauma

Responses vary, and many people recover over time. Persistent functional disruption calls for professional assessment. (NIMH, n.d. R16)

2.4 Placebo-controlled evidence for anxiety-related treatment

Carpenter and colleagues synthesized 41 randomized trials comparing cognitive behavioral therapy with psychological or pill placebo. The combined target-symptom effect was Hedges' g=0.56, varying by diagnosis and analysis population. This standardized between-group difference does not mean 56% of patients recovered. (Carpenter et al., 2018 R59)

Scope of evidence: average effects of treatments containing several procedures differ from the mechanisms transmitting those effects. A WRRA account of expectation or avoidance updating must connect intervention content and actual changes in data.

2.5 Original comparisons of procedures for obsessive-compulsive disorder

Foa and colleagues compared exposure and ritual prevention, clomipramine, their combination, and pill placebo in 122 adult outpatients. At 12 weeks, all three active conditions performed better than placebo. Findings are interpreted within the study's procedures, population, and assessment period. (Foa et al., 2005 R60)

Scope of evidence: comparative interventions strengthen discussion separating short-term consequences from later trajectories. Observed effects do not become validation of a particular WRRA residue equation.

2.6 A systematic review of post-traumatic phenomena

Lewis and colleagues reviewed 114 randomized trials of psychological treatments for adult PTSD, involving 8,171 participants. Evidence for trauma-focused CBT, EMDR, and other approaches was evaluated by treatment and comparator. Differences in evidence quantity and quality should not be merged into one psychotherapy effect. (Lewis et al., 2020 R61)

Scope of evidence: residue–avoidance connections are candidate structures for explaining these phenomena. Treatment-effect evidence and direct verification of those candidate connections are recorded separately.

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: symptom trajectories, current danger, avoidance, and checking. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: distinguish current danger, experiential residue, and short-term behavioral consequences.

WRRA separately records threat-related inputs, persisting experiential influences, current danger, and consequences of avoidance or checking. Reduced discomfort after avoidance motivates examining its effect on later choices. Continuing actual danger must not be explained solely by past residue.

4.1 Threat and current danger — WRRA interpretation

External danger, past influences, bodily state, and interpretation are separated. Unpleasant responses are assessed for information and function rather than all being labeled errors.

4.2 Obsessive thoughts and compulsive behavior — WRRA interpretation

WRRA proposes possible explanations without reducing every obsessive-compulsive phenomenon to one reinforcement equation.

4.3 Paths after trauma — WRRA interpretation

Event information, present cues, avoidance, and support are separated. Event severity alone does not determine an individual's later path; continuing current danger must also be distinguished.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned outputs: repeated discomfort and choices across situations.

6 Falsification conditions and interpretive limits

Falsification condition: responses sufficiently explained by present danger are not classified as maladaptive residue.

Different clinical phenomena are not reduced to one avoidance equation. Behavior may reduce anxiety, reflect habit, or serve another purpose. Exposure or memory manipulations involving people needing treatment are not undertaken on the basis of these hypothetical calculations. This chapter's achievement is specifying research terms and alternative explanations.

6.1 Distinguishing research from intervention

A response-loop model does not establish treatment effectiveness. New interventions require suitable comparisons, risk assessment, and consent. WRRA specifies whether intervention changes input, residue activation, or choice boundaries. Symptoms, function, persistence, and burden must be measured together.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Explaining threat and avoidance requires distinguishing current danger, earlier experience, and repeated action.

8 Next questions and evidence

Next research question: Does the connection between experience history and avoidance persist after separating current danger?

Chapter references: R14, R15, R16, R59, R60, R61.

Chapter 36 Depression, Mood Changes, and Low Energy and Motivation

1 Scene and question

Depression can affect thought, sleep, activity, and daily function as well as emotion, with biological, psychological, and environmental factors involved. (NIMH, n.d. R17) Bipolar mood and energy changes require assessment distinct from ordinary daily fluctuations. (NIMH, n.d. R18)

Chapter question: In what sequence do activity, reward expectations, and mood trajectories change?

2 Starting points in established research

2.1 Reward expectations and actual experience

The book's model design separately records anticipated outcomes, reported enjoyment during activity, and the time and effort needed to begin. This does not establish three variables as depression's causes. Collapsing them into one low-resource value obscures candidate paths, so expectation, experience, action, and subsequent updating are studied separately.

2.2 Mood trajectories and living conditions

Daily fluctuations and persistent clinical courses are different observations. Time records of sleep, activity, medication, and events can investigate maintenance conditions. (NIMH, n.d. R17)

2.3 Assessing bipolar-related changes

Marked changes in mood, energy, sleep, and activity require professional differentiation and assessment. Increased activity alone is therefore not judged recovery. (NIMH, n.d. R18)

2.4 Comparative evidence for acute antidepressant treatment

Cipriani and colleagues compared efficacy and acceptability of 21 antidepressants using 522 trials and 116,477 adults receiving acute major-depression treatment. Drugs showed better response rates than placebo overall, with uncertainty across comparisons and study-bias concerns. Certainty ranged from moderate to very low depending on the comparison. (Cipriani et al., 2018 R6232802-7))

Scope of evidence: acute adult monotherapy findings are not directly extended to bipolar disorder or long-term maintenance. Treatment response differs from confirmation of WRRA's proposed expectation–experience–action paths.

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: trajectories of mood, activity, sleep, and function. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: separate reward expectations, costs, bodily conditions, and environmental conditions.

Low activity is not explained solely as deficient willpower or resources. WRRA examines reward expectations, effort costs, bodily state, sleep, environment, and earlier experiences. Similar reports of low energy and motivation need not respond identically to the same intervention if maintenance pathways differ.

4.1 Mood trajectories and living conditions — WRRA interpretation

Measurement intervals are specified and reported averages connected to time-specific changes. Correlation does not immediately become a treatment target.

4.2 Assessing bipolar-related changes — WRRA interpretation

Assessment boundaries consider function, risk, and personal goals together. The general term state transition does not erase disorder-specific mechanisms.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned outputs: individual temporal changes in activity and mood.

6 Falsification conditions and interpretive limits

Falsification condition: consider multiple paths if one resource term cannot explain the course.

Increased behavior is assessed alongside function, safety, and mood rather than automatically called recovery. This conceptual model does not guide diagnosis or medication selection. Proposed repeated observations examine what changes first and what follows to distinguish cause from consequence.

6.1 Testing mechanistic explanations

If expectation and behavior both change after intervention, timing and alternatives must be examined. Paths changing expectations, costs, or environmental opportunities can be compared. WRRA supplies their specification; actual effect sizes require data. Calibrated values are distinguished from outputs after rules are fixed.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Low activity and mood changes are investigated over time rather than fixed to a single cause.

8 Next questions and evidence

Next research question: Which changes in sleep, activity, or expectations precede subsequent functional changes?

Chapter references: R17, R18, R62.

Chapter 37 Addiction-, Psychosis-, Neurodevelopmental-, and Neurocognitive-Related Phenomena

1 Scene and question

These are distinct research targets. Addiction involves difficulties regulating use and its consequences; psychosis involves changes in experience and reality judgment; neurodevelopmental and neurocognitive phenomena involve developmental paths, cognition, and everyday function. Grouping them here does not assume a common cause.

Chapter question: Can a common grammar address different clinical phenomena while preserving mechanistic differences?

2 Starting points in established research

2.1 Processes in addiction

Research examines reward, learning, cues, craving, regulation difficulties, and consequences together. Neural–environmental interactions require investigation. (NIDA, n.d. R29)

2.2 Psychotic experiences and assessment

Hallucinations, delusions, and changes in thought and activity require assessment alongside suffering, function, and medical conditions. Unfamiliar beliefs alone are not diagnostic; clinical context matters. (NIMH, n.d. R19)

2.3 Neurodevelopmental differences

ADHD research concerns persistent inattention and hyperactivity–impulsivity symptoms; autism-spectrum research examines varied social-communication and behavioral characteristics. Individuals and support needs vary within each category. (NIMH, n.d. R30; NIMH, n.d. R31)

2.4 Reading schizophrenia-maintenance evidence by outcome

Ceraso and colleagues reviewed 75 randomized trials with 9,145 participants. In the subset of 30 trials and 4,249 participants comparing relapse at 7–12 months, rates were 24% with medication maintenance and 61% with placebo. The review also evaluated movement-related adverse effects, sedation, and weight gain. (Ceraso et al., 2020 R63)

Scope of evidence: these are group outcomes within specified follow-up and trial conditions. They do not determine an individual's course or long-term outcomes. Benefits and burdens are separate outputs, a distinction retained in WRRA applications separating symptoms, function, and resources.

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: phenomenon-specific professional assessments, trajectories, and functional data. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: specify different mechanisms and pathways under common questions.

Schizophrenia and other mental disorders require professional assessment and care and cannot be diagnosed with one WRRA residue term. (NIMH, n.d. R19) Other conditions likewise require separating current responses, persistent differences, illness and medication, and environmental support. Neurodivergent differences are not all defined as defects to eliminate.

WRRA's shared grammar standardizes questions: What are actual inputs and present states? What influence remains? Where do functional difficulties arise? The mechanisms and values answering them differ by phenomenon. Separate domain models and clinical validation have not yet been performed for this manuscript.

4.1 Processes in addiction — WRRA interpretation

Current inputs, use history, available actions, and costs are recorded separately. Addiction is not defined as one moral or willpower deficiency.

4.2 Psychotic experiences and assessment — WRRA interpretation

WRRA can study input sources, interpretation, confidence, and action connections without identifying causes from one residue value.

4.3 Neurodevelopmental differences — WRRA interpretation

Sensory environments, task demands, strengths, and difficulties are distinguished while person–environment fit is studied.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned outputs: phenomenon-specific explanatory models and application scopes.

6 Falsification conditions and interpretive limits

Falsification condition: abandon a single-mechanism claim if identical coefficients cannot explain distinct phenomena.

6.1 Neurocognitive function and research scope

Cognitive changes have multiple possible causes, not age alone. Functional trajectories, health, medication, and support require examination. WRRA separates measurement of current state from estimation of persistent change, using different data and rules for different phenomena. This chapter connects research questions; it does not reproduce all conditions with one clinical model.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Sharing a grammar of questions still requires separate mechanisms and validation for distinct clinical phenomena.

8 Next questions and evidence

Next research question: Can separate small models be constructed with fixed phenomenon-specific inputs and states?

Chapter references: R13, R19, R28, R29, R30, R31, R63.

Chapter 38 Psychotherapy, Biological Treatment, and Environmental Intervention

1 Scene and question

Psychotherapy includes several approaches to emotional, cognitive, and behavioral difficulties. Type and suitability depend on the problem and personal circumstances, and may complement medication or other treatments. Effects are assessed through outcomes such as symptoms, function, and quality of life. (NIMH, n.d. R20)

Chapter question: How does evidence of treatment effects differ from evidence for their mechanisms?

2 Starting points in established research

2.1 Locations of change in psychotherapy

Cognitive-behavioral and relational or interpersonal approaches have different questions and procedures. Evidence must be checked for each target problem and method. (NIMH, n.d. R20)

2.2 Biological treatments and paths

Medication, brain stimulation, and related interventions are clinical domains with specific indications, evidence, and risks. This book provides no new prescription or procedure-selection criteria. (Spielman et al., 2020, §16.2 R1)

2.3 Effects of environments and services

Housing, financial support, and work or school adjustments can change people's circumstances. When combined with psychological intervention, their contributions should be investigated separately.

2.4 Treatment effects require specified comparators

Cuijpers and colleagues synthesized 409 trials with 52,702 participants on CBT for depression. Outcomes were better than controls such as usual care or waiting lists, while differences from other psychotherapies were small and nonsignificant in most sensitivity analyses. Interpretation depends on what was compared and when. (Cuijpers et al., 2023 R64)

Scope of evidence: data assessing a treatment differ from data assessing WRRA connections. Explanatory paths can be compared when intervention locations, intermediate changes, and final function are actually measured.

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: intervention content, comparison conditions, and symptom and functional changes. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: distinguish intervention location, mediation paths, and outcomes.

WRRA provides a framework for recording where intervention acts. Psychological interventions may change interpretations and behavioral rules, environmental interventions may alter choices and burdens, and biological treatments may change bodily or neural processes. Each actual mechanism requires its own evidence; this is not a fixed one-to-one mapping for every treatment.

Treatment effectiveness and correctness of a WRRA explanation require separate evidence. Improved outcomes do not establish the expected mediator. Any new WRRA-based intervention receives the same effect and risk criteria as existing treatments. This book announces no new treatment efficacy.

4.1 Locations of change in psychotherapy — WRRA interpretation

WRRA records intended changes to interpretation, action rules, relational inputs, or present resources. Identically named treatments still require records of actual content and dose.

4.2 Biological treatments and paths — WRRA interpretation

WRRA connects intervention location, actual change, and functional outcomes. Biological change differs from confirmation of a proposed psychological mechanism.

4.3 Effects of environments and services — WRRA interpretation

External resources and choices are not misrecorded as internal ability. Successful maintenance of function is evaluated with its support conditions.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned outputs: effect estimates and tests of mechanisms.

6 Falsification conditions and interpretive limits

Falsification condition: revise a mechanism explanation if outcomes improve without change in the designated mechanism.

6.1 Effects, mechanisms, and persistence

Before–after differences alone rarely establish intervention causality. Comparison conditions, natural course, selection, and attrition must be examined. Symptom reduction, functional improvement, burdens, and persistence over time are separated. WRRA mechanism hypotheses receive the same rigor as effect evaluation, and unsupported paths are revised.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Evidence of improved treatment outcomes differs from evidence establishing the mediating mechanism.

8 Next questions and evidence

Next research question: Are improved outcomes and changes in the hypothesized mediation path both observed?

Chapter references: R1, R20, R64.

Chapter 39 Applications in Education, Work, Health, and Technology Environments

1 Scene and question

Schools examine retention and transfer; workplaces examine performance and cooperation; health management examines sustained behavior; ergonomics studies person–device interaction. Similar failures may arise from limited knowledge, interface problems, time constraints, or authority restrictions.

Chapter question: How should designs reducing execution burdens across people and environments be evaluated?

2 Starting points in established research

2.1 Retention and transfer in education

Practice accuracy, delayed memory, and transfer to new problems are different achievements. Original research has examined retrieval tests' contribution to long-term retention. (Roediger and Karpicke, 2006 R32)

2.2 Workplace and organizational design

Errors require examining knowledge, unclear instructions, authority, and information transmission as alternative causes.

2.3 Health behavior and everyday technology

Intentions may not produce sustained actions without time, place, and support. (Ajzen, 1991 R4790020-T))

2.4 An experiment separating immediate performance from delayed memory

Roediger and Karpicke had students restudy prose or take recall tests without feedback and then compared memory. Repeated study was advantageous after 5 minutes, while prior retrieval testing showed advantages after 2 days and 1 week. Familiarity during study and later retention were therefore different indicators. (Roediger and Karpicke, 2006 R32)

Scope of evidence: findings concern the materials and testing conditions used. WRRA distinguishes current assistance from retained structure without assuming identical effect sizes across every subject and learner.

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: tasks, interfaces, records,

resources, and authority. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Transformation specification: design where burdens are placed across the person and environment.

WRRA asks what must be remembered, what can be recorded externally, and which transmission structures can be reused. Good information placement reducing errors can be compared with designs demanding more personal memory. Minimal computation is assessed within an explicit combined person–environment boundary.

4.1 Retention and transfer in education — WRRA interpretation

Practice-provided information differs from independently maintained residue. Under matched study time and task conditions, structures and outcomes can be compared.

4.2 Workplace and organizational design — WRRA interpretation

Inputs, staff relationships and records, resources, and permission boundaries are specified. Removing checks needed in exceptional cases requires reassessing preserved function. Safety-critical redundancy may be part of minimum cost.

4.3 Health behavior and everyday technology — WRRA interpretation

Intention differs from execution conditions. Reminder and recording-tool effects differ from changed personal habits. Recommending a health intervention requires its own evidence; plausible structure alone does not establish effectiveness.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned outputs: changes in performance, errors, and costs.

6 Falsification conditions and interpretive limits

Falsification condition: cost reduction alone does not establish success if functional or safety criteria fail.

Improved educational or work performance does not justify every form of surveillance. Psychological data collection requires explicit purposes, authority, and uses. Other applications, including forensic and sports psychology, retain their own standards and ethics. This manuscript organizes possibilities without separately establishing empirical application effects.

6.1 Extending to forensic, sports, and environmental fields

Memory and judgment accuracy in forensic psychology, skill and pressure in sports, and space–behavior relationships in environmental psychology can be investigated. Shared grammar preserves each field's data and criteria. Concrete achievements require disclosure of variables actually connected and outputs produced, with failures and boundaries retained for independent review.

7 Current verdict

The proposed WRRA applications in education, work, and health environments are INFERRED. Empirical confirmation of effects is OPEN.

Chapter in one sentence: Applied minimal computation evaluates function, burden, and error within a combined person–environment boundary.

8 Next questions and evidence

Next research question: Do external records and improved interfaces reduce both errors and burden?

Chapter references: R32, R47.

Part 10 WRRA Computational Models and Integrative Research

Chapter 40 Constructing State Variables and Transformation Rules

1 Scene and question

Computing an observed phenomenon requires deciding what to retain as state. Recording every experience enlarges data without automatically improving explanation; discarding necessary traces makes different responses to the same current input difficult to explain. Minimal computation seeks structures satisfying functional criteria between these problems.

Chapter question: How is the minimum state needed for the next update and observation determined?

2 Starting points in established research

Starting points for memory, motivation, and measurement connect to the literature in Chapters 15, 21, and 4. The state variables and functions selected here are the author's model design, not equations already estimated by external experiments.

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: candidate states, measurement units, and target outcomes. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Source of the work: the rules, calculations, or validation procedures are the author's WRRA application. Core originates in Choi (2026, W1); the earlier interdisciplinary work is Choi and Choi (2026, W2). Executed values and code are in Appendix D.

Transformation specification: fix updating rules, observation rules, and coefficient sources.

4.1 Specify the question and system boundary first

Adding states without specifying outputs easily complicates a model. Explaining one trial's choice and a day's activity require different inputs and intervals. Internal processing alone or external records and support must also be specified. Minimality is assessed under that goal and boundary.

4.2 States needed for the next update

Different outcomes under identical current input motivate candidate states carrying historical differences. The aim is a small summary preserving necessary distinctions. States producing the same present action are not automatically merged; later conditions may require distinguishing them. Sufficiency must be validated within the specified task and error range.

4.3 Implement updating and observation separately

Separating state change from indicators makes multiple data sources easier to connect. Deterministic updates may have noisy observations, and updates themselves may be stochastic. Combining different uncertainties into one noise term requires stating the simplification's scope. Neural-indicator noise is not equated with intrinsic behavioral-choice probability.

4.4 From existing research to a WRRA application

The examples separate current input, residue, resources, relational coefficients, possible actions, and observation rules. State-updating functions differ from functions reading states as reports or actions. A probability distribution may express several outcomes from one state, but outputting a distribution does not identify every uncertainty source.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned output: an executable model specification.

6 Falsification conditions and interpretive limits

Falsification condition: restrict interpretations of individual coefficients as real entities if different coefficients generate identical outcomes.

Coefficient sources are marked as measurements, calibration to verified values, or hypothetical settings. Reproducing calibration data is legitimate consistency performance, but does not establish performance elsewhere. Doubling residue scale and halving its output coefficient while retaining outputs can make the two coefficients separately unidentifiable; this is recorded.

6.1 Variable counts and identifiability

Several coefficient combinations may generate one output curve, preventing separate determination. More repetitions under unchanged conditions may not resolve this. Other inputs or observations may be needed. WRRA evaluates both minimum variable count and what the data can distinguish.

6.2 Implementation-check sequence

Match units, check permissible resource and probability ranges, and inspect extreme-condition outputs. Check reproduction under identical inputs before comparing data. These checks concern implementation consistency, not empirical validation. Labeling every coefficient as hypothetical, measured, or calibrated supports later review.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Computational models specify necessary states, updates, and observations and examine alternatives producing identical outputs.

8 Next questions and evidence

Next research question: If one state is removed while required function is preserved, do computation and identifiability hold?

Chapter references: W1, W2.

Chapter 41 Computational Example One: Maintaining a Memory Trace

1 Scene and question

Consider a name learned yesterday that does not immediately come to mind today. Before explaining all memory, we first compute a rule maintaining one trace without new input. The following is a hypothetical example addressing that question alone.

Chapter question: Which trace values does a fixed maintenance rule actually produce?

2 Starting points in established research

Chapter 15 distinguishes encoding, retention, and retrieval; Chapter 16 addresses alternative explanations of forgetting. This example preserves those distinctions while computing one maintenance rule.

3 Validation inputs

Execution inputs: hypothetical initial value 1, retention rate 0.85, and update counts. These are hypothetical settings, not human measurements or constants calibrated from the literature. Values and code are provided here and in Appendix D.

4 WRRA-specific transformation

Source of the work: the rules, calculations, or validation procedures are the author's WRRA application. Core originates in Choi (2026, W1); the earlier interdisciplinary work is Choi and Choi (2026, W2). Executed values and code are in Appendix D.

Transformation specification: multiply the preceding trace by the retention rate to compute the next trace.

The smallest example maintains one trace at a fixed proportion per update. Starting at 1, each next value is 0.85 times the previous value while there is no new input. These are hypothetical settings rather than measured or calibrated constants. An update has not been mapped to actual seconds or days.

4.1 Complete input and rule specification

Inputs are initial trace 1, retention ratio 0.85, and specified update counts. New input is zero at every step. The model therefore concerns dimensionless trace maintenance, not new learning or retrieval success. Real time and neural implementation remain unmapped.

4.2 What cannot be known without an observation rule

The model gives no grounds for equating a halved trace with halved recall probability. Cues and testing methods can yield different outcomes from the same trace. Recall rates require a separate trace-to-observation function. This preserves the distinction between numerical execution and reproduction of memory phenomena.

5 Outputs and execution record

Execution record: fixed inputs and Appendix D code computed trace values and sensitivity to retention rate. These are actually executed hypothetical-model outputs, distinct from empirical observations.

The results are 1.000000 after 0 updates, 0.850000 after 1, 0.614125 after 3, 0.443705 after 5, and 0.196874 after 10. The structure implements different remaining traces as update counts change despite equal initial input. Trace values are not recall probabilities; retrieval cues and observation rules must be added to compute those.

5.1 Changing the retention rate

Holding other conditions fixed and changing retention to 0.95 yields 0.598737 after 10 updates, versus 0.196874 at 0.85. This directly demonstrates rule sensitivity, not the discovery of two kinds of people. Empirical comparison requires calibration from human data and specified observation rules.

6 Falsification conditions and interpretive limits

Falsification condition: numerical errors are implementation failures; mismatch with actual data requires revision of the maintenance model.

6.1 Which data change the model?

If delay-specific data do not fit one rate, intervening experience and interference should be examined. Multiple trace components or condition-dependent rates may be introduced if needed. Before adding variables, specify which difference each term must reproduce. Compare simpler alternatives calibrated to the same data as well as more complex models.

7 Current verdict

The achievement is actual numerical execution of a maintenance rule, not reproduction of human forgetting curves. Actual data could calibrate the rate and fix retrieval rules for testing at other delays. If one rate cannot explain repeated measurements, alternatives such as interference or multiple maintenance modes are examined.

7.1 Assessing reproduction performance

Reproducing table values from identical inputs shows consistency between code and rules. Reproducing a known recall curve with a data-calibrated rate could add consistency and explanatory performance. A subsequently fixed model deriving unmeasured-delay outcomes would provide predictions. The present example executes hypothetical settings and sensitivity calculations.

COMPUTATIONAL PASS is restricted to execution and reproduction of the specified hypothetical rules. Empirical fit to human behavior and neural mechanisms is OPEN. Mathematically derived relationships receive EXACT only with assumptions and derivation scope stated; numerical success is not expanded into proof of the whole Core.

Chapter in one sentence: A hypothetical maintenance rule reproduces trace values; connecting them to recall requires additional rules.

8 Next questions and evidence

Next research question: Do retention and observation rules fitted to actual memory data work at other delays?

Chapter references: R1.

Chapter 42 Computational Example Two: Failure Experience and the Next Choice

1 Scene and question

Consider the hypothetical choice of trying again after five failures at the same task. Separating remaining effects of failure from currently available resources allows condition-specific choices to be computed for the same history.

Chapter question: Which choice probabilities does a fixed model connecting failure residue and current resources produce?

2 Starting points in established research

Chapter 13 distinguishes learning and behavioral consequences; Chapter 21 addresses motivation and costs; Chapter 24 addresses resource conditions. Coefficients here are hypothetical, not estimated in those studies.

3 Validation inputs

Execution inputs: a hypothetical failure sequence and resource values 1.0 and 0.3. These are hypothetical settings, not human measurements or constants calibrated from the literature. Values and code are provided here and in Appendix D.

4 WRRA-specific transformation

Source of the work: the rules, calculations, or validation procedures are the author's WRRA application. Core originates in Choi (2026, W1); the earlier interdisciplinary work is Choi and Choi (2026, W2). Executed values and code are in Appendix D.

Transformation specification: connect failure-residue updating and resource conditions to choice probabilities.

Initial residue is 0, each failure adds 0.4, and the previous residue retains a proportion of 0.8. Challenge probability applies a logistic function to a score consisting of baseline 1.2 plus 0.9 times resources minus 1.1 times residue. Every coefficient is hypothetical; action-availability conditions are assumed equal.

4.1 Failure inputs and action boundaries

Failure counts are externally supplied. Feedback through which chosen actions change later failure probabilities is not implemented. Resources 1.0 and 0.3 are hypothetical conditions without one-to-one correspondence to measured fatigue. Available actions are held constant while challenge-choice probability varies.

5 Outputs and execution record

Execution record: Appendix D's inputs and code computed choice probabilities by failure count and resource condition. Fixed points, cessation of new failures, and latent-scale changes were also checked. The values below are actual executions of the hypothetical model.

Before failure, challenge probabilities are 0.890903 for resources 1.0 and 0.813057 for 0.3. After five failures, residue is 1.344640 and probabilities are 0.650422 and 0.497724. Thus present resources alter outputs under identical experiential residue.

5.1 Conditional limit under continuing failure

With identical failures continuing, the update's fixed point is residue 2: 0.8 times 2 plus 0.4 equals 2. At that state, probabilities are 0.475021 for resources 1.0 and 0.325195 for 0.3. This is a mathematical property of the specified rule, not a universal final value of actual behavior.

5.2 Stopping new failure input

After five failures, five maintenance-only updates yield residue 0.440612 and challenge probability 0.834150 at resources 1.0. This implements decay under zero new input, not successful experience. It is not an actual treatment or recovery effect, and the time unit remains unspecified.

6 Falsification conditions and interpretive limits

Falsification condition: revise the signs and rules if effects persist in the opposite direction after actual conditions are matched.

6.1 Identical outputs and different internal scales

Doubling residue scale and failure increments while halving the residue coefficient in the output equation gives identical choice probabilities. Choice data alone therefore cannot independently establish residue's absolute magnitude and output sensitivity. Fixing a scale reference or adding observations is necessary. This identifiability issue should be checked before using a model.

7 Current verdict

Failure's negative effect follows from the chosen rule, not a finding that everyone reduces challenges after failure. A fixed model must be compared with actual choices to assess direction and magnitude. After calibration, unmeasured-condition outputs count as that fixed model's predictions. The present verdict concerns hypothetical computation.

COMPUTATIONAL PASS is restricted to execution and reproduction of the specified hypothetical rules. Empirical fit to human behavior and neural mechanisms is OPEN. Mathematically derived relationships receive EXACT only with assumptions and derivation scope stated; numerical success is not expanded into proof of the whole Core.

Chapter in one sentence: Probabilities connecting failure residue and resources were computed; fit to human behavior remains future work.

8 Next questions and evidence

Next research question: After calibration to actual trials, can choice distributions under new resource conditions be derived?

8.1 Moving to actual data

Data can match current difficulty and reward information while investigating previous-feedback effects. Coefficients are calibrated, fixed, and evaluated using person- or condition-level splits. Opposite effects require revising signs or update rules. Specifying a direction in hypothetical computation must not be reported as an empirical discovery.

Chapter references: R1.

Chapter 43 Connecting Neural Activity, Bodily Responses, Experience Reports, and Behavior

1 Scene and question

Chapters 41 and 42 computed behavior-related numbers but did not use neural activity as input or output. They therefore do not complete neural implementation. Integration requires putting neural and physiological data into actual computation and explaining measurement timing and error.

Chapter question: How can neural, bodily, and behavioral data on different timescales be connected?

2 Starting points in established research

NIMH's RDoC is a framework addressing multiple analytical levels, including molecules, cells, circuits, physiology, behavior, and self-report. (NIMH, n.d. R7)

2.1 A study combining state trajectories and intervention

Sauerbrei et al. (2020) examined motor cortex and thalamic inputs during reaching in mice, reporting changes in cortical activity and movement following perturbations and input manipulation. This original study illustrates adding input and intervention evidence to state trajectories in a particular motor system. It does not validate control of human psychology generally or validate WRRA. [R68]

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: temporally aligned neural, physiological, behavioral, and self-report data. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

3.1 Aligning timescales

Connecting second-scale choices and slower physiological indicators requires specifying when each arises and which interval it reflects. Ignoring delays may change the apparent direction of a connection. Delays selected after seeing outcomes are labeled exploratory. Fixed delays and observation rules must also be checked in other data.

4 WRRA-specific transformation

Source of the work: the rules, calculations, or validation procedures are the author's WRRA application. Core originates in Choi (2026, W1); the earlier interdisciplinary work is Choi and Choi (2026, W2). Executed values and code are in Appendix D.

Transformation specification: specify level-specific observation functions and connect them to an updating model.

4.1 Candidate common states and separate observations

Neural activity, physiology, behavior, and experience reports may read a common latent state or different states. Assuming they are copies of one value from the outset obscures disagreements. WRRA specifies which inputs and present states each measurement depends on. Collecting data together differs from connecting mechanisms.

4.2 From existing research to a WRRA application

Even using such data, WRRA must specify its own states, updating rules, and observation paths. Introducing an external framework does not complete implementation inside WRRA.

4.3 Models that read states and models that change them

Successful state-space classification is a result about observation. Control additionally requires specifying how an action or intervention changes UPDATE and how that change passes through the RENDERER into phenotype and measurements. Two models fitting nonintervention records equally well may produce different intervention outcomes if their intervention-dependent update rules differ. This is a logical identifiability example, not a newly executed simulation. [W1, §2.3; R6, R68; suggestion provenance I3]

The WRRA application separately records classification accuracy, next-state computation error, future prediction loss, and error in goal changes following intervention. A failed control model does not automatically invalidate a useful classifier. Conversely, classification accuracy alone does not close a control contract. This distinction already appears in Core 1.0 §2.3; the revision makes it operational for neural-data studies.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned outputs: correspondences across data and reproduction across conditions.

6 Falsification conditions and interpretive limits

Falsification condition: revise integrated scope and rules if connecting functions fail under other conditions.

6.1 Contributions of additional indicators

Reduced output error after adding a neural indicator counts within its scope, but additional information and evaluation leakage must be checked. More instruments do not always yield a better minimal model. Costs and required distinctions are compared, specifying goals for which extra measurements help.

6.2 Conditions for causal integration

Changing a circuit and examining state and behavior can add pathway evidence. Interventions may affect other components, however, requiring specific designs and comparisons. WRRA records an integrated mechanism only when intervention, internal updating, and observation connections are actually computed. This manuscript contains no neural intervention or analysis of human neural data.

6.3 Verdicts for separate contracts

The present explanatory mapping is INFERRED. The new design has not been fitted to neural data or validated by intervention; its empirical verdict is OPEN. Budgets, observation errors, temporal alignment, and intervention scope must be declared before assigning contract-specific results. Success in interpretation, computation, prediction, or control is not transferred automatically to another contract.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Integrating neural, bodily, and behavioral accounts requires actual internal computation including temporal alignment and observation error.

8 Next questions and evidence

Next research question: Do additional physiological indicators improve the distinguishability of a behavior-only model?

A possible study collects choices, heart rate, and neural signals during the same task. After alignment, it compares whether indicators read common or distinct states. Improved explanation after adding an indicator does not prove causality. Manipulation data and independent evaluation should be connected where possible.

Chapter references: R6, R7.

Chapter 44 Model Calibration, Reproduction, Prediction, and Falsification

1 Scene and question

Chapter question: Under which distinct contracts should calibration, reproduction, and prediction be judged?

2 Starting points in established research

The chapter starts from the distinctions among research findings and measurements developed earlier. External results and the WRRA application's specified scope are recorded separately.

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: calibration data, fixed rules, and evaluation data. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Source of the work: the rules, calculations, or validation procedures are the author's WRRA application. Core originates in Choi (2026, W1); the earlier interdisciplinary work is Choi and Choi (2026, W2). Executed values and code are in Appendix D.

Transformation specification: trace data roles and transformation sources.

Models may be revised using verified measurements. Recording which data determined what makes calibration legitimate theory construction. Reproducing known outcomes through WRRA's

own rules counts as consistency and explanatory performance; familiar numbers do not erase that achievement.

A fixed model deriving values for unmeasured conditions provides predictions. Another theory producing the same value does not negate WRRA's own calculation. Comparisons specify beforehand whether accuracy, cost, explanatory scope, or interpretability is assessed. Every study need not excel on every criterion.

4.1 Label data roles

Distinguish data seen during construction, used to determine coefficients, and evaluated after fixing rules. Multiple roles for the same data must be disclosed. Reproducing known findings can be an independent purpose. Reporting that achievement separately from new-condition performance avoids treating calibration itself as improper.

4.2 Decide what to compare

Scope, output error, data requirements, computational cost, and interpretability are different measures. WRRA need not always outperform every alternative to contribute meaningfully. Assessment asks which structural connection serves which purpose. Existing models receive the same inputs and error criteria, without credit for mechanisms they did not implement.

5 Outputs and execution record

The current output is an explanatory specification of phenomena, variables, and validation paths. Planned outputs: consistency, explanatory performance, and conditional predictions.

6 Falsification conditions and interpretive limits

Falsification condition: revise or reject the claim after repeated mismatches outside prespecified tolerances.

Falsification conditions must not move after results are seen. Tolerances reflect measurement error and goals and are specified beforehand; failures are recorded. Failure of a residue rule supports rejecting that rule. Rejecting every Core application simultaneously requires separate argument. Comparison models receive the same standard.

6.1 Distinguish structural and coefficient falsification

Wrong magnitudes, opposite condition effects, and failure to reproduce historical influences under the same present conditions call for different revisions. Coefficient calibration differs from changing the state distinctions themselves. Neither success nor failure is converted into a verdict on the whole Core beyond the claim's scope.

6.2 Preserve revision history

Revising hypotheses and code is normal research. Earlier inputs, rules, and failures are retained and changed terms identified. Repeated fitting to the same data is not described as a first evaluation after freezing. WRRA's explanatory and predictive ownership identifies its own computations while accurately crediting other theories.

6.3 Failures determine the next design

If residue is unhelpful, current measurements may already contain enough historical influence. Redundancy of one residue term differs from all residue being unnecessary. Distinguishing observation and model turns failures into new questions: this is the research role of fail-closed practice.

7 Current verdict

Current verdict: INFERRED for the WRRA correspondences and explanatory structure; OPEN for empirical fit of the transformation.

Chapter in one sentence: Calibration, reproduction of known values, and fixed-model predictions are assessed through their respective data roles and falsification conditions.

8 Next questions and evidence

Next research question: Which claims hold when calibration and evaluation roles are fixed?

Chapter references: W1, W2.

Chapter 45 Achievements and Open Research Questions

1 Scene and question

Chapter question: Which explanations and computations are complete, and what is the next unit of research?

2 Starting points in established research

The chapter starts from the distinctions among research findings and measurements developed earlier. External results and the WRRA application's specified scope are recorded separately.

3 Validation inputs

The concepts and observational evidence from existing research are identified in this chapter's references and Appendix E. Planned inputs for the WRRA application: the earlier conceptual

synthesis and actual execution records. Measurements and experiments without a separate execution record are research plans, not actual data collected for this manuscript.

4 WRRA-specific transformation

Source of the work: the rules, calculations, or validation procedures are the author's WRRA application. Core originates in Choi (2026, W1); the earlier interdisciplinary work is Choi and Choi (2026, W2). Executed values and code are in Appendix D.

Transformation specification: classify the scopes of explanation, computation, and empirical validation.

4.1 Accumulating connections across disciplines

Explaining a social boundary and implementing a neural circuit need different amounts of evidence. Achievements at each level are recognized while actual connections expand step by step. This is why common carriers and minimal computation are not predefined as one substance or score. A shared structure preserving necessary differences is central.

4.2 Following one scene through to completion

Recall Chapter 1's person who stops speaking before a presentation. Connect that scene to Chapter 42's hypothetical computation. This is not an analysis of actual participants. It is an explanatory application reading challenge choice as choosing to speak again when an opportunity is provided. Suitability for actual presentations remains untested.

Starting from existing explanations

Established distinctions separate earlier experience, current performance conditions, and actual behavior. Chapter 14 distinguishes acquisition from current performance; Chapter 15 distinguishes trace retention from access. Chapter 21 does not substitute goals and intentions for all behavior. These distinctions were needed before introducing WRRA. Its role is to show where each enters a scene's computation, not claim them as new discoveries.

The explanation “earlier failure prevented speaking” still leaves gaps. What counted as failure? How much influence remains until the next opportunity? How do current conditions enter? Were having no speaking opportunity and remaining silent despite an opportunity counted identically? Without these specifications, different explanations may label the same silence without showing computationally how they differ.

Turning explanation into an execution specification

Chapter 42 externally supplies failure counts. Their retained influence is residue, updated as 0.8 times the preceding residue plus 0.4 for a new failure, starting from 0. Current conditions use

dimensionless resources 1.0 and 0.3. Available actions and speaking opportunities are assumed equal.

The output is challenge-choice probability, not speech content or success. Chapter 42 adds 0.9 times resources to baseline 1.2, subtracts 1.1 times residue, and applies the logistic function. These are hypothetical coefficients showing separate entry points for residue and present conditions, not estimates from presentation research. The function is unchanged from Chapter 42 and Appendix D.

Reading the executed values

Before failure, probabilities are 0.890903 at resources 1.0 and 0.813057 at 0.3. Five failures produce residue 1.344640 and probabilities 0.650422 and 0.497724. Equal residue yields different outputs under different current resources. Within this model, past influence alone cannot determine current choice probability.

Stopping new failures and applying five maintenance updates yields residue 0.440612 and probability 0.834150 at resources 1.0. Neither successful experience nor counseling was included. This is the specified maintenance rule under no additional failure, not a recovery rate for presentation anxiety or a treatment effect of support.

Questions revealed by the connecting specification

First, silence counts alone cannot determine whether differences reflect earlier experience or present conditions. Requiring separate inputs reveals additional records needed. Second, resources change outputs but remain unmapped to sleep duration, preparation time, or support quantity. Real application must identify the chosen condition and its numerical mapping.

Third, without guaranteed opportunities the current boundary assumption fails: opportunity availability must first be recorded, then choice conditional on an opportunity analyzed. Fourth, choice probabilities do not establish absolute residue magnitude. Doubling residue scale while halving its output coefficient preserves probabilities in Chapter 42. Identical outputs therefore do not guarantee a unique account of internal state.

The explanatory contribution is a single specification distinguishing observed actions, past influence, present conditions, and choice boundaries, while tracing which assumptions change which outputs. It is a constructive achievement preserving established distinctions and revealing gaps in execution and interpretation. It is neither a new psychological law nor evidence of better fit than existing models.

What remains unconnected

The computation contains no neural activity or bodily responses. It does not specify residue's circuit or physiological implementation, or how reports relate to model values. Action-dependent changes in subsequent failures are also unimplemented. These lead to Chapter 43's observation paths and temporal alignment. What is complete is choice computation under declared hypothetical conditions; empirical behavioral–neural connections remain future research.

5 Outputs and execution record

Execution record: phenomena and questions across 45 chapters were mapped to a common execution grammar. The hypothetical models in Chapters 41 and 42 and supplementary Appendix D calculations were executed. Combined output: the manuscript and two hypothetical computational examples.

The manuscript maps general psychology's main questions onto WRRA 1.0 execution structure. Present and residue, input and relationships, body and environment, resources and rendering connect disciplinary questions. This is the present conceptual achievement.

5.1 The book's organization and connections

Across Parts 1–10, chapters connect concepts, examples, research distinctions, and WRRA correspondences. Part 1 introduces questions, variables, causality, measurement, and ethics. Later parts read sensation and memory through social relationships and clinical research using the same grammar while preserving topic-specific differences.

6 Falsification conditions and interpretive limits

Falsification condition: strong achievements are not claimed from nonexistent data or unperformed computations.

7 Current verdict

Two hypothetical examples were actually computed: trace maintenance and choice probabilities involving failure residue and resources. This manuscript performed no human-participant analysis, neural-data connection, or clinical-effect validation. Chapter explanations therefore do not become complete computational models of actual brains or validated treatment theories.

7.1 Scope of executed calculations

Basic outputs were supplemented by changed retention, stopped failure inputs, fixed points, and scale-transformation checks. Changes, standard deviation, and standard error for learning-example scores were also computed. These are not actual human responses. They establish structural execution and numerical reproduction, not natural neural mechanisms or clinical effects.

Conceptual connections are INFERRED; executions in Chapters 41 and 42 are COMPUTATIONAL PASS within their stated scope; validation using human and neural data is OPEN. No separate experiment or new calculation was added in this chapter.

Chapter in one sentence: Present achievements are conceptual connections and hypothetical computations; the next step is closing the data and rules for one task.

8 Next questions and evidence

Next research question: Can one task's data, execution rules, and verdict ledger form one reproducible package?

The next study selects a task and fixes variables and measurement. Behavioral data can calibrate the model, neural and physiological measurements can be connected, and scope can expand. Development of WRRA neuropsychology accumulates explanation and computation under the same questions. Reproduction of known outcomes and derivation under new conditions are recorded as distinct achievements.

8.1 A small unit for the next study

The next step fixes definitions of input, residue, current resources, and output in one behavioral task. Suitable public data may be reanalyzed; missing conditions may motivate a new design. Calibration to measurements is legitimate. After examining outcomes, rules can be fixed and assessed under other conditions.

8.2 Leaving a research program

Readers should be able to trace inputs, executed transformations, outputs, and revision conditions. Consistency with known phenomena, structural explanation, fixed-model prediction, and actual control are recorded separately. Through such records, WRRA neuropsychology can develop from a glossary into a reproducible research program.

Chapter references: W1, W2.

8.3 Three connections to test first

First, test whether specified UPDATE and RENDERER rules reconstruct condition-specific judgment and action distributions from identical sensory stimuli and distinct present states. Second, test the timing among state, action, and subsequent input when actions alter sensory conditions. Third, assess state-estimation performance and explanations of intervention effects under separate contracts. Costs and provenance are common recording requirements across these tests.

Appendix C4 credits the suggestions. Sources: R65–R70 and R41. These designs remain OPEN.

Appendix A Standard Explanation of WRRA 1.0

This explanation preserves the definitions in reference W1 and draws on W2's introduction. Neuropsychological statements are this book's applications, not changes to Core 1.0 itself.

A1 WRRA's name and starting point

WRRA means Wonsik Reality-Renderer Architecture. This account adapts the frozen Core 1.0 and existing book explanations for neuropsychology readers. It begins by distinguishing information, relationships and rules, present state, execution, and observation in complex phenomena. Particular cosmological equations and channel counts are not compulsory in every application.

A2 Minimal computation

Minimal computation uses the states and processing needed to meet declared function, expression, stability, or survival criteria. It does not mean unconditionally computing less. Removing required distinctions until function fails is deficiency; necessary margins and redundancy may belong to minimum cost. Costs and preserved criteria must be specified first.

A3 Common carrier

A common carrier is a reusable transmission or updating structure shared by distinguishable channels. Shared infrastructure does not make information owners, paths, or purposes identical. Networks reused by several functions are candidates in neuropsychology. Until sharing is demonstrated, these remain correspondence hypotheses, not predetermined neurotransmitters or brain regions.

A4 Phenotype and observation

Internal information is not itself an observation. Present state becomes phenotype through execution and rendering rules under boundary conditions. An observable is the value obtained by reading that phenotype through a particular device and procedure. Distinctions among neural activity, behavior, and experience reports apply this principle. Even if phenotype and observable are implemented together, the connecting measurement procedure remains necessary.

A5 The full execution structure

Core 1.0's common structure comprises SOURCE, RELATION and LAW, STATE and RESIDUE, BOUNDARY, COMMON CARRIER, UPDATE, RENDERER, PHENOTYPE, and OBSERVABLE, RECORD, and LEDGER. These are not necessarily neural-processing stages traversed once in a line. Applications specify feedback, parallel processing, and repeated updates. What corresponds to each element is central.

A6 Fixed present and residue

For a past influence to act now, its effects must remain in present state, residue, or accessible records. A model may read past records if their present location and access paths are specified. Residue includes learned skills, familiarity, trust, and adaptation as well as injury or negative experiences. Persistent benefits and burdens are measured separately.

A7 Relationships and boundaries

Relationships specify how elements influence others under rules. Boundaries specify system interiors and exteriors and permitted or blocked flows. Physical bodily boundaries, task action restrictions, and social authority have different implementation levels. Constraints are not all collapsed into one boundary variable. System scope is fixed to avoid counting external support as internal capacity.

A8 Ownership and record ledgers

Equal numbers represent different variables when their owners differ. Ownership identifies whose information, resources, memory, and outputs are represented. Personal reports, another's estimates, neural measurements, and behavior records do not substitute for each other. Ledgers record values, units, times, sources, transformations, and outcomes. Accounting for physical conserved quantities differs from documenting measurement responsibility.

A9 Interpretation, computation, prediction, and control

Interpretation reconstructs or explains targets using present or recorded information. Computation derives outcomes from specified inputs and rules. After a model is set using verified values, its derivation for unmeasured targets or conditions counts as that model's prediction. Control selects actions or interventions changing later states and function. Reproduction of known

results counts as consistency and explanation; absence of new numbers does not remove its meaning.

A10 Fail-closed claims and separation of applications

When inputs, rules, errors, or validation are insufficient, only completed work receives a verdict and the remainder stays open. Failure of an application differs from abandoning the entire Core. Revising neuropsychological coefficients or state definitions changes a domain application, not automatically Core 1.0. Independent prediction is not required for every explanation. Each achievement is judged after presenting validation inputs, WRRA-specific transformation, outputs, and falsification conditions.

A11 Reading this book alongside the Core 1.0 paper

Foundational source W1: Wonsik Choi, *WRRA Core 1.0: A Domain-Agnostic Execution Architecture*, version 1.0.0, frozen 7 September 2026. Archival DOI · Full text · Official source repository.

Core 1.0 supplies the domain-independent execution grammar. This book proposes a neuropsychology application, or Domain Profile, using that grammar. The table is the book's explanatory crosswalk; it is not an addition to the original paper's frozen application registry. In particular, the examples in Chapters 41 and 42 demonstrate specified hypothetical rules, not empirical validation of the Core's constitutional invariants.

Core 1.0 source section	Connection in this book	Concrete application and present limit
§1.1 Minimum Computation	Chapters 2, 11, 14, and 17	Specify required accuracy, stability, and task function before comparing memory, time, or search cost. Comparative minimality is not established by the two numerical examples.
§1.2 Common Carrier	Chapters 5, 6, and 11	Identify a shared transmission/update structure while preserving channel identity. A shared neural implementation remains a hypothesis.
§1.3 Phenotype; §2 required elements	Chapters 1, 2, 22, and 43	Distinguish internal state, implemented action, experience report, and measured signal; specify rendering and observation paths.
§2.1 Fixed-Present Principle	Chapters 15, 16, 41, and 42	Earlier influences enter through presently instantiated traces, residue, or accessible records. The two hypothetical examples execute maintenance and updating rules.

Core 1.0 source section	Connection in this book	Concrete application and present limit
§2.2 Ownership Principle	Chapters 2, 4, 16, and 20	Identify whose experience, record, state, and output each quantity represents; avoid double counting residue within present state. Technical ownership does not replace privacy rights.
§2.3 Fail-Closed Boundary Principle	Chapters 3, 4, 44, and 45; Appendix F	Keep interpretation, computation, empirical fit, and unresolved connections separate. COMPUTATIONAL PASS applies only to the computations actually executed.
§3 Core and Domain Profiles; §7 version control	Chapter 2; Appendices A–C	Field variables, coefficients, observation functions, and datasets belong to the application. Revising them does not automatically revise Core 1.0.
§6 and §6.1 minimum conformance	Chapter 40; Appendices B–D	Check sources, states, boundaries, carriers, updates, rendering, observation, ownership, and records explicitly. Missing implementations are disclosed rather than marked complete.

Readers may follow the route Core §§1–3 → this book's Chapter 2 and Appendix B → Chapters 41–43 and Appendix D → Chapter 45 and Appendix F. This traces definitions into an application specification, executed outputs, and the verdict on what has actually been connected. The DOI identifies an archival source; it is not a peer-review or empirical-validation certificate.

Prediction terminology: this book records an unmeasured-condition output of a fixed model as that model's conditional prediction. Core §2.3 additionally specifies prediction performance through reduced loss or uncertainty for a future target. Those performance targets, comparison criteria, and loss or uncertainty measures must be specified separately when tested. This distinction does not make independent prediction a prerequisite for explanatory value or originality.

Appendix B Terminology Correspondences

These are proposed research correspondences, not universally one-to-one identities between WRRA and disciplinary terminology.

Core element	Proposed neuropsychological correspondence	Distinction to examine
SOURCE	Sensory and bodily inputs; task information	External input versus internal state
RELATION LAW	Connections and permissible transition rules	Statistical versus causal relationships
STATE	Current neural, bodily, and task states	Complete state versus measured components
RESIDUE	Experiential influences active now	Residue maintenance versus retrieval access
BOUNDARY	Bodily, task, and social constraints	Physical versus rule-based boundaries
COMMON CARRIER	Reused transmission and updating structures	Shared infrastructure versus signal identity
UPDATE	Learning, adaptation, and state updating	Current response versus lasting change
RENDERER	Rules from internal state to expression	State updating versus output rules
PHENOTYPE	Implemented actions and responses	Possible versus actual outcomes
OBSERVABLE	Self-reports, behavior, and physiological measurements	Measurement versus experience itself
RECORD LEDGER	Experience records and measurement/computation ledgers	Personal memory versus external records

Appendix C Measurement and Variable Specifications

Models record ownership, units, timing, measurement methods, errors, and possible duplication when defining variables. The following are examples of specifications to prepare, not actual research data.

Item	What to define
Present state	State immediately before response, observation time, and scope
Experiential residue	Which experience's influence; maintenance and activation indicators
Available resources	Concrete dimensions such as time, sleep, and support; measurement methods
Relationships and boundaries	Influence direction, possible actions, and external conditions
Updating coefficients	Hypothetical, measured, or calibrated values
Output rules	Functions connecting state to reports and choices
Missingness and exclusions	Timing, prespecified criteria, and sensitivity analyses
Falsification thresholds	Evaluation measures, tolerances, and repeated-failure criteria

Initial work may use neutral tasks without strong emotional stimulation. Person-level repeated trials record experience order and resources separately while controlling current difficulty. Sample sizes and tolerances are fixed beforehand based on expected effects, measurement variation, and purpose. No actual recruitment or data collection occurred for this manuscript.

Identifiability checks examine whether different coefficient combinations yield the same outputs. Unobserved residue requires fixed scale and sign references. Adding a neural indicator does not by itself establish its identity with a latent variable; this requires separate validation.

C2 Core 1.0 application checklist

This is a Domain Profile specification to complete for each task, not an established neural-mechanism model. The reference is W1 §§0–3; Chapter 2 §4.8 summarizes the three invariants. Concrete implementation and empirical verdicts remain separate.

SOURCE — Distinguish stimuli, prior instructions, external information, and interventions; record their origins and presentation times.

RELATION / LAW — Specify permitted influences and delays linking input, state, action, and subsequent input.

STATE — Define the memory, attention, bodily, and other states assumed to exist immediately before response, their owners, and estimation methods.

RESIDUE / CARRY — Specify currently retained effects of earlier experience and their maintenance or decay, without counting the same effect twice in STATE.

BOUNDARY — Define which person, task, device, and environmental processes are included, available actions, and open information pathways.

COMMON CARRIER — Declare which channels reuse which transmission/update scaffold while preserving channel identity, ownership, and destination. A shared model data structure and a biological carrier are separate hypotheses.

UPDATE / TAKEOVER / RESET — Specify next-state rules, intervention effects, resets, and carryover between trials.

RENDERER — Define how current state and boundary produce judgment reports, choices, or actions.

PHENOTYPE — Define instantiated task-level outputs, distinct from latent state and merely possible outputs.

OBSERVABLE — Define sensor values, behavioral logs, reported scales, and measurement errors. Experience and reports of experience are not identical.

RECORD / LEDGER — Preserve timing, provenance, costs, versions, and ownership; define which stored records are accessible in the present.

After fixing rules, declare target errors and budgets. Evaluate interpretation, computation, prediction, and control separately; unresolved data or causality leaves the relevant contract OPEN. Explanation, calibration, and reproduction of known results count within their declared scope. Absence of a novel prediction alone does not negate architectural or explanatory contributions. [W1, §1.1, §2.3, §3]

C3 An example of testing an exploratory connection

No established relation between zeta zeros and calcium signals is asserted here. The connection is an exploratory topic from a relayed informal suggestion, recorded in I4. R70 reviews the spatial and temporal diversity of calcium signaling; it supplies measurement context, not evidence for a zeta correspondence.

A test must first specify which quantities of which zeros are compared with which calcium-signal features. Mapping dimensionless quantities to Hz requires an explicit timescale and

transformation. Control sample selection, indicator kinetics, sampling, filtering, noise, null models, arbitrary matching, and multiple comparisons. Similarity obtained using scales chosen after inspecting the data remains exploratory. A claim about performance on new data requires fixing the transformation and evaluating that claim on separate data.

Statistical similarity, a generating mechanism, and intervention effects are separate contracts. Proposing calcium as a common carrier additionally requires specifying what is shared or transported across which channels and with what ownership. The zeta–calcium correspondence and its neural-mechanism link remain OPEN. No new numerical test or biological validation is reported here. [W1, §1.2, §2.3; R70]

C4 Suggestion provenance and contribution record

An identifier accompanies each incorporated external concept, finding, or suggestion; this record states its contribution and verification status. Academic claims cite authors, years, and R identifiers; W1 identifies the WRRA reference. The five proposals relayed by the user on 22 October 2026 motivated conversion of the following informal discussions into testable application specifications. Receiving a suggestion does not establish its originator’s verification or endorsement.

I1 — The supplied review describes an account of hierarchical predictive processing by the handle luria. Contribution: the prompt to strengthen comparison theories and variable mappings in Chapter 10. The original URL and posting date were not verified. The academic account instead cites the independently checked Rao and Ballard (1999), R66. The handle is not identified with neuropsychologist A. R. Luria.

I2 — The supplied review attributes an active-inference discussion to herasynthesist. Contribution: the suggestion to specify the perception–action loop and differences from WRRA. The original URL, date, and wording remain unverified. The academic description relies on Friston et al. (2017), R67.

I3 — The supplied review attributes the distinction between finding a state space and controlling it to vina. Contribution: prompting the operational contract distinctions in Chapter 43. Candidate link retained in earlier working notes: https://www.moltbook.com/post/2db36673-5c40-4186-97b6-0a84be19570b . The original content, authorship, and date were not independently verified in this revision. This is not treated as a direct quotation; the account is reconstructed using Core §2.3 and R68.

I4 — The supplied review raises an exploratory zeta–calcium topic. Earlier working notes associated it with luria; this attribution remains unverified. Candidate link: https://www.moltbook.com/post/10119dd9-4063-4f24-b625-7144084fc446 . Original content,

authorship, and date were not independently verified. Its contribution is the motivation for the test specification in C3, not evidence for an established scientific relation.

Author’s application design — This revision maps the five proposals onto Core 1.0’s eleven elements, specifying ownership, present residue, the common carrier, RENDERER, outputs, and falsification conditions. Predictive coding, active inference, source monitoring, and relayed ideas are not represented as originating in WRRA. Verification of the original posts would allow I1–I4 to be completed with authors, titles, dates, stable links, and precise contributions.

Appendix D Numerical Results and Reproduction Code

Numerical display rule: ratios, probabilities, and calculated outputs are rounded to six decimal places. Counts, hypothetical input constants, and integer statistics in explanatory prose retain their original notation. Internal calculations are not rounded, and displayed values are not used for subsequent calculations.

These hypothetical models were executed using only Python's standard library, without external data or random numbers. Values are dimensionless model outputs, not human observations.

D1 Memory-trace retention

Update count	Trace value
0	1.000000
1	0.850000
3	0.614125
5	0.443705
10	0.196874

D2 Failure residue and challenge probability

Failures	Residue	Resources 1.0	Resources 0.3
0	0.000000	0.890903	0.813057
1	0.400000	0.840238	0.736916
2	0.720000	0.787178	0.663292
3	0.976000	0.736217	0.597822
4	1.180800	0.690214	0.542676
5	1.344640	0.650422	0.497724

Retention and response coefficients are hypothetical. Resource effects were set positive and failure effects negative, so output direction itself is not an empirical discovery. Columns compare two conditions with identical residue, not one person's observed before–after recovery.

D3 Reproduction code

```
import math


# Case 1: hypothetical retention, not measured recall probability
for n in (0, 1, 3, 5, 10):
    trace = 0.85 ** n
    print(n, f"{trace:.6f}")


# Case 2: hypothetical failure residue and challenge probability
def probability(residue, resource):
    score = 1.2 + 0.9 * resource - 1.1 * residue
    return 1.0 / (1.0 + math.exp(-score))


residue = 0.0
for failures in range(6):
    high = probability(residue, 1.0)
    low = probability(residue, 0.3)
    print(failures, f"{residue:.6f}",
          f"{high:.6f}", f"{low:.6f}")
    residue = 0.8 * residue + 0.4


# Additional calculations in the expanded manuscript
import statistics
changes = [4, 2, 6, 2, 6]
print("mean", f"{statistics.mean(changes):.6f}")
print("sample_sd", f"{statistics.stdev(changes):.6f}")
print("mean_se", f"{statistics.stdev(changes) / math.sqrt(len(changes)):.6f}")
print("retention_0.95_at_10", f"{0.95 ** 10:.6f}")
print("fixed_point", f"{0.4 / (1 - 0.8):.6f}")
print("fixed_high", f"{probability(2.0, 1.0):.6f}")
print("fixed_low", f"{probability(2.0, 0.3):.6f}")
r = 0.0
for _ in range(5):
    r = 0.8 * r + 0.4
for _ in range(5):
    r = 0.8 * r
print("after_no_new_failure_5", f"{r:.6f}", f"{probability(r, 1.0):.6f}")


# A rescaled hidden trace can give identical observable probabilities
r1 = r2 = 0.0
max_difference = 0.0
for _ in range(6):
    p1 = probability(r1, 1.0)
    score2 = 1.2 + 0.9 - 0.55 * r2
    p2 = 1.0 / (1.0 + math.exp(-score2))
    max_difference = max(max_difference, abs(p1 - p2))
    r1 = 0.8 * r1 + 0.4
    r2 = 0.8 * r2 + 0.8
print("scale_change_max_difference", f"{max_difference:.6f}")
```

The verdict concerns execution of hypothetical rules and numerical reproduction. It is not derivation of natural constants from primitive assumptions or validation of clinical effects. Reexecuting the code with the same inputs must reproduce the rounded tables.

D4 Changes for a statistics exercise

Hypothetical participant	Change
1	4
2	2
3	6

Hypothetical participant	Change
4	2
5	6

The five independent changes have mean 4 points, sample standard deviation 2, and standard error 0.894427. These are hypothetical inputs illustrating distinct statistics, not actual participants' data.

D5 Sensitivity and identifiability calculations

Condition	Output
Trace after 10 updates at retention 0.95	0.598737
Fixed point of failure residue	2.000000
Challenge probability at fixed point, resources 1.0	0.475021
Challenge probability at fixed point, resources 0.3	0.325195
Residue after 5 failures then 5 updates with no new failure	0.440612
Maximum probability difference after doubling residue scale and halving its coefficient	0.000000

The retention comparison shows hypothetical-rule sensitivity, the fixed point is a property of the update, and scale equivalence exposes latent identifiability. The extended reproduction code verifies all values.

Appendix E References and Thematic Reading

Literature inclusion cutoff: 22 October 2026. Individual checking dates retain the actual checking records. Sources checked on 18 October are included by this cutoff. Inclusion cutoff and source-checking dates are different information.

W identifies earlier WRRA works; R identifies external references. External sources support validation, reinforcement, and disciplinary definitions. Introducing a theory or mechanism does not establish that it was computed within WRRA. This is a list of principal sources, not an exhaustive disciplinary bibliography.

[W1](https://doi.org/10.5281/zenodo.22650956) Choi, Wonsik. *WRRA Core 1.0: A Domain-Agnostic Execution Architecture*. Version 1.0.0, frozen 7 September 2026, including application lineage. https://doi.org/10.5281/zenodo.22650956. Official full text: https://github.com/Wonsik-Choi-janefather/wrra-core/blob/main/WRRA_Core_1.0_EN.md. Foundational source for Core definitions and Appendix A. Appendix A11 maps the original sections to this book.

[W2](https://github.com/Wonsik-Choi-janefather/how-life-runs-v2.7.3) Choi, Wonsik, and Choi, Jeongin. *How Life Runs* [생명은 어떻게 실행되는가]. Publication-final version v2.7.3, revised 21 September 2026. See Chapter 1 for WRRA explanation and execution structure. https://github.com/Wonsik-Choi-janefather/how-life-runs-v2.7.3

R1 Spielman, R. M., Jenkins, W. J., & Lovett, M. D. Psychology 2e. OpenStax, 2020. https://openstax.org/details/books/psychology-2e — Introductory support for basic concepts. Section numbers in the text connect to the section URLs below; this differs from direct evidence for particular experiments.

[R2](https://www.apa.org/topics/neuropsychology) American Psychological Association. Neuropsychology. https://www.apa.org/topics/neuropsychology — Scope of established neuropsychology.

[R3](https://doi.org/10.1126/science.aac4716) Open Science Collaboration. Estimating the reproducibility of psychological science. Science, 349, aac4716, 2015. https://doi.org/10.1126/science.aac4716

[R4](https://doi.org/10.1113/jphysiol.1973.sp010273) Bliss, T. V. P., & Lømo, T. Long-lasting potentiation of synaptic transmission in the dentate area of the anaesthetized rabbit following stimulation of the perforant path. Journal of Physiology, 232, 331–356, 1973. https://doi.org/10.1113/jphysiol.1973.sp010273

[R5](https://www.ninds.nih.gov/health-information/public-education/brain-basics) National Institute of Neurological Disorders and Stroke. Brain Basics. https://www.ninds.nih.gov/health-information/public-education/brain-basics

[R6](https://doi.org/10.1016/j.tics.2005.12.004) Poldrack, R. A. Can cognitive processes be inferred from neuroimaging data? Trends in Cognitive Sciences, 10, 59–63, 2006. https://doi.org/10.1016/j.tics.2005.12.004

[R7](https://www.nimh.nih.gov/research/research-funded-by-nimh/rdoc/units) National Institute of Mental Health. RDoC Units of Analysis. https://www.nimh.nih.gov/research/research-funded-by-nimh/rdoc/units

[R8](https://www.ninds.nih.gov/sites/default/files/2025-05/understanding-sleep.pdf) National Institute of Neurological Disorders and Stroke. Brain Basics Understanding Sleep. https://www.ninds.nih.gov/sites/default/files/2025-05/understanding-sleep.pdf

[R9](https://doi.org/10.1037/h0045925) Bandura, A., Ross, D., & Ross, S. A. Transmission of aggression through imitation of aggressive models. Journal of Abnormal and Social Psychology, 63, 575–582, 1961. https://doi.org/10.1037/h0045925

[R10](https://doi.org/10.1136/jnnp.20.1.11) Scoville, W. B., & Milner, B. Loss of recent memory after bilateral hippocampal lesions. Journal of Neurology, Neurosurgery & Psychiatry, 20, 11–21, 1957. https://doi.org/10.1136/jnnp.20.1.11

[R11](https://doi.org/10.1126/science.185.4157.1124) Tversky, A., & Kahneman, D. Judgment under Uncertainty Heuristics and Biases. Science, 185, 1124–1131, 1974. https://doi.org/10.1126/science.185.4157.1124

[R12](https://doi.org/10.1056/NEJM199801153380307) McEwen, B. S. Protective and damaging effects of stress mediators. New England Journal of Medicine, 338, 171–179, 1998. https://doi.org/10.1056/NEJM199801153380307

[R13](https://www.nimh.nih.gov/health/topics) National Institute of Mental Health. Health Topics. https://www.nimh.nih.gov/health/topics

[R14](https://www.nimh.nih.gov/health/topics/anxiety-disorders) National Institute of Mental Health. Anxiety Disorders. https://www.nimh.nih.gov/health/topics/anxiety-disorders

[R15](https://www.nimh.nih.gov/health/topics/obsessive-compulsive-disorder-ocd) National Institute of Mental Health. Obsessive-Compulsive Disorder. https://www.nimh.nih.gov/health/topics/obsessive-compulsive-disorder-ocd

[R16](https://www.nimh.nih.gov/health/topics/post-traumatic-stress-disorder-ptsd) National Institute of Mental Health. Traumatic Events and Post-Traumatic Stress Disorder. https://www.nimh.nih.gov/health/topics/post-traumatic-stress-disorder-ptsd

[R17](https://www.nimh.nih.gov/health/topics/depression) National Institute of Mental Health. Depression. https://www.nimh.nih.gov/health/topics/depression

[R18](https://www.nimh.nih.gov/health/topics/bipolar-disorder) National Institute of Mental Health. Bipolar Disorder. https://www.nimh.nih.gov/health/topics/bipolar-disorder

[R19](https://www.nimh.nih.gov/health/topics/schizophrenia) National Institute of Mental Health. Schizophrenia. https://www.nimh.nih.gov/health/topics/schizophrenia

[R20](https://www.nimh.nih.gov/health/topics/psychotherapies) National Institute of Mental Health. Psychotherapies. https://www.nimh.nih.gov/health/topics/psychotherapies

[R21](https://www.apa.org/ed/precollege/topss/science) American Psychological Association. High School Psychology is Science. https://www.apa.org/ed/precollege/topss/science

[R22](https://www.cos.io/initiatives/prereg) Center for Open Science. Preregistration. https://www.cos.io/initiatives/prereg

[R23](https://doi.org/10.1080/00031305.2016.1154108) Wasserstein, R. L., & Lazar, N. A. The ASA Statement on p Values Context Process and Purpose. The American Statistician, 70, 129–133, 2016. https://doi.org/10.1080/00031305.2016.1154108

[R24](https://www.hhs.gov/ohrp/regulations-and-policy/belmont-report/read-the-belmont-report/index.html) National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research. The Belmont Report, 1979. https://www.hhs.gov/ohrp/regulations-and-policy/belmont-report/read-the-belmont-report/index.html

[R25](https://www.apa.org/ethics/code) American Psychological Association. Ethical Principles of Psychologists and Code of Conduct. https://www.apa.org/ethics/code — Checked 18 October 2026.

[R26](https://doi.org/10.1126/science.1067020) Kandel, E. R. The molecular biology of memory storage a dialogue between genes and synapses. Science, 294, 1030–1038, 2001. https://doi.org/10.1126/science.1067020

[R27](https://www.ninds.nih.gov/health-information/patient-caregiver-education/brain-basics-genes-work-brain) National Institute of Neurological Disorders and Stroke. Brain Basics Genes At Work In The Brain. https://www.ninds.nih.gov/health-information/patient-caregiver-education/brain-basics-genes-work-brain

[R28](https://www.nia.nih.gov/health/memory-loss-and-forgetfulness/memory-forgetfulness-and-aging-whats-normal-and-whats-not) National Institute on Aging. Memory Problems Forgetfulness and Aging. https://www.nia.nih.gov/health/memory-loss-and-forgetfulness/memory-forgetfulness-and-aging-whats-normal-and-whats-not

[R29](https://nida.nih.gov/publications/drugs-brains-behavior-science-addiction) National Institute on Drug Abuse. Drugs Brains and Behavior The Science of Addiction. https://nida.nih.gov/publications/drugs-brains-behavior-science-addiction

[R30](https://www.nimh.nih.gov/health/publications/attention-deficit-hyperactivity-disorder-what-you-need-to-know) National Institute of Mental Health. Attention Deficit Hyperactivity Disorder What You Need to Know. https://www.nimh.nih.gov/health/publications/attention-deficit-hyperactivity-disorder-what-you-need-to-know

[R31](https://www.nimh.nih.gov/health/publications/autism-spectrum-disorder) National Institute of Mental Health. Autism Spectrum Disorder. https://www.nimh.nih.gov/health/publications/autism-spectrum-disorder

[R32](https://doi.org/10.1111/j.1467-9280.2006.01693.x) Roediger, H. L., III, & Karpicke, J. D. Test enhanced learning taking memory tests improves long term retention. Psychological Science, 17, 249–255, 2006. https://doi.org/10.1111/j.1467-9280.2006.01693.x

[R33](https://www.nimh.nih.gov/health/publications/eating-disorders) National Institute of Mental Health. Eating Disorders What You Need to Know. https://www.nimh.nih.gov/health/publications/eating-disorders

[R34](https://www.nimh.nih.gov/health/topics/borderline-personality-disorder) National Institute of Mental Health. Borderline Personality Disorder. https://www.nimh.nih.gov/health/topics/borderline-personality-disorder

R35 Logothetis, N. K., Pauls, J., Augath, M., Trinath, T., & Oeltermann, A. (2001). Neurophysiological investigation of the basis of the fMRI signal. Nature, 412, 150–157. https://doi.org/10.1038/35084005 — Original study comparing monkey visual-cortex neurophysiology and BOLD; not direct validation for every human task.

R36 Garfinkel, S. N., Seth, A. K., Barrett, A. B., Suzuki, K., & Critchley, H. D. (2015). Knowing your own heart: Distinguishing interoceptive accuracy from interoceptive awareness. Biological Psychology, 104, 65–74. https://doi.org/10.1016/j.biopsycho.2014.11.004 — Original study distinguishing interoceptive performance accuracy, subjective sensibility, and metacognitive awareness.

R37 Raja, S. N., et al. (2020). The revised IASP definition of pain: Concepts, challenges, and compromises. Pain, 161(9), 1976–1982. https://doi.org/10.1097/j.pain.0000000000001939 — Definition and commentary distinguishing pain experience and nociception, not an experiment on treatment effects.

R38 Pashler, H. (1994). Dual-task interference in simple tasks: Data and theory. Psychological Bulletin, 116(2), 220–244. https://doi.org/10.1037/0033-2909.116.2.220 — Review of dual-task interference and alternative processing-bottleneck explanations.

R39 Heitz, R. P. (2014). The speed-accuracy tradeoff: History, physiology, methodology, and behavior. Frontiers in Neuroscience, 8, 150. https://doi.org/10.3389/fnins.2014.00150 — Review of speed–accuracy trade-offs and measurement methods.

R40 Baddeley, A. (2003). Working memory: Looking back and looking forward. Nature Reviews Neuroscience, 4, 829–839. https://doi.org/10.1038/nrn1201 — Theory and research review of working-memory maintenance, processing, and model organization.

R41 Johnson, M. K., Hashtroudi, S., & Lindsay, D. S. (1993). Source monitoring. Psychological Bulletin, 114(1), 3–28. https://doi.org/10.1037/0033-2909.114.1.3 — Theory and research review of content recognition, source judgments, and errors.

R42 Gross, J. J. (1998). The emerging field of emotion regulation: An integrative review. Review of General Psychology, 2(3), 271–299. https://doi.org/10.1037/1089-2680.2.3.271 — Integrative review distinguishing situation, attention, cognitive change, and response regulation.

R43 Hagger, M. S., et al. (2016). A multilab preregistered replication of the ego-depletion effect. Perspectives on Psychological Science, 11(4), 546–573. https://doi.org/10.1177/1745691616652873 — Multilaboratory preregistered replication under the specified tasks and procedures, not falsification of all self-regulation.

R44 Putnick, D. L., & Bornstein, M. H. (2016). Measurement invariance conventions and reporting: The state of the art and future directions for psychological research. Developmental Review, 41, 71–90. https://doi.org/10.1016/j.dr.2016.06.004 — Methodological review of measurement invariance across groups and times.

R45 Haggard, P. (2017). Sense of agency in the human brain. Nature Reviews Neuroscience, 18, 196–207. https://doi.org/10.1038/nrn.2017.14 — Review of agency concerning personal actions and their outcomes.

R46 Wood, W., & Rünger, D. (2016). Psychology of habit. Annual Review of Psychology, 67, 289–314. https://doi.org/10.1146/annurev-psych-122414-033417 — Review of context, repetition, habits, and goal-directed behavior.

R4790020-T) Ajzen, I. (1991). The theory of planned behavior. Organizational Behavior and Human Decision Processes, 50(2), 179–211. https://doi.org/10.1016/0749-5978(91)90020-T — Theory and research concerning attitudes, norms, perceived control, intentions, and action; not validation of the whole WRRA framework.

R48 Ostrom, E. (2009). A general framework for analyzing sustainability of social-ecological systems. Science, 325(5939), 419–422. https://doi.org/10.1126/science.1172133 — Analytical framework separating resources, actors, rules, and interactions rather than asserting one universal cooperation cause.

R49 Lazic, S. E. (2010). The problem of pseudoreplication in neuroscientific studies: Is it affecting your analysis? BMC Neuroscience, 11, 5. https://doi.org/10.1186/1471-2202-11-5 — Methodology concerning pseudoreplication through confusion of independent samples and repeated observations.

R50 Vabalas, A., Gowen, E., Poliakoff, E., & Casson, A. J. (2019). Machine learning algorithm validation with a limited sample size. PLOS ONE, 14(11), e0224365. https://doi.org/10.1371/journal.pone.0224365 — Methodology and simulations on validation procedures and leakage-related performance bias in small samples.

R51 Cohen, S., & Wills, T. A. (1985). Stress, social support, and the buffering hypothesis. Psychological Bulletin, 98(2), 310–357. https://doi.org/10.1037/0033-2909.98.2.310 — Review distinguishing direct social-support effects and stress-buffering hypotheses.

R52 Steinberg, L. (2008). A social neuroscience perspective on adolescent risk-taking. Developmental Review, 28(1), 78–106. https://doi.org/10.1016/j.dr.2007.08.002 — Theory and research review of developmental and social contexts in adolescent risk-taking; not classification of individual judgment by age alone.

R53 Barrett, L. F., Adolphs, R., Marsella, S., Martinez, A. M., & Pollak, S. D. (2019). Emotional expressions reconsidered: Challenges to inferring emotion from human facial movements. Psychological Science in the Public Interest, 20(1), 1–68. https://doi.org/10.1177/1529100619832930 — Review of context and generalization limits in inferring emotion from facial movement.

R54 Deci, E. L., Koestner, R., & Ryan, R. M. (1999). A meta-analytic review of experiments examining the effects of extrinsic rewards on intrinsic motivation. Psychological Bulletin, 125(6), 627–668. https://doi.org/10.1037/0033-2909.125.6.627 — Meta-analysis distinguishing intrinsic-motivation effects by reward type and conditions; it does not establish that all rewards are always harmful.

R55 Altman, D. G., & Bland, J. M. (2005). Standard deviations and standard errors. BMJ, 331(7521), 903. https://doi.org/10.1136/bmj.331.7521.903 — Statistical explanation distinguishing observational variation, estimator uncertainty, standard deviation, and standard error.

R56 Bouton, M. E. (2004). Context and behavioral processes in extinction. Learning & Memory, 11(5), 485–494. https://doi.org/10.1101/lm.78804 — Selective behavioral review supporting accounts of extinction and context, not one universal maintenance mechanism for all human memories.

R57 Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. Econometrica, 47(2), 263–291. https://doi.org/10.2307/1914185 — Choice problems and theory construction distinguishing reference points, value, and probability weights; no established identity with WRRA residue.

R58 Kroenke, K., Spitzer, R. L., & Williams, J. B. W. (2001). The PHQ-9: Validity of a brief depression severity measure. Journal of General Internal Medicine, 16(9), 606–613. https://doi.org/10.1046/j.1525-1497.2001.016009606.x — Original measurement validation comparing scores, interviews, and functioning, not individual causal-mechanism identification from scores alone.

R59 Carpenter, J. K., Andrews, L. A., Witcraft, S. M., Powers, M. B., Smits, J. A. J., & Hofmann, S. G. (2018). Cognitive behavioral therapy for anxiety and related disorders: A meta-analysis of randomized placebo-controlled trials. Depression and Anxiety, 35(6), 502–514. https://doi.org/10.1002/da.22728 — Meta-analysis of randomized placebo-controlled trials; interpret variation by diagnosis, analysis population, and intervention.

R60 Foa, E. B., et al. (2005). Randomized, placebo-controlled trial of exposure and ritual prevention, clomipramine, and their combination in the treatment of obsessive-compulsive disorder. American Journal of Psychiatry, 162(1), 151–161. https://doi.org/10.1176/appi.ajp.162.1.151 — Original adult outpatient OCD study comparing specific procedures and duration; generalization to all ages and interventions is separate.

R61 Lewis, C., Roberts, N. P., Andrew, M., Starling, E., & Bisson, J. I. (2020). Psychological therapies for post-traumatic stress disorder in adults: Systematic review and meta-analysis. European Journal of Psychotraumatology, 11(1), 1729633. https://doi.org/10.1080/20008198.2020.1729633 — Systematic review of adult PTSD psychotherapies, distinguishing treatment types, comparators, and study quality.

R62 32802-7) Cipriani, A., et al. (2018). Comparative efficacy and acceptability of 21 antidepressant drugs for the acute treatment of adults with major depressive disorder: A systematic review and network meta-analysis. The Lancet, 391(10128), 1357–1366. https://doi.org/10.1016/S0140-

6736(17)32802-7 — Network meta-analysis of acute adult major-depression monotherapy; not extended to long-term course, bipolar disorder, or individual prescribing.

R63 Ceraso, A., et al. (2020). Maintenance treatment with antipsychotic drugs for schizophrenia. Cochrane Database of Systematic Reviews, 8, CD008016. https://doi.org/10.1002/14651858.CD008016.pub3 — Systematic review of schizophrenia maintenance treatment, considering relapse, adverse effects, follow-up, and outcome-specific evidence quality.

R64 Cuijpers, P., Miguel, C., Harrer, M., Plessen, C. Y., Ciharova, M., Ebert, D., & Karyotaki, E. (2023). Cognitive behavior therapy vs. control conditions, other psychotherapies, pharmacotherapies and combined treatment for depression: A comprehensive meta-analysis including 409 trials with 52,702 patients. World Psychiatry, 22(1), 105–115. https://doi.org/10.1002/wps.21069 — Comparative meta-analysis of CBT for depression, distinguishing controls, sensitivity analyses, and follow-up.

R65 — Kok, P., Jehee, J. F. M., & de Lange, F. P. (2012). Less Is More: Expectation Sharpens Representations in the Primary Visual Cortex. Neuron, 75(2), 265–270. https://doi.org/10.1016/j.neuron.2012.04.034

R66 — Rao, R. P. N., & Ballard, D. H. (1999). Predictive coding in the visual cortex: a functional interpretation of some extra-classical receptive-field effects. Nature Neuroscience, 2, 79–87. https://doi.org/10.1038/4580

R67 — Friston, K., FitzGerald, T., Rigoli, F., Schwartenbeck, P., & Pezzulo, G. (2017). Active Inference: A Process Theory. Neural Computation, 29(1), 1–49. https://doi.org/10.1162/NECO_a_00912

R68 — Sauerbrei, B. A., Guo, J.-Z., Cohen, J. D., Mischiati, M., Guo, W., Kabra, M., Verma, N., Mensh, B., Branson, K., & Hantman, A. W. (2020). Cortical pattern generation during dexterous movement is input-driven. Nature, 577, 386–391. https://doi.org/10.1038/s41586-019-1869-9

R69 — Shenhav, A., Botvinick, M. M., & Cohen, J. D. (2013). The expected value of control: An integrative theory of anterior cingulate cortex function. Neuron, 79(2), 217–240. https://doi.org/10.1016/j.neuron.2013.07.007

R70 — Berridge, M. J., Lipp, P., & Bootman, M. D. (2000). The versatility and universality of calcium signalling. Nature Reviews Molecular Cell Biology, 1, 11–21. https://doi.org/10.1038/35036035

R65–R70 were checked on 22 October 2026 using publisher records, author repositories, or original-paper metadata, abstracts, and text. R65 and R68 are original empirical studies; R66 and R67 are computational theories; R69 is a theoretical integration; R70 reviews calcium signaling. Informal suggestions I1–I4 are distinguished from academic sources in Appendix C4.

R1 section-specific verification paths

R1 §1.1 — what is psychology — https://openstax.org/books/psychology-2e/pages/1-1-what-is-psychology

R1 §1.2 — history of psychology — https://openstax.org/books/psychology-2e/pages/1-2-history-of-psychology

R1 §1.3 — contemporary psychology — https://openstax.org/books/psychology-2e/pages/1-3-contemporary-psychology

R1 §2.2 — approaches to research — https://openstax.org/books/psychology-2e/pages/2-2-approaches-to-research

R1 §2.3 — analyzing findings — https://openstax.org/books/psychology-2e/pages/2-3-analyzing-findings

R1 §3.1 — human genetics — https://openstax.org/books/psychology-2e/pages/3-1-human-genetics

R1 §3.2 — cells of the nervous system — https://openstax.org/books/psychology-2e/pages/3-2-cells-of-the-nervous-system

R1 §3.4 — the brain and spinal cord — https://openstax.org/books/psychology-2e/pages/3-4-the-brain-and-spinal-cord

R1 §3.5 — the endocrine system — https://openstax.org/books/psychology-2e/pages/3-5-the-endocrine-system

R1 §4.1 — what is consciousness — https://openstax.org/books/psychology-2e/pages/4-1-what-is-consciousness

R1 §4.3 — stages of sleep — https://openstax.org/books/psychology-2e/pages/4-3-stages-of-sleep

R1 §5.1 — sensation versus perception — https://openstax.org/books/psychology-2e/pages/5-1-sensation-versus-perception

R1 §5.3 — vision — https://openstax.org/books/psychology-2e/pages/5-3-vision

R1 §5.6 — gestalt principles of perception — https://openstax.org/books/psychology-2e/pages/5-6-gestalt-principles-of-perception

R1 §6.2 — classical conditioning — https://openstax.org/books/psychology-2e/pages/6-2-classical-conditioning

R1 §6.3 — operant conditioning — https://openstax.org/books/psychology-2e/pages/6-3-operant-conditioning

R1 §6.4 — observational learning modeling — https://openstax.org/books/psychology-2e/pages/6-4-observational-learning-modeling

R1 §7.1 — what is cognition — https://openstax.org/books/psychology-2e/pages/7-1-what-is-cognition

R1 §7.2 — language — https://openstax.org/books/psychology-2e/pages/7-2-language

R1 §7.3 — problem solving — https://openstax.org/books/psychology-2e/pages/7-3-problem-solving

R1 §7.4 — what are intelligence and creativity — https://openstax.org/books/psychology-2e/pages/7-4-what-are-intelligence-and-creativity

R1 §7.5 — measures of intelligence — https://openstax.org/books/psychology-2e/pages/7-5-measures-of-intelligence

R1 §8.1 — how memory functions — https://openstax.org/books/psychology-2e/pages/8-1-how-memory-functions

R1 §8.3 — problems with memory — https://openstax.org/books/psychology-2e/pages/8-3-problems-with-memory

R1 §9.1 — what is lifespan development — https://openstax.org/books/psychology-2e/pages/9-1-what-is-lifespan-development

R1 §9.2 — lifespan theories — https://openstax.org/books/psychology-2e/pages/9-2-lifespan-theories

R1 §9.3 — stages of development — https://openstax.org/books/psychology-2e/pages/9-3-stages-of-development

R1 §10.1 — motivation — https://openstax.org/books/psychology-2e/pages/10-1-motivation

R1 §10.4 — emotion — https://openstax.org/books/psychology-2e/pages/10-4-emotion

R1 §11.4 — learning approaches — https://openstax.org/books/psychology-2e/pages/11-4-learning-approaches

R1 §11.7 — trait theorists — https://openstax.org/books/psychology-2e/pages/11-7-trait-theorists

R1 §11.8 — cultural understandings of personality — https://openstax.org/books/psychology-2e/pages/11-8-cultural-understandings-of-personality

R1 §12.1 — what is social psychology — https://openstax.org/books/psychology-2e/pages/12-1-what-is-social-psychology

R1 §12.3 — attitudes and persuasion — https://openstax.org/books/psychology-2e/pages/12-3-attitudes-and-persuasion

R1 §12.4 — conformity compliance and obedience — https://openstax.org/books/psychology-2e/pages/12-4-conformity-compliance-and-obedience

R1 §14.1 — what is stress — https://openstax.org/books/psychology-2e/pages/14-1-what-is-stress

R1 §15.2 — diagnosing and classifying psychological disorders — https://openstax.org/books/psychology-2e/pages/15-2-diagnosing-and-classifying-psychological-disorders

R1 §16.2 — types of treatment — https://openstax.org/books/psychology-2e/pages/16-2-types-of-treatment

Source checks: bibliographic details and relevant content for R35–R55 and R1 sections were checked on 19 October 2026 using public publishers, scholarly databases, and the official textbook. Undated institutional references mean no publication year was inferred. This list is not a systematic-review search. Clinical effect evidence R59–R64 is read by target and comparison conditions; broader clinical review remains subsequent work.

Thematic reading

For basic concepts, consult the specified R1 sections; for particular experiments and phenomena, use the original studies and reviews linked in the text. Clinical assessment connects to R58; anxiety, obsessive-compulsive, and trauma-related interventions to R59–R61; depression and schizophrenia treatment comparisons to R62–R64. Institutional guidance, experiments, and meta-analyses differ in evidence type and scope.

Main-text claims and their evidence

The following guide locates principal strengthened claims. Read the corresponding paragraphs for populations, methods, results, and scope. Presenting a research finding and validating a WRRA application receive separate verdicts.

Chapter 4 — R3 — Scope of a replication project

Chapter 5 — R4 — Animal experiments on connection changes

Chapter 6 — R35 — Original comparison of BOLD and neural activity

Chapter 13 — R56 — Extinction versus complete deletion

Chapter 14 — R9 — Findings of the classic observational-learning experiment

Chapter 15 — R10 — Functional inference from lesion cases

Chapter 16 — R41 — Scope of source-monitoring review

Chapter 18 — R57 — From observed choices to theory

Chapter 20 — R44 — Measurement conditions for score comparisons

Chapter 22 — R53 — Inferring emotions from facial movements

Chapter 23 — R43 — Replication tests of a resource model

Chapter 34 — R58 — Scale scores and clinical interviews

Chapter 35 — R59, R60, R61 — Placebo-controlled anxiety treatment; OCD procedure comparisons; systematic review of post-trauma treatment

Chapter 36 — R62 — Acute depression medication comparisons

Chapter 37 — R63 — Outcome-specific schizophrenia maintenance evidence

Chapter 38 — R64 — Specifying treatment comparators

Chapter 39 — R32 — Immediate performance versus delayed memory

Source verification and scope of use

The relevant content of R56–R64 was checked on 19 October 2026 in publishers or scholarly databases, using bibliographic records, abstracts, or full texts. These sources support particular claims; the book as a whole is not a systematic review. Clinical studies are presented as findings at their publication dates with populations, comparators, and follow-up specified.

Institutional introductions explain the scope of conditions and treatments; specific-effect statements connect to original studies or systematic reviews. Paragraphs on medication or psychotherapy effects do not substitute for individual treatment choices or clinical validation of WRRA models. Unperformed mechanistic connections are marked as future research.

Appendix F Common Writing Contract for the WRRA Academic Series

F1 Purpose of the common writing contract

The WRRA series starts from established research in each discipline and reconstructs phenomena through minimal computation, common carriers, and phenotype. Biology, psychology, physics, and other fields share an order for reading claims and evidence. The aim is a common explanatory and validation contract, not identical numbers of parts and chapters.

This extends the organization of *How Life Runs*: established research, WRRA reconstruction, comparative reading, interpretation–prediction boundaries, and next steps with references. Neuropsychology applies the format below.

F2 Common book structure

The preface states audience, purpose, and present achievements. The introduction briefly explains WRRA and guides readers through familiar disciplinary classifications and explanatory sequences. Chapters follow F3's eight stages. The final part organizes domain computations, synthesis, and further research.

Shared appendices include standard Core 1.0 explanation, terminology correspondences, input and variable specifications, execution records and code, references, and this contract. Core definitions remain distinct from applications; definitions from the same Core version are not arbitrarily altered between volumes.

F3 Eight stages for every chapter

Sections 1–8 form the common stages. Supplementary headings use parent-section numbers such as 1.1 and 1.2. They are not additional stages; each chapter includes as many as its sections require.

1 Scene and question: begin with an observable phenomenon or concrete problem and delimit the explanatory question. Distinguish factual from hypothetical scenes.

2 Starting points in established research: present established mechanisms, major theories, measurement scope, and open questions. Existing achievements are not rewritten as WRRA discoveries.

3 Validation inputs: distinguish observations, constants, datasets, literature, and hypothetical settings. Record sources, units, timing, errors, and calibration roles. Missing data are labeled acquisition plans.

4 WRRA-specific transformation: specify task-relevant SOURCE, LAW/RELATION, STATE/RESIDUE, BOUNDARY, COMMON CARRIER, UPDATE, RENDERER, and observation paths. Select the structures needed for explanation and computation rather than mechanically adding every term. Equations, code, and actual connections distinguish external citation from internal implementation.

5 Outputs and execution record: record actual numerical, explanatory, and model outputs. Unexecuted outputs are planned outputs. State environment, code, inputs, and tolerances and distinguish empirical from hypothetical data.

6 Falsification conditions and interpretive limits: specify which observations or computations challenge which claims. State limits of interpretation, reproduction, and prediction; distinguish coefficient and structural changes; disclose alternatives and identifiability.

7 Current verdict: judge each claim after presenting inputs, transformations, outputs, and falsification conditions. Do not assign one blanket grade to a whole chapter or discipline. End with one sentence expressing the chapter's point.

8 Next questions and evidence: specify data to obtain and work to execute and connect chapters, references, and DOIs. Disclose absent raw data or code.

F4 Meaning of claim-level verdicts

EXACT: a relationship mathematically derived or proved under stated assumptions. Numerical agreement alone is insufficient.

EMPIRICAL PASS: a claim supported within specified actual data, comparison criteria, and tolerances. State data scope and calibration roles.

COMPUTATIONAL PASS: successful execution and reproduction under fixed inputs, rules, and checks. Fit to actual phenomena requires separate evidence.

INFERRED: a structure or interpretation proposed from existing evidence without direct verification of that connection or completed computation.

FAIL: failure of the claim or execution check under prespecified conditions. Preserve failure scope and reproduction procedure.

OPEN: a concrete question and verification path remain, with current evidence insufficient to close the conclusion.

NOT ESTABLISHED: required evidence is not established, so the claim cannot yet be asserted. Unperformed research is not success; absence of failure evidence is not automatically FAIL.

These labels are not one ranking. One model may have EXACT mathematical properties, COMPUTATIONAL PASS implementation, and OPEN empirical fit. Verdicts are scoped to individual claims.

F5 Reproduction, calibration, and prediction as achievements

Revising a model using verified constants and observations is legitimate construction. The ledger records which data determine which rules or coefficients. Reproducing known values through its own structure counts as reality consistency and explanation.

Originality is assessed through WRRA structure, transformations, connections, and explanation, not numerical novelty alone. Independent prediction is not mandatory for theoretical standing or originality. Once a model is fixed using verified values, deriving unmeasured quantities through its own structure counts as that fixed model's WRRA prediction.

Other theories producing equal numbers do not erase calculations actually performed within WRRA. External theories support validation and reinforcement; connections not implemented or computed internally are not marked fully integrated. Comparison models receive the same standards for data, error, and claim scope.

F6 Validation methods vary by field

Biology and neuropsychology use observation, experiment, and analysis; physics uses measurement, mathematical derivation, and model computation as appropriate. Mathematical claims require assumptions, proofs, and counterexample conditions; social sciences require sampling, measurement, and identification conditions. Fields need not fit one experimental type.

The common format makes evidence comparable. It does not fill missing data with terminology or force different objects and timescales into one numerical scale.

F7 Claim-ledger fields

Connect claim identifier and statement, target and scope, input sources and versions, roles of constants and coefficients, WRRA transformations and code locations, outputs and errors, falsification conditions, verdicts, revision histories, and next tests. Separate subclaims where evidence types differ.

Neuropsychology example NP41 uses hypothetical initial trace 1 and retention 0.85, updating the next r as 0.85r to obtain 0.196874 after 10 updates. Reexecution outside the table's rounding tolerance fails the implementation check. Its verdict is COMPUTATIONAL PASS; fit to human memory is OPEN.

Example NP42 uses hypothetical failures and resource values, combining residue updates with logistic output to calculate choice probabilities. Latent-scale transformations preserving observed probabilities are also checked. Implementation reproduction, latent identifiability, and fit to human behavior receive separate verdicts.

F8 Manuscript-completion checks

Check that questions connect to inputs, external evidence differs from WRRA work, and execution records differ from plans. Inputs, transformations, outputs, and falsification conditions precede conclusions. Literature alone does not complete implementation; hypothetical calculations are not observations.

Titles, contents, and cases may vary across disciplines, but stage order and verdict meanings remain common. Record Core, domain-model, and editorial versions separately to track later revisions.

F9 In-text citations and evidence scope

Place author–year citations and identifiers immediately beside sentences or short paragraphs stating existing concepts or findings. Specify textbook sections, article DOIs, or institutional pages. Do not leave readers to connect claims to a bibliography on their own.

Where established research and author applications share a paragraph, separate the statements and label WRRA interpretation or proposed design. Do not generalize beyond species, samples, tasks, and measurement scope. Reviews and meta-analyses are not individual experiments; institutional guidance is not evidence establishing a particular intervention's effect.

Hypothetical inputs, computations, and reproduction code are distinguished from external experimental data. Additional citations do not automatically change empirical verdicts. Unsupported connections are narrowed or recorded as hypotheses and subsequent work.

Index

Available resources: Chapters 2, 11, 24, 42 / Attention: Chapter 11 / Boundaries: Chapter 2, Appendix A / Common carrier: Chapters 2, 5, Appendix A / Consciousness: Chapter 12 / Development: Chapters 25–27 / Emotion: Chapters 22–23 / Falsification: Chapter 44 / Learning: Chapters 13–14 / Memory: Chapters 15–16, 41 / Minimal computation: Chapters 2, 11, 39 / Motivation: Chapter 21 / Neuroimaging: Chapters 6, 43 / Personality: Chapter 28 / Phenotype: Chapter 2, Appendix A / Present state: Chapters 2, 40 / Relationships: Chapters 30–33 / Residue: Chapters 2, 16, Appendix A / Sensation: Chapter 9 / Core 1.0 paper: Appendix A11.